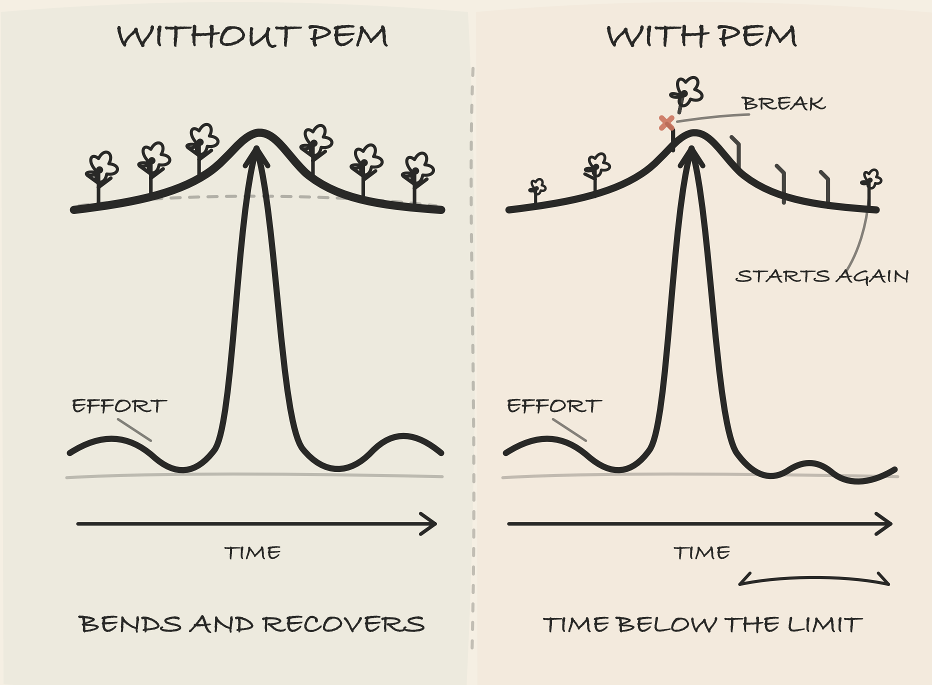
UNDERSTANDING PEM

What the Research Says in 2026



THE MANUAL YOUR BODY DIDN'T COME WITH

A Synthesis of Current Research on PEM, Long COVID, ME/CFS, and POTS

JANE SMORODNIKOVA

UNDERSTANDING PEM

What the Research Says in 2026

The Manual Your Body Didn't Come With

A Synthesis of Current Research on PEM, Long COVID, ME/CFS, and POTS

JANE SMORODNIKOVA

Independently published · 2026

Jane Smorodnikova, founder and CEO of Welltory. 2026. The author's affiliation does not represent an endorsement by Welltory of any content herein.

Contents

A Note from the Author

Hi. I'm Jane Smorodnikova, and I founded Welltory — the app that provides general-wellness insights based on signals such as heart-rate variability, sleep, and activity. Over the past year I've also spent much of my time studying post-infectious chronic conditions: Long COVID, ME/CFS, POTS, and the broader group of conditions often described as energy-limiting conditions.

This book grew out of research I had already begun and a decision I made at the start of 2026 to turn that work into a practical guide.

For most of Welltory's history we built general-wellness tools primarily for people who wanted to understand signals such as HRV, stress, sleep, activity, and recovery. When we looked carefully at who was actually using the product in 2025, we found that a meaningful share were people living with chronic conditions, including Long COVID, ME/CFS, POTS, and fibromyalgia.

Some told me they looked at Welltory's general-wellness trends — including HRV, sleep, activity, and heart-rate patterns — alongside their own symptom records and conversations with their healthcare providers. They described using this information to reflect on patterns such as changes in energy after activity, delayed worsening of symptoms, heart-rate changes on standing, and differences between better and more difficult days.

These were user-led uses for which we had not designed the product.

That experience led me to study the research and patient experience around these conditions more closely and to write down what I found.

Patients and clinicians often pay attention to pacing, delayed symptom changes, and the relationship between activity and recovery.

Separately, many people already track everyday signals such as heart-rate variability, sleep, activity, and heart rate for general wellbeing.

I became interested in what the published research could — and could not — tell us about the relationship between those everyday signals and the experiences people describe. These signals are not diagnostic or disease-monitoring tools, but they may provide additional general-wellness context for personal observation and conversations with qualified healthcare providers.

To approach the subject responsibly, I spent most of the last year reading peer-reviewed studies, rehabilitation guidelines, patient-led research, clinician interviews, and long-form recovery archives. What you are reading now is the practical, science-forward map I developed during that process: an explanation of how these conditions are currently understood, how patients experience them, where the evidence is stronger or weaker, and which questions may be useful to explore with qualified healthcare professionals.

I'm sharing it because it would be strange not to. If I spent a year trying to understand the science and lived experience behind an illness you live with, keeping the summary to myself would waste most of the value of the work. So this book is published for a general audience and available to anyone on the same terms — it isn't a benefit of any subscription, and you don't have to use Welltory to read it.

The part where the lawyers make me speak carefully

Below are the boundaries this book runs inside. Read them to know what you are holding, or skip to Chapter 1 if you already know how disclaimers work.

Who I am, and where Welltory sits in this. I wrote this book as an independent author. It is not a Welltory product and not a medical service, and my affiliation does not mean Welltory endorses anything in it. As for the app: The Welltory app does not diagnose, predict, monitor, prevent, treat, or mitigate any disease or medical condition, or determine whether a person has one.

What this book will not do. It cannot diagnose you, replace a clinician who knows your history, prescribe a medication, or change a dose. It cannot promise recovery on a schedule. And whatever the answer turns out to be, it is not “try harder” — most people reading this have already tried harder than they should have, and effort alone is not what these illnesses respond to.

What the information is good for. Understanding concepts that may be relevant to your experience; keeping clearer records of symptoms, activity, and change over time; preparing sharper questions for qualified healthcare professionals; and reading the benefits and the limits of the self-observation tools people discuss. Rather than telling you which treatment is right for you, it helps you describe your experience clearly enough to work that out with someone qualified.

Catherine. There is a woman named Catherine whose story runs through this book. She is not a real patient. She is a composite, assembled from the experiences of hundreds of people with Long COVID, ME/CFS, POTS, and similar conditions, drawn from clinical studies, rehabilitation literature, published interviews, and patient community archives. Any resemblance to a specific individual is coincidental. She is in the book because a single case, carefully built, lets us illustrate patterns that are hard to see in aggregate numbers. Her story should not be understood as a clinical case report or as representing the experience of every person with these conditions.

How the book was built. The chapters draw on four sources. First, clinical and research literature — several hundred peer-reviewed studies, rehabilitation guidelines, and consensus statements, with particular weight given to the evidence base that informed the UK's NICE guideline revision in 2021 and the NIH-funded RECOVER initiative in the United States. Second, medical and scientific review focused on general educational and scientific accuracy; where patient literature and that clinical reading of the evidence diverged, I followed the clinical reading. That review was not a clinical approval and did not establish a standard of care. This review reflects general educational and scientific accuracy, not a clinical position, medical recommendation, or treatment recommendation of Welltory. Third, patient-led research — the work of communities that documented these illnesses before the medical system recognized them. Fourth, long-form patient interviews, including the Raelan Agle Recovery Stories archive and other patient-facing channels, whose voices fill some of the remaining gaps in the clinical literature.

Throughout the book I quote, with attribution, from the published work of many clinicians, researchers, and writers — including Bateman, Van Ness, Stevens, Systrom, Davis, Putrino, Peluso, Medinger, Altmann, Ballering, and others. These excerpts are clearly attributed and set apart from the surrounding text, so you can encounter these voices alongside my synthesis rather than only my summary.

About the data. Research cited here was current as of April 2026. The field of post-infectious chronic conditions is very active, as findings replicate, fail to replicate, or are refined. Some of the specific numbers in this book will be outdated within a year. However, the underlying patterns — how PEM behaves, how POTS subtypes differ, how autonomic dysregulation presents — have appeared consistently across a decade of ME/CFS literature. Their mechanisms, classification, and measurement nevertheless remain active areas of research. For research updates, the US Patient-Led Research Collaborative and the ME Association in the UK publish accessible summaries.

You will notice that the numbers in the literature are often wide. When this book cites the prevalence of autonomic dysfunction in Long COVID, the range runs from roughly a third to over eighty percent. That spread is an accurate description of cohorts grouped by diagnosis rather than by how their bodies respond to exertion. Measuring different groups with different tools produces different numbers. Tighter numbers require a phenotypically uniform group. That is why the work now runs through an observational cohort we are following, selected for a shared post-exertional malaise phenotype rather than a chart label. Everyday signals, read as general-wellness context rather than as anything diagnostic, may make it possible to see patterns that disappear when you average a room full of people whose underlying mechanisms differ. This book is the map drawn before that observation begins. It represents the best synthesis the literature offers in the spring of 2026, and the next book will be written on the data we gather. The observation has received IRB approval, and as I write this it is still open.

Everything else is the book.

— Jane Smorodnikova

Some sections of this book discuss research conducted or supported by Welltory on recovery patterns, HRV, activity, sleep, and energy-limiting conditions. These references describe exploratory research and research hypotheses. They do not describe a current consumer feature of the Welltory app unless expressly stated otherwise. The Welltory app does not diagnose, predict, monitor, prevent, treat, or mitigate Long COVID, ME/CFS, POTS, PEM, dysautonomia, or any other disease or medical condition. Any wellness trends shown in the app are provided for general wellbeing context, lifestyle reflection, and conversations with a qualified healthcare provider.

Introduction

When your body speaks, and no one listens

Catherine was the person people could always call. The friend you rang for a ten-minute chat about nothing much, or when you needed someone to pick up the kids from soccer practice on short notice. She was an elementary school teacher in the suburbs of Columbus, Ohio, with two sons, eight and eleven, a husband named Dan, and an old yellow Lab named Charlie. On an ordinary Sunday morning with coffee and the sound of children arguing about cereal, everything was in its place.

Then in March 2021, she contracted COVID. Not a bad case. She handled it at home. Her doctor wrote “mild illness, no complications” on the chart.

Six weeks later, she could not shower standing up.

Not because it hurt. Not because she was dizzy, although she was dizzy too. Her body — the one that had stood, walked, worked, and raised children without ever asking for attention — had started charging her for things that used to be free. Standing at the sink. Walking to the mailbox. Answering a text message when her head felt full of static and her fingers had stopped feeling like hers.

Her labs were normal.

The primary care doctor looked at the results and gave a small, honest shrug. Her cardiologist found no pathology. A neurologist said they should keep an eye on it. One physician gently suggested she might benefit from talking to a therapist. Catherine did not take offense. She had already started wondering the same thing herself. Maybe she was doing something wrong. Not trying hard enough. Or perhaps not wanting it enough.

This was the second illness. The first had a medical name: COVID. The second had no name on any chart. It was the slow dismantling of her trust in her own body.

Catherine is a woman, but this illness does not choose by sex. Post-infectious chronic conditions affect men, women, and teenagers. Women are diagnosed more often, for reasons that are partly biological and partly about which complaints the medical system takes seriously; that is a longer conversation the book will return to. Among patients are former athletes, programmers, nurses who caught the infection in the first wave of a surge, teenagers who lost a year of school, retirees who cannot understand why they never bounced back from what their doctors called a light case. If Catherine's specific story is not yours, that is to be expected. She is one possible portrait. Not the only one.

The conditions this book is about

Catherine has Long COVID. The book, however, reaches further. It addresses a family of conditions that share a particular kind of patient experience: delayed crashes after effort, a diagnostic system that looks in the wrong places, wide swings between good days and bad days, and symptoms that are difficult or impossible to see in ordinary lab tests. The umbrella term now used across US health policy is *Infection-Associated Chronic Conditions* (IACC). It appears in Department of Health and Human Services usage and sits at the centre of the 2024 National Academies of Sciences, Engineering, and Medicine report that HHS commissioned to define Long COVID. These include:

- Long COVID, the largest cohort and the entry point for many readers
- Myalgic Encephalomyelitis / Chronic Fatigue Syndrome (ME/CFS), the community whose decades of work the rest of this field stands on

- Postural Orthostatic Tachycardia Syndrome (POTS) and other forms of post-infectious dysautonomia
- Mast Cell Activation Syndrome (MCAS) triggered or unmasked by infection
- Post-Lyme, post-mononucleosis (post-EBV), post-SARS, post-Ebola, and post-infectious chronic conditions more broadly
- Fibromyalgia with a post-exertional malaise component, when the triggering event can be traced

All of these conditions share a symptom pattern called *post-exertional malaise* — a worsening of symptoms after physical, cognitive, or emotional effort, most often arriving 48 to 72 hours later. It can come sooner, from about a day out. The 48-to-72-hour window is the one this book plans around — the payback window, named and explained in Chapter 6. PEM is the thread the book follows, because it is what makes these conditions behave differently from the illnesses medicine is most confident about. The family is sometimes called the *energy-limiting conditions* in patient literature. We use that phrase too, because it describes what it feels like from the inside.

Long COVID made this family visible to people outside it. ME/CFS patients had been living in this territory for decades. POTS patients, too. Long COVID did not create the terrain; it lit up the room that already existed and brought several million new people into it.

Several mechanisms, acting at once

If this is your first attempt to understand what happened to your body, the first thing worth knowing is that there is no single answer, and anyone who tells you otherwise is simplifying in a way that will cost you time.

Post-infectious chronic conditions are multifactorial. What that means in practice is that several distinct biological mechanisms are usually acting at the same time and interacting with each other, and no one of them can correctly be described as “the cause.” Based on the published evidence as of 2026, the mechanisms most consistently implicated are: immune dysregulation, autonomic nervous system dysfunction, microcirculatory and endothelial disturbances, changes in mitochondrial activity, and over-sensitization. We cover each of these mechanisms in more detail in Chapter 1.

These mechanisms overlap and amplify each other. They are present in different proportions in different patients, which is why one patient’s illness looks different from another’s, and why a particular treatment will help one person and do nothing for the next. Complexity of this kind is inherent to the category, not a defect in the science that has tried to describe it.

The practical consequence for you is that you need a comprehensive, individualized assessment. Not a search for The One Thing.

Why your tests are normal

A normal blood panel does not mean your body is working 100% properly; it means the specific things the panel measured fell within the reference range on the day the blood was drawn. Most routine panels look at electrolytes, basic kidney and liver markers, blood counts, and a handful of hormones. They were designed to catch a different kind of illness — acute injury, organ failure, infection, anemia, diabetes — not the regulatory failures characteristic of this family of conditions.

Autonomic dysregulation does not produce a lab marker. A tilt-table test or a ten-minute stand test with a heart-rate monitor can catch it; a fasting blood draw cannot. Microvascular dysfunction often leaves gross organ function intact. Reduced oxygen extraction at the tissue level can coexist with a normal pulse oximeter reading and a normal resting ECG.

Changes in pain processing do not show up on an MRI of the joint that hurts.

So this is the first thing to understand: normal test results and normal function are not equivalent to one another. One is about what the lab measured. The other is about how your body works in life. A cardiologist can be right that your heart is not diseased, and you can be right that something is wrong with how it behaves when you stand up. Both can be true. Usually both are.

The patient community saw this first

The term *Long COVID* was coined by patients, in a Twitter thread in May 2020. They named their illness before the medical system recognized it existed (Callard & Perego, 2021). The same community, working with patient-led researchers, produced one of the first detailed maps of Long COVID symptoms: an international survey of 3,762 people that documented 203 symptoms across essentially all body systems, published in *EClinicalMedicine* in 2021 (Davis et al., 2021). That map predated most institutional descriptions by a year or more.

The point here is practical, not sentimental. In conditions where medicine has not caught up, and especially in conditions where medicine's ordinary instruments cannot see what the patient is living through, the patient community becomes a primary source of data, not a footnote to research. This book treats it that way.

This book also treats the ME/CFS community as the foundation on which the rest of this understanding is built. When Long COVID patients began describing delayed crashes, they were describing post-exertional malaise — a phenomenon ME/CFS patients had named, measured, and lived with for three decades before COVID existed. When POTS patients explained to Long COVID patients how to test for orthostatic intolerance at home, they were passing along knowledge they had organized across twenty years. If the last four years have shortened the learning curve for Long COVID patients, it is largely because a community that had been dismissed for decades was ready with a map.

What you will find in this book

The book is arranged as a path, from understanding to action.

- First, the mechanics. Why the body behaves as it does after infection. How blood work can look ordinary while your life does not. Why five apparently unrelated diagnoses on your chart — POTS, IBS, fibromyalgia, migraine, anxiety — are often the same regulatory problem seen from five different angles. Without this layer, practical advice tends to sit at a distance from the body it is trying to help.
- Then, tools. How to measure autonomic function at home in ten minutes and what to do with what you find. Ways of managing an energy budget that is smaller than it used to be — not through willpower, but through arithmetic. Choices in food, sleep, and movement that keep tomorrow within reach.
- Then, navigation. How to convert months of lived experience into a single page of data a clinician can read in two minutes. Which questions to ask. Which tests to request. When to look for a different specialist, and how to tell.
- Finally, life. Grief for the version of yourself that existed before. Relationships with people who see you laughing on Friday and cannot understand why you cannot get out of bed on Saturday. Psychological support as a tool — a way of handling an unfamiliar body's new demands — rather than as a suggestion that the illness is imagined.

Because knowledge without a next step is just anxiety with more detail.

In the appendices, you will find a symptom diary template designed around a 48-hour window, a one-page summary for clinician visits, and a short glossary of terms that will come up repeatedly.

One thing before we go further

Medicine has not caught up with this illness. Not because physicians are bad at their work — most are trying in conditions that make good work hard — but because post-infectious chronic conditions do not fit the template that modern medicine is organized around. The template looks for a problem in one organ, with a clear test and a specific treatment. These conditions are not that shape. They are organism-level dysfunctions of regulation, not localized diseases. A good cardiologist can tell you honestly that your heart is structurally fine and be exactly right. It was never the question that needed asking.

This book is not a case against medicine, but a guide for moving inside it with better equipment. The physician who takes your case seriously, refers you to a dysautonomia clinic, and agrees to read a printed summary of your symptom history, is an ally worth keeping. What you need is not a doctor who knows everything — no such doctor exists — but a way of thinking about your body, language for speaking with clinicians, and tools that work now, while the larger system catches up.

Catherine's story did not end with that shower. She found answers — not all at once, not without losses, and not without anger at how long it took. But the first step was not a medication and not a specialist. The first step was understanding what had changed.

That is where we will begin.

Optional self-observation prompt

Tonight or tomorrow morning, open a note on your phone or take out a piece of paper. For the next two weeks, keep a simple record. For each day, note three things: what you did (only what was non-trivial — a shower, a work call, a short walk, a trip to the store), how you felt at the time, and how you felt the next morning and the morning after that. Some crashes arrive the same evening. Many arrive a day or two later. The delay itself is part of the illness; the pattern is what you are looking for.

At the end of two weeks, read the record from the beginning. Look for days where the crash was out of proportion to the activity. Note activities that cost more than they seemed to at the time. Pay attention to what lying down did for you, and how long it took to make a difference.

This is not busywork. It is the start of your own manual, the one the hospital discharge papers did not include.

Sources

- Callard F, Perego E. How and why patients made Long Covid. *Social Science & Medicine*. 2021;268:113426. DOI: 10.1016/j.socscimed.2020.113426
- Davis HE, Assaf GS, McCorkell L, et al. Characterizing Long COVID in an international cohort: 7 months of symptoms and their impact. *EClinicalMedicine*. 2021;38:101019. DOI: 10.1016/j.eclinm.2021.101019

Chapter 1. Fatigue is not the right word

Three months after her mild COVID, Catherine tried to explain to her husband Dan what was happening to her. He had just come home from work. He was sitting at the kitchen counter with a mug of coffee, looking at her with the expression she had learned to recognize across years of marriage — a mix of worry and bewilderment. “Maybe just start small,” he said. “A short walk. Some fresh air.” He wasn’t being cruel. He was simply living in a world where tiredness gets fixed by rest or by a change of scene.

Catherine used to live in that world too.

She didn’t live there anymore.

“Fatigue” may be one of the most overloaded words in medicine. It covers the healthy person at the end of a long day, the runner after a marathon, the depression that makes mornings feel impossible, anemia, or an underactive thyroid. But it also covers what happens in Long COVID, ME/CFS (Myalgic Encephalomyelitis/Chronic Fatigue Syndrome), and similar post-infectious conditions. One word, very different machinery underneath. That confusion has cost many patients months, sometimes years, of the wrong advice.

To understand what’s happening in your body, it helps to start with how your state differs from what most people call fatigue. The difference isn’t a matter of degree; you aren’t just “more tired than usual.” The mechanics have changed. The bill now arrives on a different schedule, and for reasons that don't show up on a standard blood panel.

Three kinds of tired, and why the difference matters

Imagine three people at the end of a long week.

The first is the person having a normal tired day. She spent more energy than usual today, but sleep, food, and a slow Sunday will refill what she spent. The ordinary fatigue of healthy people has several components running in the background: adenosine building up in the brain as a signal that it's time to sleep, circadian rhythms dipping in the late afternoon regardless of what you did, and small chemical shifts at the level of muscles and nerves. None of it is broken. Rest puts it back.

The second has been off her feet for a long stretch — maybe after surgery, maybe after a long bedridden illness, maybe just a year of not moving much. Her heart has to work harder to do what used to be easy, her muscles protest sooner, standing feels like effort. This is deconditioning. It's real, and it's tiring. But it has a kind plot: start moving again, in careful steps, and within weeks the body rebuilds what it lost. Deconditioning fixes itself by use.

The third person looks, from the outside, like she might be in one of those first two categories. She is often told so. Underneath, something else is going on. For her, the more she does today, the worse she will be in two days. Rest doesn't quite rebuild her the way it used to, and movement doesn't bring her fitness back. Instead, a walk that would have been easy a year ago can leave her flattened for a week. The blood work comes back clean, the cardiologist sees nothing wrong with her heart, the neurologist suggests observation. She leaves each appointment with the quiet sense that she is failing a test she can't study for.

That third state is what this book is about. It is not laziness, not deconditioning, not anxiety presenting itself in disguise. It is what happens when the systems that produce energy, regulate blood flow, manage inflammation, and coordinate recovery stop lining up the way they used to. The cost of a task isn't felt as you do it. It shows up later.

This is the part the standard advice gets wrong. "Just start with a short walk" works beautifully for deconditioning. Applied to the third state, it can send someone backward for weeks.

What's happening inside — a multi-factor picture

For years, the most common explanation offered to patients was that their mitochondria — organelles in the cell that help make energy — were the root of the problem. The picture was tidy: the cell's energy units were damaged, and everything else followed from that. Tidy, but too narrow.

The picture that is now emerging is more complicated. In people with post-infectious conditions like Long COVID, ME/CFS, and similar syndromes, several distinct biological processes appear to act at the same time and to reinforce each other:

- **Immune dysregulation.** The immune system doesn't fully stand down after the infection. Low-grade inflammation persists; in some patients there are signs of possible autoimmune reactions. The body stays in a state somewhere between fighting and recovering, which consumes resources and alters how other systems respond to load.
- **Autonomic nervous system dysregulation.** The part of the nervous system that runs heart rate, breathing, blood pressure, digestion, and temperature on autopilot stops responding to everyday demands the way it used to. Standing up, eating, a warm shower: seemingly small prompts receive large responses. This is the subject of Chapter 2.
- **Microcirculatory and endothelial changes.** The lining of small blood vessels (the endothelium) shows signs of damage in some patients, and blood flow through the smallest vessels can be uneven.

There is also an active hypothesis — not yet fully confirmed — that in some people, very small clots may be obstructing the capillaries (the tiniest blood vessels), leaving tissues slightly short of oxygen even when the heart and lungs look fine.

- **Changes in cellular energy metabolism.** In some studies, mitochondrial function appears to be altered, with reduced synthesis of ATP (Adenosine Triphosphate), the primary energy currency of the cell. Within mitochondria, the most efficient metabolic pathway to produce ATP is via the enzyme Complex V (also called ATP synthase). Complex V uses "proton power" to synthesize ATP, producing roughly 30 molecules of ATP per molecule of glucose. However, increased proton leakage across the mitochondrial membrane can badly affect this process, and at the whole-body level, a shift is seen toward much less efficient energy pathways such as glycolysis, which yields just two ATP per glucose.
- **Central sensitization.** The brain's and spinal cord's processing of pain and sensory information changes. Signals that used to pass unnoticed become loud. This is the subject of Chapter 3.

Added to these processes are individual predispositions — connective tissue features (hypermobility, EDS traits), prior autonomic sensitivity, genetic factors — which don't cause the illness but can make some bodies more vulnerable to tipping into it after a trigger.

None of these mechanisms is "the" cause. They overlap. They amplify each other. Which is why two patients with the same diagnosis can have different symptom profiles, and why the treatment that helps one may not help another.

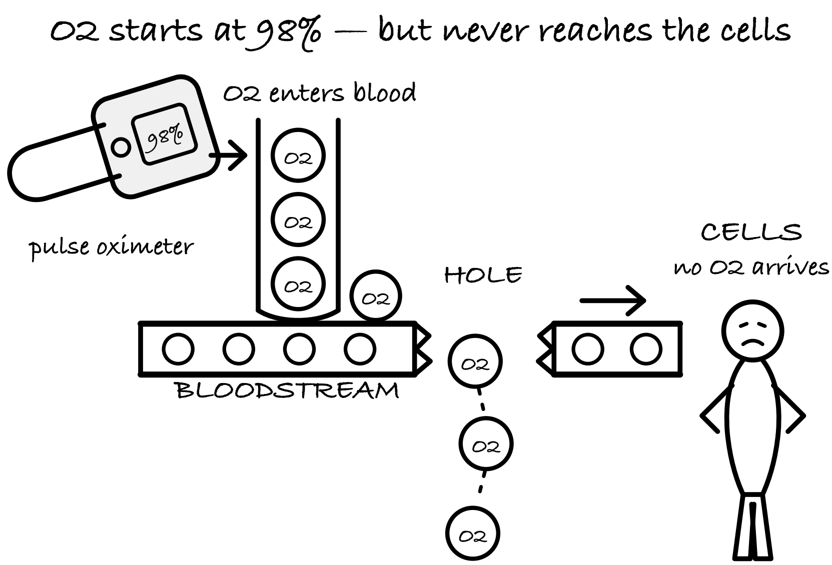
The most important consequence of the multi-factor view is this: if you were told earlier that your fatigue was "just mitochondrial" and that some supplement should fix it, it's not that you failed the supplement. The model was too small.

Oxygen can arrive and still not be used

Many patients with post-infectious conditions hear, from more than one specialist: "your oxygen saturation is normal, your lungs sound clear, your resting heart looks fine." All of that can be true and your body can still be starving for oxygen when it's asked to work.

The reason is due to a distinction that standard tests tend to flatten. Oxygen delivery and oxygen use are not the same step.

A pulse oximeter on your fingertip confirms that oxygen is being loaded onto your red blood cells. It says the mail arrived at the mailbox. It doesn't say whether anyone opened the envelope — whether that oxygen is reaching tissues, and whether those tissues can take it up and turn it into usable energy.



A pulse oximeter reports delivery. It says nothing about use.

Research on Long COVID and ME/CFS has found that delivery isn't usually the main problem. Extraction is. Invasive cardiopulmonary studies in Long COVID patients with clean lungs, clean chest imaging, and clean resting cardiac function, have shown significantly impaired oxygen extraction at the tissue level compared with healthy controls [1]. Gez Medinger and Danny Altmann, who write about Long COVID for a general audience, summarize the finding this way:

"Patients with Long COVID in whom no abnormalities were detected in chest imaging, pulmonary function or resting heart function had significantly impaired oxygen extraction compared with healthy controls. The essential delivery mechanism for oxygen transfer simply wasn't working as well as it should."

— Gez Medinger & Danny Altmann, *The Long COVID Handbook*

Across studies from 2021 to 2024, the "oxygen pulse" — a measure of how much oxygen the body can deliver per heartbeat and how efficiently the tissues extract it — came in significantly below expected values in Long COVID patients; in one large cohort, roughly 41% of patients had a peak oxygen uptake (VO_2 peak) below 84% of the predicted level [1] [2]. In practical terms this means that: the heart pumps fine at rest, the lungs pass the standard tests, and the body still collapses on a flight of stairs because the step between blood and cell isn't efficient anymore.

Here is a line worth borrowing for the next conversation with your doctor. "My resting tests are normal, but I have exertional intolerance with delayed worsening. The failure shows up under load and keeps unfolding for days afterward." That sentence is closer to the physiology than anything the word "fatigue" captures.

The second-day drop

If there's one test that shattered the "you're just out of shape" story, it was the two-day cardiopulmonary exercise test (CPET).

The logic is plain. A healthy person does a maximal exercise test one day and again the next. The two results look about the same. Being tired from yesterday shaves a little off today's numbers, but the performance is reproducible. Low ceiling, same building.

In ME/CFS, the two days often don't match. Patients can perform close to expected on day one. On day two, their oxygen uptake at the ventilatory threshold — the point at which the body starts to shift toward less efficient, anaerobic-heavy energy production — drops by roughly 30% compared with the first day [3]. Deconditioning does not behave like that; being out of shape lowers how high you can go, but it doesn't cause the ground floor to drop overnight.

The mechanism behind the two-day drop isn't simple "ATP hasn't had time to refill." It's closer to several systems failing to recover together. After a heavy effort, the autonomic nervous system doesn't reset its regulation of heart rate and vascular tone the way it should. Small amounts of muscle damage from an unfamiliar load produce inflammatory signaling that lingers. Tissue-level oxygen extraction, already reduced, is even less efficient under repeated demand. The body's ability to rapidly produce and use energy in response to a load is constrained, and the constraint deepens on day two [4].

The broader phenomenon has a name: post-exertional malaise, or PEM. Current clinical guidance describes it as a worsening of symptoms beginning hours to a day or two after the triggering activity, lasting days or weeks [5]. The U.S. National Academies (formerly IOM) identified PEM as a hallmark feature of ME/CFS in their 2015 review:

"The committee concluded that post-exertional malaise — a worsening of symptoms following even minor physical, cognitive, or emotional exertion — is a cardinal feature of ME/CFS, not an optional or peripheral symptom."

— paraphrased from the National Academies of Sciences, Engineering, and Medicine, *Beyond Myalgic Encephalomyelitis/Chronic Fatigue Syndrome: Redefining an Illness* (2015)

The practical consequence is uncomfortable. How you feel in the moment is a poor guide. Your body sends the bill late. A brisk walk that felt fine at 11 a.m. on Tuesday can turn into a fogged-out, heavy-limbed, can't-string-a-sentence-together Thursday, and the connection between the two is not visible from the outside.

This is also why exercise programs built for deconditioning can hurt people with PEM. Adding load to a system that reliably breaks down from load isn't the same kind of rehabilitation.

> **SAFETY —** A formal two-day CPET can itself trigger a serious and sometimes lasting worsening. For many people it is not worth doing simply to prove a point to a skeptical clinician. There are safer routes to the same information, covered in Chapter 5 and Chapter 9.

Here's another line worth keeping in your pocket for doctors, partners, and relatives: "**I can sometimes do the task once. My problem is that I cannot reliably recover from it.**"

Why the body doesn't return to normal

If all of this started with an infection, why didn't the body reset when the acute illness ended?

There isn't a single settled answer. Several mechanisms are under active study, and they are not mutually exclusive.

Viral persistence. Fragments of SARS-CoV-2 — and in some cases evidence of live virus — have been detected in tissues in the gut, lymph nodes, and nervous system months after a negative swab [6]. When the immune system continues to detect pieces of a pathogen, it stays partly activated longer than it should. That continued activation is expensive: it alters metabolic regulation, affects how other systems respond to load, and keeps inflammatory signaling elevated in the background. More recently, researchers have named the molecular hook: the spike protein itself, circulating in plasma months after infection in some patients, behaves as a danger signal the innate immune system keeps reading as "pathogen still here," which helps explain why the background activation does not fully stand down [9].

Immune dysregulation. A number of inflammatory markers remain elevated for months in some Long COVID patients [7]. A body sitting in that kind of low-grade signaling for that long doesn't feel normal, regardless of how clean the standard panels look.

Endothelial changes. The lining of small blood vessels can be damaged after certain infections, and that damage affects how blood is distributed through tissues and how efficiently cells receive oxygen [8]. A related hypothesis — microclots, very small clots in the smallest vessels, not visible on standard coagulation tests — has drawn a great deal of attention; it is a promising line of research but not yet fully confirmed.

All three of these can coexist in the same person. That is part of why no single "cure" has emerged yet: a treatment that works on one mechanism may do little for a patient in whom a different mechanism dominates.

What this changes for you

Understanding the machinery changes the questions you ask.

Instead of "why am I so weak?", a better question is: "what systems are currently overloaded?" Instead of "shouldn't I be able to do this by now?", "what is this task going to cost me 48 to 72 hours from now?"

A shower is the classic example, and the one most patients recognize. From the outside, it looks like a single task. To a body that isn't extracting oxygen well, isn't regulating blood flow smoothly, and is carrying a background inflammatory load, it's more like five stacked tasks: standing, heat exposure, arm movements overhead, sensory input, and recovery afterwards. Each of those pulls from the same constrained pool. The real cost isn't any one of them — it's the combination, and the combination is what the body is billing you for over the next two days.

The first shift this makes possible is small but important: stop judging a task by how trivial it looks from the outside, and start judging it by how many systems it loads at once. A short phone call plus standing plus concentrating is not a small task on a strained system; it is three loads on one circuit. Once you can see that, pacing stops being mysterious. It becomes bookkeeping.

None of this makes you weaker than other people. It means your body is running a higher overhead cost to do the same work, and that cost keeps accruing after the work is done.

Optional self-observation prompt

Start an activity diary with a 48-hour window. Not "how am I feeling today?" — but "what did I do one, two, and three days ago?"

Pick one ordinary activity: a shower, a walk to the mailbox, a work call, twenty minutes on a laptop. Write down the activity. Then, that same evening, the next morning, and the morning after that, write a line or two about how you feel: energy, head, heart, sleep, pain, mood. Do this for two weeks. You don't need to be precise. You need to be consistent.

What you're building is a map of the lag between effort and consequence. That map is the single most useful diagnostic tool most patients never learn to make. It is also the best thing you can bring to the next appointment with a sympathetic clinician — because patterns across days are something medicine knows how to read, while a single-day "how do you feel?" is not.

Sources

1. Singh I, Joseph P, Heerdt PM, Cullinan M, Lutchmansingh DD, Gulati M, Possick JD, Systrom DM, Waxman AB. Persistent Exertional Intolerance After COVID-19: Insights From Invasive Cardiopulmonary Exercise Testing. *Chest*. 2022;161(1):54–63. DOI: 10.1016/j.chest.2021.08.010
2. Durstenfeld MS, Sun K, Tahir P, Peluso MJ, Deeks SG, Aras MA, Grandis DJ, Long CS, Beatty A, Hsue PY. Use of Cardiopulmonary Exercise Testing to Evaluate Long COVID-19 Symptoms in Adults: A Systematic Review and Meta-analysis. *JAMA Network Open*. 2022;5(10):e2236057. DOI: 10.1001/jamanetworkopen.2022.36057
3. Keller BA, Pryor JL, Giloteaux L. Inability of myalgic encephalomyelitis/chronic fatigue syndrome patients to reproduce VO_2peak indicates functional impairment. *Journal of Translational Medicine*. 2014;12:104. (Updated series from the Workwell Foundation group, including Stevens S, Davenport T, Snell C, Van Ness JM.) DOI: 10.1186/1479-5876-12-104
4. Stevens S, Snell C, Stevens J, Keller B, Van Ness JM. Cardiopulmonary exercise test methodology for assessing exertion intolerance in myalgic encephalomyelitis/chronic fatigue syndrome. *Frontiers in Pediatrics*. 2018;6:242. DOI: 10.3389/fped.2018.00242
5. National Institute for Health and Care Excellence. *Myalgic encephalomyelitis (or encephalopathy)/chronic fatigue syndrome: diagnosis and management* (NICE guideline NG206). London: NICE; 2021. https://www.nice.org.uk/guidance/ng206
6. Stein SR, Ramelli SC, Grazioli A, Chung JY, Singh M, Yinda CK, Winkler CW, Sun J, Dickey JM, Ylaya K, et al. SARS-CoV-2 infection and persistence in the human body and brain at autopsy. *Nature*. 2022;612(7941):758–763. DOI: 10.1038/s41586-022-05542-y

7. Davis HE, Assaf GS, McCorkell L, Wei H, Low RJ, Re'em Y, Redfield S, Austin JP, Akrami A. Characterizing long COVID in an international cohort: 7 months of symptoms and their impact. *EClinicalMedicine*. 2021;38:101019. DOI: 10.1016/j.eclinm.2021.101019
8. Nalbandian A, Sehgal K, Gupta A, Madhavan MV, McGroder C, Stevens JS, Cook JR, Nordvig AS, Shalev D, Sehrawat TS, et al. Post-acute COVID-19 syndrome. *Nature Medicine*. 2021;27(4):601–615. DOI: 10.1038/s41591-021-01283-z
9. Swank Z, Senussi Y, Manickas-Hill Z, Yu XG, Li JZ, Alter G, Walt DR. Persistent circulating severe acute respiratory syndrome coronavirus 2 spike is associated with post-acute coronavirus disease 2019 sequelae. *Clinical Infectious Diseases*. 2023;76(3):e487–e490. DOI: 10.1093/cid/ciac722

Chapter 2. The thermostat that won't turn off

Around the fourth month, Catherine noticed a pattern. Her heart would start pounding. Not when she was nervous, but when she stood up from the couch. Not when she thought about something worrying, but when she walked to the sink to wash a dish. Lie down, and within a few minutes it was gone. She told her primary care doctor. He wrote a referral to a cardiologist and, almost in passing, mentioned an anxiety disorder. Catherine walked out of the office with the quiet sense that she had been speaking one language and had been heard in a different one.

She wasn't wrong. Her problem simply didn't have a name yet — at least not one her doctor had reached for.

When the heart starts pounding for no obvious reason, the first guess is almost always anxiety. It's a reasonable guess from the outside. The sensations line up: fast pulse, pressure in the chest, a feeling that something is about to go wrong, the sense that the body is sounding an alarm. The brain looks for an explanation and grabs the nearest one.

But there is a simple way to check that explanation. Anxiety doesn't care whether you are sitting or standing. If your symptoms appear when you get up and settle when you lie down, the first suspect shouldn't be your mind. It should be physics; more precisely, gravity, and how your body handles it.

That distinction matters, because the two problems need entirely different approaches. Confusing them has cost many patients years.

What the autonomic nervous system does — and why it matters here

Your autonomic nervous system (ANS) is the background layer of regulation that runs your heart rate, your blood pressure, the tone of your blood vessels, your body temperature, your digestion, your sweating, and dozens of other processes you never consciously think about. You don't decide to narrow your blood vessels when you stand up from the couch. It happens on its own, if the system is working well.

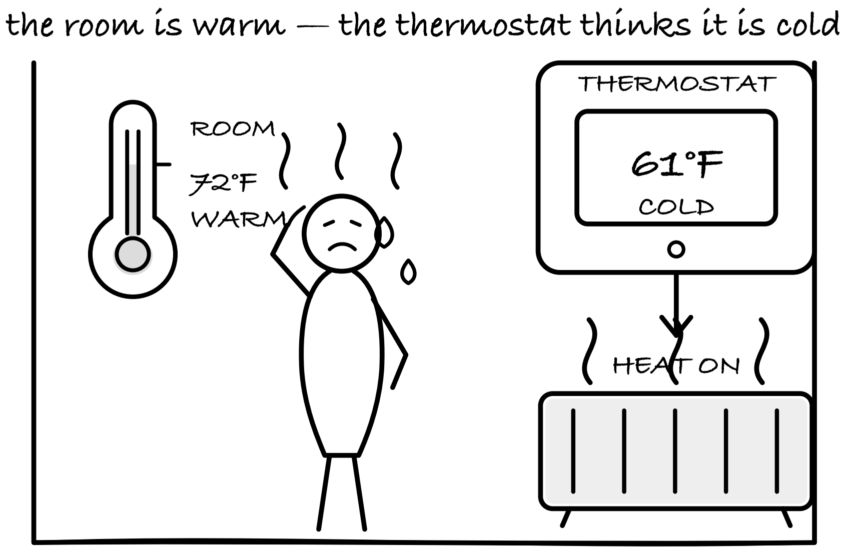
The ANS has two main branches. The sympathetic branch is the mobilizing one: the "go and respond" mode that raises heart rate, raises blood pressure, and routes blood to the muscles. The parasympathetic branch is the recovery one: the "rest and digest" mode that slows the heart, lowers blood pressure, and turns digestion back on. In a healthy person, these two branches trade off smoothly, many times a minute, depending on what the moment asks for. Think of them as a dynamic balance rather than a pair of switches.

There is one note worth carrying with you. The ANS does not pump your blood; your heart pumps. The ANS *regulates*. It adjusts the tone of your blood vessels, the rate and force of the heartbeat, and the distribution of blood between regions, depending on posture, effort, temperature, and the state of digestion. The difference is small in words and large in consequences.

When this regulation is disrupted — a state clinicians call **dysautonomia** — the balance comes apart. Heart rate jumps when it shouldn't. Vessels don't narrow when they should. The signal to digest and the signal to mobilize can fire at the wrong times or at the same time. The organs themselves are usually intact; what has lost its footing is the regulation that coordinates them.

The thermostat metaphor

Picture a household thermostat that reads 61°F when the room is already at 72°F, and keeps blasting the heat because the sensor thinks the room is cold. The furnace isn’t broken. The radiators aren’t broken. The wall is the right color. The sensor is reading the wrong number, and the whole house pays for that one small error.



The heating works. The sensor is what is wrong.

That is a usable picture of dysautonomia. The hardware isn’t destroyed. Rather, the signal the hardware is responding to is off. The ANS touches so many organs at once that a single misread gets amplified into what looks, from the outside, like a dozen different problems.

This is one reason the diagnostic picture gets messy. A cardiologist who checks the heart at rest sees a normal heart. Reflexes test normal on the neurologist’s exam. And a lab drawing blood at 10 a.m. in a quiet clinic finds normal chemistry. Each test is accurate. None of them is asking what your body looks like when it's actually being asked to stand, digest a meal, or recover from a warm shower.

What happens when you stand up

When a healthy person stands, gravity moves roughly 500–800 mL of blood downward into the legs and abdomen within the first minute or two [1]. A working autonomic system reacts almost instantly: small vessels in the legs narrow, the calves contract and squeeze blood back upward, the heart ticks up a few beats, and the brain keeps receiving a steady supply. The whole sequence is invisible to you. It happens in seconds.

In **postural orthostatic tachycardia syndrome (POTS)** — one of the conditions most often tied to post-infectious illness — this sequence stumbles. The vessels don't narrow quickly enough, or the circulating blood volume is low to begin with, or the nerves that should give the vessels their instructions are damaged and slow. Blood pools in the lower body, and less returns to the heart. The heart compensates the only way it can — it speeds up, sometimes dramatically. In POTS, the clinical marker is a sustained rise in heart rate of 30 beats per minute or more within ten minutes of standing (40 bpm in adolescents), without a matching drop in blood pressure [2]. POTS itself comes in several clinically distinct forms, with different mechanisms and different implications for treatment; we'll take that apart properly in Chapter 4.

This is why standing still is often harder than walking. When you walk, the muscles of the calves and thighs act as a pump, pushing blood back toward the heart. When you stand in place — in a checkout line, at the kitchen sink, in the shower — that pump is off. The whole job falls on the ANS, and if the ANS can't carry it, the body lets you know: fast pulse, lightheadedness, nausea, pressure behind the eyes, the feeling that you are about to faint.

This is not anxiety. This is hemodynamics (the physics of how blood moves through vessels).

A 2007 study looked specifically at the question of whether the exaggerated heart-rate response in POTS was caused by anxiety, and concluded that it was not [3]. And yet, in a large community survey of POTS patients carried out in 2019, 77% of respondents reported being told by clinicians, before the correct diagnosis was reached, that their symptoms were psychiatric [4]. More than three out of four!

Three states worth telling apart

One of the genuinely useful ideas for people living with these conditions is that the nervous system doesn't have one dial; rather, it has several recognizable states. The clearest framing we've seen for patients comes out of polyvagal theory, which organizes states of the nervous system into three rough modes: safe and engaged, mobilized, and shutdown.

A brief caveat before we use this language. Polyvagal theory is a practical model that helps people describe what they experience, and we'll use its vocabulary in this chapter because it works for that purpose. Scientific debate continues about whether the underlying neuroanatomy operates exactly as the theory describes. We're using it here because it helps patients communicate their states, not because it is settled neuroscience. With that noted:

- **Safe and engaged.** You can think, plan, digest a meal, hold a conversation without scanning the room. The system is in its day-to-day working mode.
- **Mobilized.** Fast heart rate, tight muscles, visual sharpening, a sense of urgency, sometimes a "wired on the edge of fragile" feeling. Your body has engaged its go-and-respond branch. This is what most people recognize as "stress" or "panic."

- **Shutdown.** Heavy limbs, a blank or foggy head, nausea, the sense that the lights have been dimmed, a pulling-away of energy from everything nonessential. The system has withdrawn.

Most people easily recognize the second state. Physiologically it is uncomfortable but unmistakable — it announces itself. The third state is the one that fools everyone, including the person in it. From the outside it can look like calm or apathy; from the inside it can feel like disappearance. What looks like peace is actually the body deciding that activity is too expensive right now.

In chronic post-infectious illness, people often move between the second and third states with little warning. Wired at 2 p.m. Flattened by 4 p.m. Pulse racing in the kitchen, then unable to answer a text an hour later. If you have ever asked yourself how you can be both overstimulated and half-vanished at the same time, the answer is that both states are responses from the same nervous system running different programs, and in dysautonomia they can stack on each other within a single afternoon.

Naming the state you are in, even silently, changes what you do next. A mobilized state responds to slower breathing, cooler temperature, horizontal rest, lowering the sensory input. A shutdown state responds to warmth, quiet, lying flat without pressure to "pull it together," and protecting whatever energy is left. Using the fix for the wrong state usually makes things worse.

Brain fog has a hemodynamic signature

When blood flow to the brain drops, the brain does not produce clear, unhurried thought. It produces fog.

A 2020 study measuring cerebral blood flow in people with ME/CFS found that, when upright, their brain blood flow dropped by an average of 24–28%, compared with about 7% in healthy controls. In the more severe cases, the reduction was above 40% [5]. That figure is doing real work in your life: it's the reason finding words is hard, the reason reading a page requires effort that reading never used to require, the reason an ordinary conversation can feel like manual labor.

A clinical review published the following year summarized the broader pattern in Long COVID:

"Patients with Long COVID commonly describe symptoms that are consistent with autonomic dysfunction — orthostatic intolerance, palpitations, gastrointestinal symptoms, thermoregulatory disturbances, and abnormal sweating — and objective testing frequently confirms it. Recognition matters, because these symptoms often do not respond to reassurance or generic advice, but do respond to targeted management."

— adapted from Dani M et al., *Clinical Medicine* (2021) [6]

There is an important framing to hold onto here. The reduced blood flow, the sensory overwhelm, the way a supermarket can feel like an obstacle course: none of it means you have become "delicate" or "oversensitive" in the character-trait sense. It means the system that decides how loudly to respond to incoming signals is calibrated differently right now, and that calibration is measurable.

On the vagus nerve and calming the sympathetic side

A feature that makes these conditions especially hard to live with is that the autonomic nervous system stops switching cleanly between its two branches, and the sympathetic side — the go-and-respond one — is often the one that stays most visible from the inside. It can remain elevated for long stretches, sometimes for hours or days when nothing in the surrounding world calls for mobilization. Bringing it back down is a legitimate clinical target. Many people with these conditions report that some form of vagal or parasympathetic practice — slow exhalations, cool-water exposure on the face, humming, at-home stimulation devices, paced breathing guided by a heart-rate monitor — helps them, and those reports are worth taking seriously.

It is worth being precise about where the scientific debate sits. The vagus nerve is the tenth of twelve pairs of cranial nerves, and it carries traffic in both directions. Outbound motor signals go from the brain to the throat and larynx, while parasympathetic signals are sent to the heart, the airways, and most of the digestive tract up to roughly the middle of the large intestine. Signals inbound to the brain come from those same thoracic and abdominal organs, from aortic pressure sensors, and from the skin behind the outer ear.

The popular story has two parts. That most vagal fibers carry signals inbound to the brain — toward the brain, not away from it — is anatomically correct. That most of those inbound signals come from the gut is where the story goes wrong: the lungs, the heart, and the airways contribute more of that traffic than the gut.

Polyvagal theory is a related practical vocabulary that many patients find useful for naming the nervous system states they move through. The specific neuroanatomy on which the theory is built is debated in academic physiology, but the usefulness of the vocabulary does not depend on all of those details being right. If the language helps you to name what you are experiencing, it is doing a valuable job.

Medical research has begun to probe these questions in formal trials. Pilot studies of transcutaneous auricular vagus stimulation (a small skin electrode placed over the branch of the vagus that runs through the outer ear) in Long COVID and ME/CFS cohorts [9], feasibility work on heart-rate-variability biofeedback in dysautonomia [10], and randomized studies of slow-paced breathing, which requires only a timer [11], have produced early, mixed results. The picture that emerges indicates that the target is a real clinical problem, that the mechanisms are not fully nailed down, and that which technique helps which person remains an empirical question. A reasonable position for you to take toward this is to gauge any practice by your own measured response; bring to marketing claims the same caution you would bring to any health product.

Break the loop where it's easiest

A stuck autonomic state tends to run in loops. Each loop has specific mechanics. That sounds like bad news — loops feel relentless from the inside — but it is also the useful news, because each loop can be interrupted at more than one point, and you do not need to stop all of them to feel better.

The orthostatic loop. You stand; blood pools in the lower body; less blood returns to the heart; brain blood flow drops; the sympathetic branch activates to compensate; the heart races. *Break it* by not waiting for the worst of it. Sit for kitchen tasks. Use a stool in the shower. If you must stand in place, tighten the calf muscles, cross one leg behind the other, and shift your weight. Lying down at the first wave of symptoms is one of the simplest and most effective interventions you have at your disposal: it removes the gravitational load that started the cascade.

The heat loop. Heat dilates blood vessels, especially the large vessels in the skin. Total blood volume stays the same, but it redistributes: more in the dilated skin vessels, less in the central circulation. If you are also upright, the orthostatic load lands on top of this and POTS symptoms worsen. Clinicians call this functional hypovolemia; the central circulation behaves as if volume were low, even though the blood hasn't gone anywhere. *Break it* with cooler showers rather than hot, a seat in the shower, and building in a recovery time after bathing rather than immediately asking the body to do a task.

The meal loop. A large meal draws blood toward the gut to support digestion, which is fine in a healthy system. In a system already short on central circulation, this post-meal shift can pull noticeably from the brain. *Break it* with smaller, more frequent meals, more protein relative to simple carbohydrates, and the option to lie down or recline for ten or fifteen minutes after eating, especially during flares.

The sensory loop. When the processing of incoming sensory information is calibrated high, which is common in these conditions, bright light and loud sound push the nervous system further into the mobilized state, which in turn sharpens the processing further. *Break it* by reducing input early rather than after a crash. Put on sunglasses before you walk out. Lower brightness on screens. Use earplugs or headphones in environments you know will be hard. Opt for grocery pickup instead of a full shop.

The meaning loop. You feel a sudden racing heart, a wave of what your body registers as alarm, and the next thought is "something terrible is happening" or "I'm losing my mind." That thought matters; it adds a second wave of stress on top of the physical one. *Break it* by giving the state a neutral, accurate name. "This is an orthostatic surge." "This is a shutdown wave." "My nervous system is mobilized right now." Naming the state correctly gives the brain something other than "danger" to do with the signal, which is different from dismissing the experience or pretending the body is fine.

One of the things that has become clear from listening to patients describe these states — in support groups, in clinic, in the long-form conversations that have become part of how this community shares knowledge — is that the specific mechanism changes what helps. A mobilized wave responds to slow exhalation, cool air, lying flat, and quiet. A shutdown wave responds to warmth, zero demands, and the permission to do absolutely nothing until the system comes back up. The US ME/CFS Clinician Coalition, in their clinical guidance for physicians, puts it in more measured terms:

"Autonomic dysfunction is a core feature of ME/CFS and similar post-infectious conditions, and symptoms that appear on standing, after meals, after heat exposure, or during sensory overload should be recognized as autonomic rather than reflexively attributed to anxiety. Management begins with recognition."

— paraphrased from the US ME/CFS Clinician Coalition and Bateman Horne Center clinical guidance [7]

The sentence to keep in mind is the second one. Management begins with recognition.

What this changes for you

Dysautonomia is neither a sentence nor a rare curiosity. Estimates of autonomic dysfunction in people with Long COVID range widely depending on criteria — from roughly 30% to more than 80% in different studies [6][8] — but across studies, what's consistent is that it is among the most common features of the condition, and it is among the most responsive to targeted management. That does not mean a cure. It means the floor can be raised.

Understanding that many of your symptoms are posture-dependent, heat-dependent, meal-dependent, and sensory-dependent is already information. It changes the question you are asking. Instead of "why am I like this?", the useful question becomes "what is gravity doing to my blood right now?" or "what did the last meal ask of my circulation?"

Here is an idea to carry into the next appointment: something like "*my symptoms are posture-dependent and improve when I lie down; I'd like to check orthostatic vitals and consider autonomic involvement.*" That one line reframes the conversation from a list of complaints to a specific, testable hypothesis. It gives a clinician something they know how to work with. It also, quietly, moves you from the passive position of "explaining how bad you feel" to the active one of "requesting a specific measurement." The second position is sturdier ground.

And since the thermostat metaphor is load-bearing in this chapter, it's worth ending with what it implies. A thermostat doesn't reset because someone stares at it harder. It resets when something in the feedback loop actually changes. The change doesn't have to be dramatic. But it has to be consistent.

Optional self-observation prompt

Morning, before you get out of bed: take your pulse lying down. Stay flat, breathe normally, count for a full minute, write it down. Then sit up, put your feet on the floor, and stand up slowly against a wall or next to a bed you can sit on. After two minutes of standing, take your pulse again. After five minutes, again. After ten minutes if you can do it safely, again. Write down each number with the time.

Write down any symptoms that appear alongside the numbers: lightheadedness, nausea, pressure in the head, the feeling of needing to sit down, any pain in the neck or shoulders, any visual changes. If at any point you feel that you are going to pass out, sit or lie down immediately. This is data collection, not a test of endurance.

If your heart rate rises by 30 beats per minute or more within ten minutes of standing without a significant drop in blood pressure, you have not given yourself a diagnosis, but you have given yourself evidence that is difficult for a clinician to ignore. A more careful, structured version of this test (the one several specialist clinics now use at home) is covered in Chapter 5, along with what the numbers mean under which conditions. For tomorrow, a simple morning reading is enough.

Catherine did this test for the first time on a Tuesday, in her bedroom, with a cheap finger pulse oximeter. Her lying pulse was 68. At three minutes of standing it was 102. At eight minutes it was 118. Her primary care doctor had not asked her to do this. Nobody had. She looked at the numbers, lay back down, and understood, for the first time in four months, that the thing happening to her had a shape.

Sources

1. Raj SR. Postural Tachycardia Syndrome (POTS). *Circulation*. 2013;127(23):2336–2342. DOI: 10.1161/CIRCULATIONAHA.112.144501
2. Freeman R, Wieling W, et al. Consensus statement on the definition of orthostatic hypotension, neurally mediated syncope and the postural tachycardia syndrome. *Clinical Autonomic Research*. 2011;21(2):69–72. DOI: 10.1007/s10286-011-0119-5 (See Chapter 4 reference 9 for the full author list.)
3. Masuki S, Eisenach JH, Johnson CP, Dietz NM, Benrud-Larson LM, Schrage WG, Curry TB, Sandroni P, Low PA, Joyner MJ. Excessive heart rate response to orthostatic stress in postural tachycardia syndrome is not caused by anxiety. *Journal of Applied Physiology*. 2007;102(3):896–903. DOI: 10.1152/japplphysiol.00927.2006
4. Shaw BH, Stiles LE, Bourne K, Green EA, Shibao CA, Okamoto LE, Garland EM, Gamboa A, Diedrich A, Raj V, et al. The face of postural tachycardia syndrome — insights from a large cross-sectional online community-based survey. *Journal of Internal Medicine*. 2019;286(4):438–448. DOI: 10.1111/joim.12895
5. van Campen CLMC, Verheugt FWA, Rowe PC, Visser FC. Cerebral blood flow is reduced in ME/CFS during head-up tilt testing even in the absence of hypotension or tachycardia: a quantitative, controlled study using Doppler echography. *Clinical Neurophysiology Practice*. 2020;5:50–58. DOI: 10.1016/j.cnp.2020.01.003

6. Dani M, Dirksen A, Taraborrelli P, Torocastro M, Panagopoulos D, Sutton R, Lim PB. Autonomic dysfunction in 'long COVID': rationale, physiology and management strategies. *Clinical Medicine*. 2021;21(1):e63–e67. DOI: 10.7861/clinmed.2020-0896
7. Bateman L, Bested AC, Bonilla HF, Chheda BV, Chu L, Curtin JM, Dempsey TT, Dimmock ME, Dowell TG, Felsenstein D, et al. Myalgic Encephalomyelitis/Chronic Fatigue Syndrome: Essentials of Diagnosis and Management. *Mayo Clinic Proceedings*. 2021;96(11):2861–2878. DOI: 10.1016/j.mayocp.2021.07.004 (US ME/CFS Clinician Coalition / Bateman Horne Center consensus guidance)
8. Larsen NW, Stiles LE, Miglis MG. Preparing for the long-haul: Autonomic complications of COVID-19. *Autonomic Neuroscience*. 2021;235:102841. DOI: 10.1016/j.autneu.2021.102841
9. Badran BW, Huffman SM, Dancy M, Austelle CW, Bikson M, Kautz SA, George MS. A pilot randomized controlled trial of supervised, at-home, self-administered transcutaneous auricular vagus nerve stimulation (taVNS) to manage long COVID symptoms. *Bioelectronic Medicine*. 2022;8(1):13. DOI: 10.1186/s42234-022-00094-y
10. Corrado J, Iftekhar N, Halpin S, Li M, Tarrant R, Grimaldi J, Simms A, O'Connor RJ, Casson A, Sivan M. Heart Rate Variability Biofeedback for Long COVID Dysautonomia (HEARTLOC): Results of a Feasibility Study. *Advances in Rehabilitation Science and Practice*. 2024;13. DOI: 10.1177/27536351241227261 PMCID: PMC10826406.
11. Mauro M, Cegolon L, Bestiaco N, Zulian E, Filon FL. Heart Rate Variability Modulation Through Slow-Paced Breathing in Health Care Workers with Long COVID: A Case-Control

Study. *The American Journal of Medicine*. 2025. DOI: 10.1016/j.amjmed.2024.05.021

Chapter 3. Five diagnoses, one problem

Five months after she got sick, Catherine had four diagnoses in her chart: tachycardia, irritable bowel syndrome, migraine, and an anxiety disorder. A fifth — fibromyalgia — was added a few weeks later, after an appointment with a rheumatologist. Each specialist had been attentive. Each had found something real in their own field. None of them said the thing she had been feeling for months: this was not five different illnesses. It was one problem speaking five languages at once.

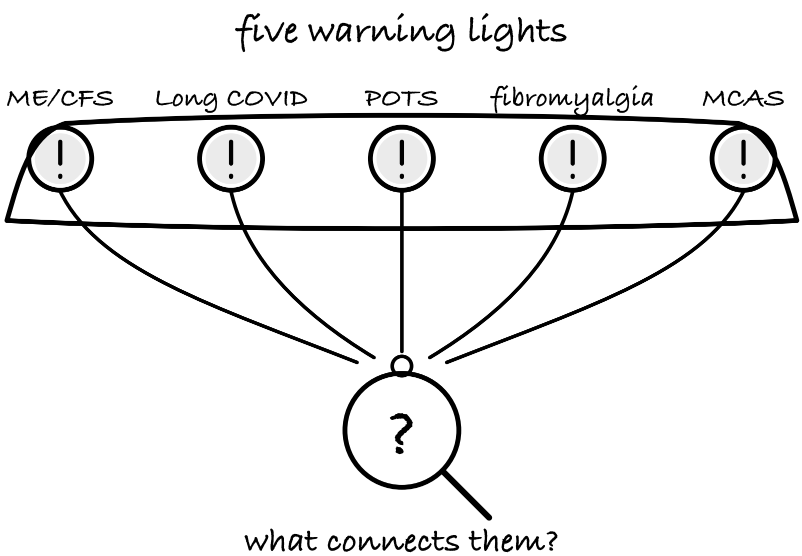
She turned out to be right.

If you have more than two or three diagnoses from the following list — POTS, migraine, irritable bowel syndrome, fibromyalgia, insomnia, mast cell activation syndrome, chronic fatigue — that is not catastrophic bad luck and it is not hypochondria. It is a pattern. And the pattern has an explanation.

One important caveat first. Each of those diagnoses can also exist on its own. IBS, migraine, and fibromyalgia are real conditions with their own mechanisms, and they are not always tied to a post-infectious process. The pattern this chapter describes applies when several of these labels appeared together after an infection, move together, and respond to the same triggers. If you have had migraines for the last ten years, that is most likely a separate story riding alongside the new one.

In a study from a Boston autonomic clinic published in January 2026, **86.6%** of patients with confirmed POTS met criteria for central sensitization syndrome — a state in which the central nervous system amplifies incoming signals [1]. Take a moment with that figure. Almost nine in ten. Numbers like that don't describe a coincidence. They describe a system that has come apart in several places at once.

Medicine often reads this picture backwards. The more diagnoses in your chart, the "more complicated" you become as a patient. But in any system with linked components, a chain of related failures usually points to one upstream problem. If five warning lights come on at once in a car, a good mechanic doesn't replace five bulbs. The mechanic checks the alternator.



Five lights on the dashboard, one thing worth checking first.

Recognizing this pattern early matters because it hands you a tool rather than a cure. It gives you the vocabulary to understand that effort comes with a delayed bill. Once you know that, you can start pacing. Before you see the pattern, you cannot pace, because you do not know what you are tracking. Clinical experience links this early recognition to better long-term outcomes, and the mechanism is plain. Fewer repeat crashes mean fewer days forced flat in bed with the curtains drawn. Less forced bed rest means less cumulative deconditioning. Halting the boom-and-bust cycle brings your nervous system to a stable baseline; as Chapter 6 explains, a body that keeps collapsing cannot repair itself, while a stable one sometimes can.

The principle has practical weight precisely because the standard route to a diagnosis takes years, and those years pass without pacing. If you are reading this five years into your illness, you have not missed a window. Protecting yourself against crashes improves your trajectory from exactly where you stand today.

A diagnosis names the alarm, not the fire

Before going further, it helps to say plainly what most medical diagnoses actually do. Most of them describe **where the alarm went off**, not what set it off.

POTS says: you feel bad when you stand. IBS says: the gut is involved. Migraine says: the head and the sensory system are involved. Fibromyalgia says: pain, sleep, and fatigue are involved. MCAS says: the immune system appears to be releasing too many mediators. These are useful labels — they let clinicians communicate, bill insurance, and pick treatments. They are not first-cause reports.

If you look at the body as a system, you can sketch a few layers where things go wrong in this kind of illness. When several layers misbehave together, you don't get five diagnoses. You get one systemic problem with five outputs.

Layer one: pumps and valves — dysautonomia

We spent the previous chapter inside this layer, so I will keep it short and cross-reference. The autonomic nervous system regulates the distribution of blood, the rate and force of the heartbeat, vascular tone, digestion, and temperature. When that regulation falters, symptoms appear in several systems at the same time.

A pulse that jumps when you stand. Nausea after a meal. Heat that brings on a headache. Sleep that breaks at three in the morning under a wave of sympathetic activation. A gut that doesn't quite know what it wants. Read as a list, that looks like five illnesses; read as a system, it is one regulatory problem wearing several faces at once.

That is why, when dysautonomia is stabilized — through compression, posture, paced exertion, sometimes medication — many patients see several apparently unrelated symptoms ease at once. The thermostat that finally responds (Chapter 2) is the same thermostat that quietly governs digestion, sleep transitions, and how the head feels in the afternoon.

Layer two: sentries with poor self-control — mast cells

Mast cells are immune cells that sit at the body's borders — in the skin, the airways, the gut, around blood vessels and around nerves. When they decide there is a threat, they release chemical **mediators**: histamine first, then prostaglandins, leukotrienes, tryptase, several cytokines. Mast cells **participate** in allergic and inflammatory reactions; they are not the cause of those reactions on their own, and they don't recognize the borders between medical specialties.

A short frame for the diagnosis itself. **Mast cell activation syndrome (MCAS)** is the clinical label for a state in which mast cells release these mediators inappropriately and produce multi-system symptoms. Common triggers are heat, cold, stretch, certain foods, alcohol, infections, hormonal shifts, and emotional stress. The diagnostic criteria are still debated and the workup is more involved than for many conditions, which is part of why MCAS is so often missed.

Histamine alone can dilate blood vessels, irritate nerves, speed up gut motility, and raise the heart rate. So the same underlying problem — mediator release in the wrong amounts at the wrong times — can show up as flushing, IBS, palpitations, migraine, anxiety, or food that has “suddenly” become intolerable.

Weinstock and colleagues compared the symptom profile of a group of 136 Long COVID patients with the profile typically seen in MCAS and found a striking overlap — the symptom lists were close enough that the team argued mast-cell-type problems should be considered as part of the Long COVID picture rather than dismissed [2]. The overlap is suggestive rather than diagnostic — formal MCAS criteria ask for more than a symptom questionnaire can deliver. What the comparison does argue is that mast-cell-type problems are common enough in this population to deserve attention rather than a polite nod.

A practical clue: if heat, a particular food, alcohol, or stress sets off several apparently unrelated symptoms at once — flushing, racing heart, gut trouble, headache — that is suggestive. A trigger that hits multiple systems at the same time often points to a shared mechanism.

Layer three: stretchy hardware — hypermobility

Connective tissue is not only joints. The same fabric makes up the walls of veins, the support around the gut, the surface of the skin. If that tissue is more stretchy than average, the veins in the lower body can hold too much blood when you are upright. Less blood returns to the heart. The heart compensates by speeding up. Less blood reaches the brain.

In one study, **31%** of patients with POTS met criteria for hypermobile Ehlers-Danlos syndrome (hEDS) — the most common form of connective-tissue hypermobility — far above the rate in the general population [3].

A small but useful detail: you don't need to be flexible right now. Many adults stiffen with age. But if as a child or teenager you could drop into the splits without training, often rolled your ankles, had unusual posture, or were called "very bendy," that is part of your history worth bringing to a clinician. It is the kind of background detail that is rarely asked for and frequently explains a lot.

Layer four: thin wires — small-fiber neuropathy

Small nerve fibers carry signals about pain and temperature, and they help control sweating and the tone of small blood vessels. When they are damaged, you can end up with both pain and dysautonomia coming from the same source.

About **41%** of patients carrying a fibromyalgia label show evidence of small-fiber polyneuropathy on skin biopsy — meaning the small fibers in their skin are reduced in density compared to controls [4]. That finding doesn't account for every fibromyalgia case, but it explains why some people present with burning pain, electric-zap sensations, unusual temperature sensitivity, and orthostatic symptoms together as a single package.

If your symptoms include burning in the feet, patches of numbness, or sharp temperature sensitivity alongside POTS or post-exertional fatigue, small-fiber neuropathy is worth raising as a separate question. It is testable — a small skin punch biopsy on the calf, sent to a specialized lab, returns a fiber-density count.

Layer five: amplification — central sensitization

Central sensitization is the layer that gets most often misunderstood, so I want to say plainly what it is. It is a measurable change in how the central nervous system — the brain and spinal cord — handles incoming signals. The signals coming in are real. What is impaired is the nervous system's ability to inhibit and damp them — what clinicians sometimes call descending pain inhibition. The volume controls have lost their grip. Patients sometimes hear "central sensitization" as a polite way of saying their symptoms are imaginary, or that they are overreacting because they are anxious. The physiology says the opposite: the hardware itself is processing incoming signals differently, and the change can be measured.

Picture a household alarm that has started sounding at the weight of a fallen leaf. The leaf is real. The leaf really did fall. The threshold for alarm has shifted, and that shift is the problem.

This is part of why the 86.6% figure from the Mathew and Novak study is so striking: POTS is supposed to be "just heart rate when standing," and yet in real bodies it travels with migraine, sound sensitivity, food sensitivity, and widespread pain [1]. A 2022 systematic review and meta-analysis in *Cephalalgia* (a scientific name of a headache) placed the pooled prevalence of migraine in POTS at about **36.8%**, with several included studies running considerably higher [5]. That overlap is not random; central sensitization shows up in both conditions, and the same hypersensitive processing that makes a fluorescent light feel painful also makes a normal heartbeat feel like an assault.

Central sensitization is a real neurophysiological state. It reaches in through inflammation, through sustained pain, through months of autonomic dysregulation — different roads, the same intersection. And it is closely linked to **interoception**, the brain's perception of what is happening inside the body. After a long illness, the brain often starts treating ordinary internal signals — a meal arriving in the stomach, the act of standing up — as warnings. It launches a defensive response before the signal has had a chance to mean anything: more sympathetic activation, worse sleep, worse cerebral perfusion. The loop closes on itself.

Interoception has an anatomical home. It is the insular cortex, deep in the brain, where signals from the heart, the gut, the lungs and the vascular system are gathered and integrated into what you consciously feel. When those signals come in from a body that is genuinely stressed — mitochondria working below capacity, immune cells still on alert, small vessels not cooperating — the insula's reading is accurate. A growing body of work calls this an interoceptive mismatch: a faithful report on an unusual input, not a failure of perception [9]. Your brain is reading your body correctly; the input itself is what has changed. That distinction matters because it separates this condition from any suggestion that the sensations are imagined.

Why the system doesn't return to normal

The five layers describe **where** things break. There is a separate question: **why doesn't the system reset** after the original infection is gone?

Science is currently working with four overlapping hypotheses. They are not competing for a single winner. They cooperate, and probably do so in different proportions in different patients.

The first hypothesis is that **something stayed inside**. Fragments of virus — and in some cases evidence of live virus — have been found in tissues months after a negative swab. The immune system continues to detect a pathogen that the standard tests can no longer see, and stays partly activated. We covered the evidence base for viral persistence in Chapter 1; here it matters because if this mechanism is operating, antiviral approaches are a reasonable line of investigation, and several trials are underway.

The second hypothesis is that **the immune system learned the wrong lesson**. Autoantibodies, chronic low-grade inflammation, mast cells whose threshold has been reset too low. Immune memory is supposed to recognize the original burglar; in this picture, the immune system is fighting the furniture. This is the hypothesis that motivates much of the discussion around low-dose naltrexone, antihistamines, and intravenous immunoglobulin — not as cures, but as ways to lower the background noise enough that the system has room to reset. ("LDN" stands for low-dose naltrexone — *naltrexone*, not naloxone; the two are often confused in patient materials and they are different drugs. The dose here is 0.5–6 mg per day, roughly a tenth of the dose used for opioid-use disorder. Compact frames for these medications come in Chapter 7.)

On the antihistamine side, the regimen most often discussed is an H1 + H2 combination — fexofenadine and famotidine at standard doses — which in a small 2023 open-label series produced full symptom resolution in about a third of the fourteen patients treated [10]. A larger placebo-controlled trial (STIMULATE-ICP, UK) is reading out in the same window.

> **SAFETY —** This is clinician-supervised territory, and the evidence base is still thin.

IVIg (intravenous immunoglobulin) is a treatment patients often ask about, and as of 2026 the evidence still sits short of a clean answer. The first randomised trial in POTS found no difference between IVIg and albumin (iSTAND, 2024) [11]; a larger NIH trial in severe post-COVID POTS finished enrolment in 2025 and should read out late 2026. Case series from Stockholm, Berlin, and Boston report benefit in selected subgroups — autoantibody-positive, small-fiber-neuropathy, infection-triggered — but without placebo control [12]. A compact frame will come when the RCT data do.

The third hypothesis is an **energy bottleneck**. Damage to the lining of small blood vessels, microcirculatory changes, mitochondria operating less efficiently than they should. The brain sends a normal command and the muscles are intact, but the fuel doesn't arrive on time and the oxygen isn't extracted as it should be. We covered the cellular-energy and microcirculation evidence in Chapter 1; in this chapter the relevant point is how an energy bottleneck *combines* with the other three.

The fourth hypothesis is the hardest one to talk about without it sounding like an insult, so let me say it plainly first: the brain participates in regulating the state of the body, and that participation does not make the symptoms imagined. If the brain regions that monitor for threat are stuck in a high-alarm setting after an infection, the result is a real cascade of autonomic and immune output — air hunger, gut trouble, sensory overload, post-exertional crashes — produced by a brain whose internal threshold for "dangerous" has shifted. Calling this "anxiety" in the colloquial sense misnames the mechanism.

These four ideas don't compete. They reinforce each other. Persistent viral signal can drive ongoing inflammation; inflammation impairs microcirculation; impaired microcirculation strains cellular energy production; the resulting interoceptive signal is read by the brain as threat; the brain's threat response increases sympathetic output, which feeds back into the loop. It is one unfortunate feedback cycle. The good news is that feedback cycles can be interrupted in more than one place.

Why your chart looks the way it does

Your body has one wiring harness. Medicine has departments.

Cardiology checks the heart. Gastroenterology checks the gut. Neurology looks for nervous-system pathology. Rheumatology checks for inflammation in joints. Each specialist sincerely answers the question their training prepared them for. Nobody is paid to step back and ask whether the wiring of the system as a whole has come undone.

In the largest survey of the POTS population — 4,835 people — the median wait from first presenting to a physician to a confirmed diagnosis was 24 months, the mean 4.9 years, and about 15% waited more than a decade. Patients saw a mean of seven physicians before someone named it [13]. Set that against the ten-minute office test that can establish the diagnosis. The gap reflects the architecture of a system that bills and treats one organ at a time, rather than the fault of any individual clinician. The same survey found women waited close to two years longer than men — 5.0 years against 3.0 — and its authors raise gender bias in the diagnostic process as a candidate explanation [13].

The more “smeared” an illness is across the body, the easier it is for it to disappear into the gaps between specialties. The chart looks chaotic. The biology, very often, is much more orderly than the chart.

How to use this knowledge

The first practical step is to make a **pattern map** rather than a longer list of complaints.

Take a sheet of paper. Down the left side, write your symptoms grouped by system — palpitations and dizziness on standing; gut symptoms; headaches; pain in the body; sleep disruption. Across the top, write the things that bring symptoms on or worsen them — standing, heat, meals, alcohol, stress — and then the things that help: lying flat, antihistamines, salt and fluids, a quiet dark room.

Then circle what moves together. If standing simultaneously raises your heart rate, your nausea, and your brain fog — circle them. If antihistamines reduce both your facial flushing and the bubbling in your gut — circle that. A heavy meal that makes you worse on several fronts at once is another kind of evidence: one trigger, many targets, almost always one shared mechanism underneath.

Bring the sheet to your next appointment and say:

"My symptoms cluster around a few mechanisms — orthostatic intolerance, mast-cell-type reactivity, and central sensitization. Could we look at shared mechanisms instead of treating each complaint separately?"

That single sentence can save you months. As Hitch, Wrench and colleagues note in their clinical handbook, a multidisciplinary, mechanism-led approach is more effective in this kind of illness than serial single-specialty care; the difficulty is that most healthcare systems are not yet organized to deliver it [6]. Bringing the map gives a clinician a way to start working at the level of mechanism even inside a single appointment.

Vaccination and reinfection — what the data say

Two questions come up in almost every Long COVID conversation, and neither has a simple answer. I'll give you what the evidence currently shows, and where the gaps still are.

The first: can vaccination help an existing case of Long COVID? Several observational studies, including a large UK analysis using ONS data [7], have reported that some patients with Long COVID notice an improvement in symptoms after vaccination. The improvements are not universal, the effect sizes vary, and the underlying mechanism is not fully worked out. One leading hypothesis is that if Long COVID is partly maintained by viral persistence, the immune response triggered by a vaccine may help clear residual virus. Another is that some of the apparent improvement reflects regression toward the mean. As of this writing, vaccination is not recommended as a *treatment* for Long COVID, but most specialists consider it safe in this population and recommend following standard schedules unless there is a specific contraindication.

The second: what happens with a reinfection? The available data are not encouraging. Population-level work has linked repeat SARS-CoV-2 infections with an increased risk of developing — or worsening — Long

COVID symptoms [8]. For someone already living with the condition, a second infection can erase months of progress. That is reason enough for sensible precautions, without panic: vaccination on schedule, attention to indoor air quality and ventilation, and the use of well-fitting respirators in crowded indoor spaces during periods of high transmission, if you choose to.

Vaccination has become a politically charged topic, and we know that. This book tries to follow the data rather than the politics. If the data change, the recommendations will too.

Optional self-observation prompt

Sketch the pattern map on a single sheet — symptoms down the left, triggers and relievers across the top. You don't have to be exact or complete. You only have to start *seeing* the pattern. What moves together is most likely linked. What improves with the same intervention is most likely sharing a mechanism. Bring that sheet to your next appointment instead of trying to retell the whole story from the beginning. The map is the story.

Sources

1. Mathew GT, Novak P. Prevalence of Central Sensitization in Postural Tachycardia Syndrome. *JAMA Network Open*. 2026;9(1):e2553694. DOI: 10.1001/jamanetworkopen.2025.53694 Case-control study of 305 patients with autonomic-testing-confirmed POTS at Brigham and Women's Faulkner Hospital; 264 (86.6%) met criteria for central sensitization syndrome.

2. Weinstock LB, Brook JB, Walters AS, Goris A, Afrin LB, Molderings GJ. Mast cell activation symptoms are prevalent in Long-COVID. *International Journal of Infectious Diseases*. 2021;112:217–226. DOI: 10.1016/j.ijid.2021.09.043
3. Miller AJ, Stiles LE, Sheehan T, Bascom R, Levy HP, Francomano CA, Arnold AC. Prevalence of hypermobile Ehlers-Danlos syndrome in postural orthostatic tachycardia syndrome. *Autonomic Neuroscience*. 2020;224:102637. DOI: 10.1016/j.autneu.2020.102637
4. Oaklander AL, Herzog ZD, Downs HM, Klein MM. Objective evidence that small-fiber polyneuropathy underlies some illnesses currently labeled as fibromyalgia. *Pain*. 2013;154(11):2310–2316. DOI: 10.1016/j.pain.2013.06.001
5. Ray JC, Pham X, Foster E, Cheema S, Corcoran SJ, Matharu MS, Hutton EJ. The prevalence of headache disorders in postural tachycardia syndrome: a systematic review and meta-analysis of the literature. *Cephalalgia*. 2022;42(11–12):1274–1287. DOI: 10.1177/03331024221095153
6. Hitch D, Wrench J, eds. 2026. *The Rehabilitation and Management of Long COVID: A Handbook for Clinical Practice.* London: Routledge. DOI: 10.4324/9781003528104
7. Ayoubkhani D, Bermingham C, Pouwels KB, Glickman M, Nafilyan V, Zaccardi F, Khunti K, Alwan NA, Walker AS. Trajectory of long covid symptoms after covid-19 vaccination: community based cohort study. *BMJ*. 2022;377:e069676. DOI: 10.1136/bmj-2021-069676
8. Bowe B, Xie Y, Al-Aly Z. Acute and postacute sequelae associated with SARS-CoV-2 reinfection. *Nature Medicine*. 2022;28(11):2398–2405. DOI: 10.1038/s41591-022-02051-3

9. Khalsa SS, Adolphs R, Cameron OG, Chapman HA, Davidson RJ, Drevets WC, Feinstein JS, et al. Interoception and Mental Health: A Roadmap. *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging*. 2018;3(6):501–513. DOI: 10.1016/j.bpsc.2017.12.004
10. Salvucci F, Codella R, Coppola A, Zacchei I, Grassi G, Anti ML, Nitisoara N, Luzi L, Gazzaruso C. Antihistamines improve cardiovascular manifestations and other symptoms of long-COVID attributed to mast cell activation. *Frontiers in Cardiovascular Medicine*. 2023;10:1202696. DOI: 10.3389/fcvm.2023.1202696 Open-label series: fourteen patients treated with fexofenadine 180 mg/day and famotidine 40 mg/day against thirteen untreated controls; symptoms resolved completely in 29% of those treated.
11. Vernino S, Hopkins S, Bryarly M, Hernandez J, Ibrahim L, Mohamed-Jabir D, Shibao CA, Raj SR, Biaggioni I, Okamoto LE. Randomized controlled trial of intravenous immunoglobulin for autoimmune postural orthostatic tachycardia syndrome (iSTAND). *Clinical Autonomic Research*. 2024;34(1):153–163. DOI: 10.1007/s10286-024-01020-9
12. Composite case-series reference: Sjögren P, Bragée B, Britton S. Successful subcutaneous immunoglobulin therapy in a case series of patients with myalgic encephalomyelitis/chronic fatigue syndrome. *Clinical Therapeutics*. 2024;46(7):597–600. DOI: 10.1016/j.clinthera.2024.05.010 **And:** Stein E, Heindrich C, Wittke K, Kedor C, Rust R, Freitag H, Sotzny F, Krüger A, Tölle M, Grabowski P, Kim L, Scheibenbogen C. Efficacy of repeated immunoadsorption in patients with post-COVID myalgic encephalomyelitis/chronic fatigue syndrome and elevated β2-adrenergic receptor autoantibodies: a prospective cohort study. *The Lancet Regional Health — Europe*. 2025;49:101161. DOI: 10.1016/j.lanepe.2024.101161 **And:** McAlpine L, Zubair AS, Joseph P, Spudich S. Case-control study of

individuals with small fiber neuropathy after COVID-19. *Neurology Neuroimmunology & Neuroinflammation*. 2024;11(4):e200244. DOI: 10.1212/NXI.0000000000200244

13. Shaw BH, Stiles LE, Bourne K, Green EA, Shibao CA, Okamoto LE, Garland EM, Gamboa A, Diedrich A, Raj V, Sheldon RS, Biaggioni I, Robertson D, Raj SR. The face of postural tachycardia syndrome — insights from a large cross-sectional online community-based survey. *Journal of Internal Medicine*. 2019;286(4):438–448. DOI: 10.1111/joim.12895 Survey of 4,835 people with a physician diagnosis of POTS: median diagnostic delay 24 months (IQR 6–72), mean 4.9 ± 7.1 years, about 15% undiagnosed beyond ten years, mean seven physicians seen before diagnosis, and a diagnostic delay of 5.0 ± 7.2 years in women against 3.0 ± 4.4 in men.

Chapter 4. Why standing is harder than walking

Catherine had run a version of this experiment before — lying down to count her resting pulse, then standing against the wall and watching the number climb, minute by minute, until her head grew heavy and the sense that something was about to go wrong arrived like a wave.

She lay back down. Three minutes later it had passed.

Catherine stared at the ceiling and thought about one thing: this happens every day. It had just never had numbers before.

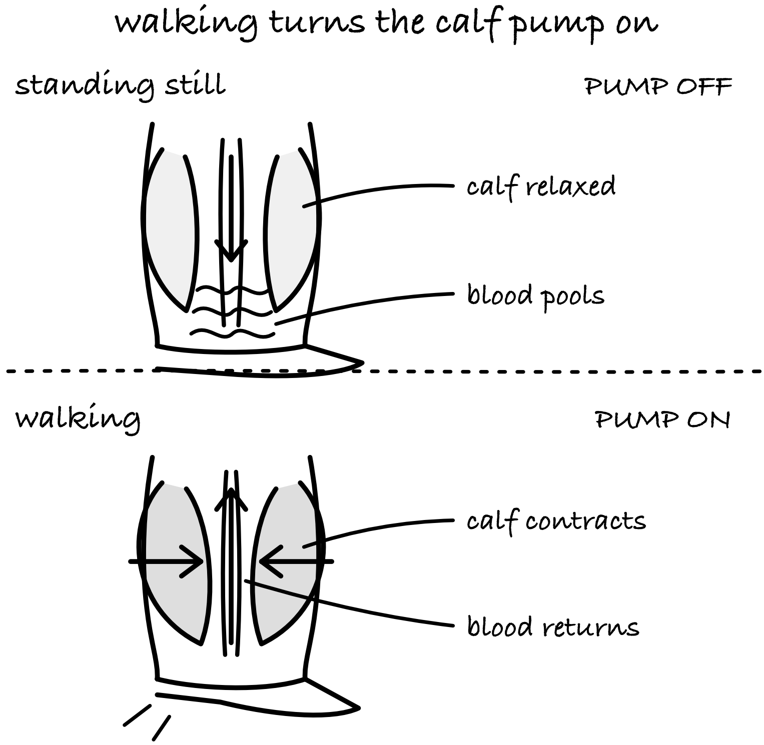
Numbers change the conversation. Not because they make the experience more real — it was already real — but because they translate the experience into a language medicine knows how to read. *"I feel awful when I stand"* is a complaint. *"My pulse rises forty-five beats in eight minutes of quiet standing, and this reproduces every morning"* is data. Same body. The second sentence opens doors the first one cannot.

This chapter answers two questions. What is actually happening in the vessels and the regulation behind POTS? And how can you measure that at home in ten minutes, with ordinary tools, so that the next time you walk into a clinic you arrive with evidence instead of adjectives?

We covered the basic orthostatic mechanism in Chapter 2 — roughly half a litre of blood redistributing downward in the first minute of standing, the sympathetic branch tightening vessel tone, the calf muscles working as a pump, the heart ticking up slightly to keep cerebral flow steady. I won't re-teach that here. What this chapter adds is the next layer: why that compensation goes wrong in several different ways, how those ways are not the same illness, and what quietly standing against a wall can show you that nothing else can.

Why standing is harder than walking

It is worth restating one mechanism from earlier here, because the rest of this chapter builds on it. Standing still is harder on the circulation than walking. Not harder on the lungs or the muscles — harder on the *return of blood to the heart*. A person with dysautonomia can sometimes cross a parking lot and arrive feeling tolerable, then fall apart waiting in a checkout line two minutes later. The body is reading a mechanical difference the person cannot feel. The line interrupts the muscle pump that walking kept active.



Walking runs a second pump. Standing still switches it off.

When you walk, the muscles of your calves and thighs contract rhythmically against the deep veins in your legs and push blood back up toward the heart. Clinicians call this the muscle pump, and in a regulation that is working correctly it carries a large share of the work of venous return. When you stand quietly — in a queue, at the kitchen sink, on a podium — the muscle pump is idle. Without it, venous return depends almost entirely on vessel tone and the small valves in the veins, both of which are exactly the things that don't work reliably in POTS. Less blood returns to the heart. The stroke volume — the amount of blood the heart pushes out with each contraction — drops. To keep enough blood moving toward the brain, the heart speeds up.

Your heart is solving a mechanical problem with the only tool it has: same task, smaller delivery per beat, so more beats.

This is also why a hot shower, a warm shop, a long day on your feet, or post-meal digestion can each stack onto the same basic hemodynamic shortage and tip what was manageable at breakfast into intolerable by lunch. (We saw one concrete version of the stacking in Chapter 1, in the shower example. Standing still is the unseen primary stressor hiding inside many other situations.)

POTS is not one illness

POTS — postural orthostatic tachycardia syndrome — is a single diagnostic label sitting on top of several different biological stories. The diagnostic criterion is simple enough, and we saw it in Chapter 2: heart rate rises by at least thirty beats per minute (forty in adolescents) within ten minutes of standing, without a matching drop in blood pressure. What that shared number hides is that the mechanism generating it differs from person to person. Treatment that helps one subtype can do nothing — or make things worse — in another.

Three classical subtypes are recognized in the clinical literature, and a fourth, more recent one has emerged from post-infectious research.

Neuropathic POTS. The most common form. Small sympathetic nerve fibres that tell blood vessels in the lower body to constrict on standing are partly damaged. Those vessels don't tighten the way they should. More blood pools in the legs. Central circulating volume drops. The heart compensates with tachycardia because that is the only lever it has left. Up to half of POTS patients show small-fibre neuropathy on skin biopsy — the same small fibres we met in Chapter 3 when we talked about fibromyalgia and pain-temperature regulation [1, 2]. Practical signs: feet that turn purple or dusky red on standing, cold hands, symptoms that track strongly with *still* standing rather than movement.

If your legs change colour when you stand up, photograph them lying down and then standing, and bring the photographs to your clinician.

Hypovolemic POTS. Some POTS patients have reduced plasma volume at baseline. In one careful comparison, blood and plasma volume ran about a fifth lower than in matched sedentary controls — a median blood volume of 60 against 71 millilitres per kilogram [3]. The cause sits upstream of hydration. The hormonal system that regulates how much fluid the body holds on to — the renin-angiotensin-aldosterone system, which normally tells the kidneys how much sodium and water to keep — has misread the situation, and the body is behaving as if it has just given blood. Practical signs: sharp worsening around the menstrual period, dry mouth on waking, and temporary, partial relief from increased fluids that never quite becomes lasting relief. (Drinking more helps because central circulating blood volume has something to work with in the short term, not because the underlying regulation has been corrected. We will come back to this in a moment.)

Hyperadrenergic POTS. A subset of patients show elevated norepinephrine in plasma when they stand — often above 400 pg/mL, sometimes above 600 [4, 5]. Norepinephrine is the neurotransmitter the sympathetic branch uses at synapses. It is not the same substance as circulating adrenaline (epinephrine), and the sensations it produces are not an "adrenaline surge" in the colloquial sense. What patients report is tremor, an internal buzzing or itch, air hunger, blood pressure that swings including episodes of actual hypertension, and a kind of wired, unsettled quality that can look, from the outside, indistinguishable from anxiety. The difference that matters in a clinical encounter is the trigger. Anxiety tracks thoughts. This tracks posture.

A small note on the air-hunger part, because it confuses everyone including clinicians. In the hyperadrenergic pattern, and in POTS more broadly, people often breathe more deeply and more frequently than the situation requires. End-tidal carbon dioxide — the CO_2 measured at the end of an exhale — can drop below the normal range. Low CO_2 constricts cerebral blood vessels, which produces light-headedness that the brain reads as "I must not be getting enough air," which drives more breathing, which drops CO_2 further. Oxygen saturation stays normal throughout. If an appointment ever leaves you with a finger pulse oximeter reading of 98% and the firm conviction that something is still wrong with your breathing, that loop is one of the things worth asking a specialist to check.

Autoimmune or post-infectious POTS. Over the past several years, and sharply since 2021, research groups have identified autoantibodies — antibodies the immune system has made against its own receptors — against the very receptors that govern heart rate and vessel tone: the adrenergic and muscarinic G-protein-coupled receptors. In one early Long COVID cohort, Wallukat and colleagues found this kind of functional autoantibody in a large share of patients tested [6]. The picture is still consolidating, and not every post-infectious POTS patient has detectable autoantibodies. But the mechanism is serious enough to take on its own terms: in this subtype, something the body built to fight an infection is interfering with the signalling that controls circulation. The settings of your circulation are being changed by signals that should not be there.

These subtypes are not exclusive categories. A person can have a neuropathic component layered with mild hypovolemia and a hyperadrenergic flare in the second half of the menstrual cycle. What helps, clinically, is asking which mechanism is speaking loudest right now. Is it the pooling feet? The pre-menstrual collapse? The tremor and high blood pressure episodes? The picture of an identifiable viral trigger and evolving autoantibody findings? That question shapes the tools that follow. We will come back to concrete tools in the next chapter. For this one, the work is the measurement.

The NASA Lean Test: ten minutes against a wall

The clinical gold standard for diagnosing orthostatic intolerance is a head-up tilt-table test. Useful, accurate, often inaccessible. Tilt tables live in specialized autonomic clinics, referrals take months, and the test itself is uncomfortable enough that many people put it off.

In the meantime a home version has earned a place in the literature, partly because specialized clinics already use it and partly because it has been compared head-to-head with tilt. The original version comes from NASA's work on orthostatic intolerance in astronauts returning from low-gravity environments, and the adapted patient version is now distributed by, among others, the Bateman Horne Center. A 2025 comparison published in *Autonomic Neuroscience* ran both tests on the same sixty patients on the same day. The overall shape of the response was similar — heart rate climbing and stroke volume falling across ten minutes — but the tests did not agree on who met the diagnostic threshold: 74 per cent of patients met the heart-rate criterion on the active stand against 98 per cent on tilt [7, 8]. That gap matters for how you read your own result. A home test that comes out positive is strong evidence worth bringing to a clinician. A home test that comes out negative does not rule POTS out, and tilt remains the more sensitive of the two. It does not replace a clinical workup. What it does is give you reproducible, measurable, bring-to-the-doctor evidence in exchange for ten minutes and a cheap pulse monitor.

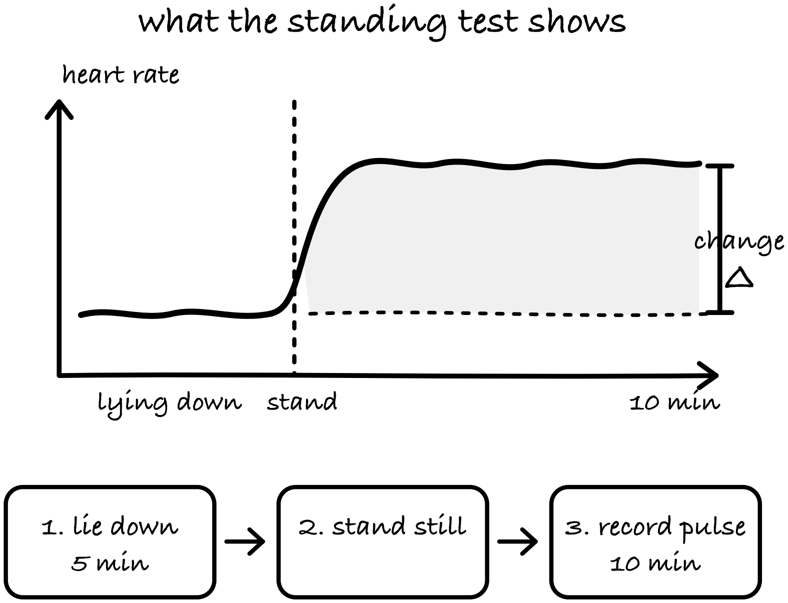
> **SAFETY —** Don't do this test alone if you tend to faint, and do it near a bed or sofa you can reach in one step. If the room begins to tilt, your vision greys out, or you feel yourself going — lie down. Immediately. This is a data-gathering exercise, not a heroism exercise. A strong symptom response during the test is information. Lie flat, wait a few minutes, the response will clear, and the minute at which the wave arrived is one of the most useful numbers you can bring to a clinician.

The physical setup matters. You stand with your shoulder blades touching a wall and your heels fifteen to twenty centimetres out from it. This particular posture switches off the calf-pump contribution almost entirely. The minor fidgeting and weight-shifting that healthy people do unconsciously while standing in line — and that patients sometimes do to stay upright — is what masks POTS in informal settings. Removing it makes the orthostatic load visible.

What you will need. A heart-rate monitor — a chest strap is more accurate than a wrist device, but a wrist device or even a cheap finger pulse oximeter works for this purpose. If you have a blood-pressure cuff, bring it too. A notebook or a phone note to write numbers down.

When to do it. First thing in the morning, before breakfast and before coffee. Note how much water you had the night before and any medications you took that morning.

The protocol is four steps.



What the ten-minute stand looks like written down.

First, lie flat on your back for five to ten minutes. Don't scroll, don't talk, don't tense. At the end of that interval, record your resting heart rate and, if you have the cuff, your blood pressure.

Second, stand up, walk over to the wall, and assume the position: shoulder blades touching, heels out fifteen to twenty centimetres. Do your best not to shift your weight, not to rise on the balls of your feet, not to tighten the calves.

Third, stand like that for ten minutes. Record your heart rate every minute, and, if you can, your blood pressure at minutes one, three, five, and ten. As you go, jot a short note about any symptoms — light-headedness, nausea, pressure in the head, coat-hanger pain (an ache across the neck and shoulders in the shape of a coat hanger, produced when the postural muscles are under-perfused; it tends to improve almost immediately on lying down), the feeling that you need to sit.

Fourth, if the symptoms overwhelm what you can tolerate — end the test early and lie down. The fact that you ended early and the minute at which you did so are data too.

How to read your numbers

A sustained rise in heart rate of 30 beats per minute or more over 10 minutes of standing — in the absence of a large drop in blood pressure — meets the standard criterion for POTS [9]. Throughout the rest of the book we will call this your *orthostatic tax*, the single most useful derived number in this population. In adolescents the threshold is 40. A drop in systolic blood pressure of 20 mmHg or more, or in diastolic pressure of 10 mmHg or more, within 3 minutes of standing is the threshold for classical orthostatic hypotension.

There is also a third, quieter group. The number-based thresholds aren't met every single time, but the symptoms are unmistakable and posture-dependent — the patient describes exactly what you would expect someone with dysautonomia to describe. Hydration status, heat, where you are in your menstrual cycle, and how much you slept the night before all shift the numbers from day to day. A single negative test in this situation does not close the question. Repeating the test on three different mornings, or during a symptom flare, often produces a clearer picture than a single encounter can.

When you bring this to your clinician, bring the numbers. A sentence like this changes the shape of the conversation: *"When I lie down, my pulse is sixty-four. After eight minutes of quiet standing it reaches one hundred and nine, with dizziness and pressure in my head. This reproduces across three separate mornings. I would like to discuss a workup for orthostatic intolerance."*

It does not sound like a complaint. It sounds like data.

Salt and water: symptom tool, not cure

One more thing before we leave the mechanics, because this is where patient guides and the biology have been in quiet disagreement for years.

Increasing daily salt intake, with enough fluids to go with it, is a well-known first-line tool in many POTS clinics. It works for some people, sometimes very well, and one important reason it helps is that temporarily raising plasma volume gives the central circulation a little more to distribute against gravity. At the hormonal level, higher sodium intake also suppresses the systemic renin–angiotensin–aldosterone axis, but this is a generic feedback response rather than a fix for the specific wiring problems seen in POTS. The reason it does not work for many others — and the reason so many patient accounts describe feeling worse on high-salt protocols — is that very few people with POTS actually have the condition because of a simple salt deficit. The three mechanisms we just walked through (damaged small nerve fibres, autoantibodies disturbing receptor signalling, a hormonal axis miscounting blood volume) are not reversed by shaking more salt onto a meal. In the long run, the kidneys and hormonal feedback loops adjust to the higher sodium load, the extra sodium and water are excreted again, and the underlying dysregulation remains largely unchanged. High-sodium intake in this setting is best understood as symptom management rather than causal treatment.

It is also something you should not improvise. The expert consensus range for POTS runs from about 3,000 to 10,000 milligrams of sodium a day, and even its lower end is roughly twice what the World Health Organization recommends as an upper limit for the general population. Sodium and salt are different numbers and easy to confuse — 3,000 milligrams of sodium is about eight grams of table salt. Chapter 5 does that arithmetic properly and says what the consensus does and does not settle. Increased circulating volume also means increased cardiac preload. Under Frank-Starling physiology, a heart with normal contractile reserve compensates by contracting more strongly — which is the reason the intervention works acutely in people whose hearts can handle it. In people whose hearts cannot — people with pre-existing hypertension, kidney disease, heart failure risk, or certain autonomic patterns in which the sympathetic response amplifies rather than compensates — the same intervention can produce worsening hypertension, volume overload, and reduced contractile efficiency.

The rule isn't "most people can add salt freely, some with risk factors cannot." The rule is: this is a cardiovascular intervention, and a clinician should be part of the decision for everyone who is considering it, not only the subset with listed risks. Describe the symptom pattern, the orthostatic numbers, and the medication list; ask for a specific recommendation about salt, fluid volume, and monitoring. Treatments that look simple on paper are sometimes the ones that need the most individual tailoring.

Specific medications for POTS — the heart-rate-lowering options, the alpha-agonists, synthetic mineralocorticoids, compression garments, physical counter-manoeuvres — are one chapter away. I want to close the book on subtypes here, and open the book on daily tools there.

The Levine protocol: when it works, when it does not

A second topic patients will hear about whenever they start reading seriously is the Levine protocol — a seven-month graded reconditioning program originally developed by Benjamin Levine and colleagues at UT Southwestern for people with POTS who are physically deconditioned [10]. The underlying logic is sound. Deconditioning does worsen POTS. Long periods in bed reduce blood volume, shrink the heart's effective working capacity, and make orthostatic tolerance still worse over time. Movement is part of the treatment.

The catch, in the context of this book, is that the original protocol was designed for POTS patients without significant post-exertional malaise. In Long COVID, ME/CFS, and post-infectious POTS, these states very often travel together. If your body reliably crashes forty-eight to seventy-two hours after effort, a reconditioning schedule that mandates weekly increases in load is pushing you into a PEM flare on a regular calendar — and in the presence of PEM, graded increases can produce lasting worsening rather than gradual improvement.

If you and your clinician decide a reconditioning program is appropriate, the safer starting approach is horizontal. Rowing machines, recumbent bicycles, exercises done lying flat on the floor — these reduce the orthostatic load while still engaging the leg muscles that contribute to venous return over time. Strengthen the calves, quadriceps, and glutes before asking the body to hold weight upright. Progress is measured in minutes added, not ten-minute jumps, and any increase happens after the body has signalled it can sustain the current level — not because a printed schedule says week four is here. If you have both POTS and significant PEM, the order matters: stabilize first, move second. That sequence is the recovery work.

Dr. Levine himself has, in more recent years, publicly discussed more cautious starting points, and several specialty clinics have developed adapted versions of the protocol specifically for patients with co-morbid PEM. If a clinician is recommending "the Levine protocol," it is reasonable to ask which version, whether PEM is being screened for, and how delayed worsening is accounted for in the schedule. A program that cannot answer those questions is a program you have a right to pause.

> **SAFETY —** Before starting any physical rehabilitation program, go through the plan with a clinician who knows your condition.

Hormones and the menstrual cycle

Most women with POTS will notice, once they start tracking, that their symptoms worsen at predictable points in the menstrual cycle — typically in the days before a period and in the first days of flow. Estrogen and progesterone influence how the body holds on to fluid, how vessels set their tone, and how sensitive the system is to orthostatic load. In a survey of over six hundred women with Long COVID collected during the research for *The Long COVID Handbook*, Gez Medinger documented this cyclical worsening as one of the most reliably reported patterns, affecting the majority of respondents [11].

If you notice a rhythm, track it. Put the harder days in the calendar in advance. Plan lighter loads for the predictable windows. Treat the pattern as biology doing what biology does. Working with your cycle is arithmetic, applied kindly.

If you are under eighteen — or parenting a teenager with POTS

Everything in this chapter applies to adolescents, with two important adjustments.

The first is diagnostic. In adolescents, the heart-rate rise threshold for POTS is forty beats per minute, not thirty, because the teenage autonomic nervous system is more reactive at baseline. Otherwise, the NASA Lean Test protocol is the same.

The second is social. Illness at forty-five affects a career and a household. Illness at fifteen affects a forming identity. The social costs — missed school, missed sports, the conversion of a teenager's peer life into time alone in a darkened bedroom — hit harder because there is less ground underneath. Teenagers with POTS routinely get labelled lazy, school-avoidant, or anxious. If your adolescent has been given one of those labels, bring the data to the school the same way you bring the data to the clinician. A heart-rate record on a standardized test tends to rewrite a conversation faster than a parental plea can.

Rehabilitation principles are the same, but medication dosing and salt recommendations in adolescents need individual adjustment with a pediatrician familiar with dysautonomia. Adolescent cardiology and adolescent autonomic medicine are now real sub-specialties — not large, but growing — and it is worth asking your pediatrician for a referral rather than assuming "wait and see" is the default.

Optional self-observation prompt

Run the NASA Lean Test yourself, with the protocol above. In the morning, before breakfast. Record your resting heart rate lying down, then your heart rate at each of ten minutes of quiet standing against a wall. Note symptoms beside the numbers. If you have a blood-pressure cuff, add blood-pressure readings at minutes one, three, five, and ten. Bring the results to your next clinical appointment.

If you are already being followed for POTS or suspected dysautonomia, run the test on three separate mornings over a week or two. Patterns travel better than single numbers, and three mornings in two weeks is both achievable and diagnostically useful.

What you will have at the end of it is what Catherine had, staring at her bedroom ceiling. The same experience you had before. Plus numbers. Plus a shape.

Sources

Numbers match the bracketed references in the chapter text. Not every entry is pointed to from the text: some are background this chapter rests on rather than support for one specific claim. 1. Thieben MJ, Sandroni P, Sletten DM, Benrud-Larson LM, Fealey RD, Vernino S, Lennon VA, Shen W-K, Low PA. 2007. Postural orthostatic tachycardia syndrome: the Mayo clinic experience. *Mayo Clinic Proceedings*, 82(3):308–313. DOI: 10.4065/82.3.308 2. Haensch C-A, Tosch M, Katona I, Weis J, Isenmann S. 2014. Small-fiber neuropathy with cardiac denervation in postural tachycardia syndrome. *Muscle & Nerve*, 50(6):956–961. DOI: 10.1002/mus.24245 3. Fu Q, VanGundy TB, Galbreath MM, Shibata S, Jain M, Hastings JL, Bhella PS, Levine BD. 2010. Cardiac origins of the postural orthostatic tachycardia syndrome. *Journal of the American College of Cardiology*, 55(25):2858–2868. DOI: 10.1016/j.jacc.2010.02.043 4. Garland EM, Celedonio JE, Raj SR. 2015. Postural tachycardia syndrome: beyond orthostatic intolerance. *Current Neurology and Neuroscience Reports*, 15(9):60. DOI: 10.1007/s11910-015-0583-8 5. Benarroch EE. 2012. Postural tachycardia syndrome: a heterogeneous and multifactorial disorder. *Mayo Clinic Proceedings*, 87(12):1214–1225. DOI: 10.1016/j.mayocp.2012.08.013 6. Wallukat G, Hohberger B, Wenzel K, Fürst J, Schulze-Rothe S, Wallukat A, Hönicke A-S, Müller J. 2021. Functional autoantibodies against G-protein coupled receptors in patients with persistent Long-COVID-19 symptoms. *Journal of Translational Autoimmunity*, 4:100100. DOI: 10.1016/j.jtauto.2021.100100 7. Bateman Horne Center. NASA Lean Test Instructions (patient handout). batemanhornecenter.org. https://batemanhornecenter.org/ 8. Uppal J, Baker JR, Hira R, Karalasingham K, Ranada S, Deol P, Sheldon RS, Raj SR. 2025. Physiological and clinical comparison of active stand and head-up tilt tests in Postural Orthostatic Tachycardia Syndrome (POTS). *Autonomic Neuroscience*, 260:103281. DOI: 10.1016/j.autneu.2025.103281 Sixty patients completed a 10-minute active stand followed by head-up tilt on the same day. Heart-rate and stroke-volume trajectories were broadly similar, but 74% met the POTS heart-rate criterion on active stand against 98% on tilt ($p<0.001$); lowering the active-stand threshold to 27 bpm narrowed the gap to 83% against 98% (p=0.02). 9. Freeman R, Wieling W, Axelrod FB, Benditt DG, Benarroch E, Biaggioni I, Cheshire WP, Chelimsky T, Cortelli P, Gibbons CH, Goldstein DS, Hainsworth R, Hilz MJ, Jacob G, Kaufmann H, Jordan J, Lipsitz LA, Levine BD, Low PA, Mathias C, Raj SR, Robertson D, Sandroni P, Schatz I, Schondorf R, Stewart JM, van Dijk JG. 2011. Consensus statement on the definition of orthostatic hypotension, neurally mediated syncope and the postural tachycardia syndrome. *Clinical Autonomic Research*, 21(2):69–72. DOI: 10.1007/s10286-011-0119-5 10. Fu Q, Levine BD. 2018. Exercise and non-pharmacological treatment of POTS. *Autonomic Neuroscience*, 215:20–27. DOI: 10.1016/j.autneu.2018.07.001 11. Medinger G, Altmann D. 2022. *The Long COVID Handbook*. London: Hutchinson Heinemann. ISBN 9781529900125. Survey of 603 women with Long COVID reporting correlation of POTS-pattern symptoms with menstrual cycle phases.

Chapter 5. Tools that work every day

Catherine had the spreadsheet open on the kitchen table. Three mornings of numbers, all in the same careful row format her husband had helped her set up. Supine, sixty-four. Two minutes standing, eighty-one. Five minutes, ninety-six. Eight minutes, one hundred and nine. The same staircase three mornings in a row, with a small variation on the third day that did not change the shape of the pattern.

She had carried the test as far as data could carry it. What she did not yet have was a sentence for the next question, which was: *what do I actually do tomorrow at seven in the morning, when I need to stand up and make breakfast for two children*.

That is what this chapter is for.

Knowing what is happening is the first step. Turning that knowledge into a repeatable morning is the second, and they are not the same kind of work. One belongs to the laboratory and the clinic. The other belongs to the kitchen counter, the glass of water beside the bed, the drawer where the compression garment lives so you remember to put it on before you stand.

This chapter covers the daily tools — hydration, sodium, compression, body position, meal structure, medication — that research and clinical experience have converged on for people with POTS and orthostatic intolerance. Many of them are unglamorous, and their plainness is part of why they work: they are the interventions you can still carry out on the days when a more ambitious plan would collapse.

One more framing point before we start, and it matters. Everything in this chapter is a symptom-management tool. None of it corrects the underlying regulation. On days when that regulation isn't getting corrected quickly, symptom management is how you keep your life moving, and that is reason enough to use these tools — alongside pacing introductions, working with a clinician on the pieces that carry real cardiovascular weight, and tracking what is changing, so that when you finally get to a specialist conversation you arrive with a record of what has and hasn't helped.

Why "drink more water" is a technically correct, practically useless sentence

The advice patients with POTS hear most often is to drink more fluids. It is correct and it is mostly useless as written, and the reason is worth understanding, because the reason is also the repair.

Plasma volume — the liquid portion of your blood — is not set by how much water you drink. It is set by the balance between water and sodium in your circulation, and it is actively regulated by the kidneys, which decide at any given moment how much of each to keep and how much to send on to the bladder. If you drink a litre of plain water, the kidneys read a drop in circulating sodium concentration and dispatch the excess water out of the body within roughly an hour or two. You have been hydrated. Your plasma volume has barely moved.

Sodium is the lever. It holds water in circulation through osmotic pressure — wherever sodium is, water follows. This is why intravenous saline produces rapid symptomatic relief in POTS: it expands plasma volume in minutes, and symptoms ease for as long as the expansion lasts [1]. It is also why the effect is temporary. The kidneys eventually excrete the added sodium, the volume returns to baseline, and you are back to whatever daily tools you were using before.

In Chapter 4 I wrote that increasing dietary salt is a cardiovascular intervention that belongs in a clinician conversation, not a DIY kitchen experiment. That remains true. Everything that follows in this chapter assumes you have had that conversation, or are about to have it, and describes how the non-medication tools actually work when you and your clinician decide they are appropriate for you.

Sodium and salt are different numbers

This is the arithmetic that costs people weeks of wrong dosing, so it is worth being slow about.

Food labels and most clinical guidelines report sodium in milligrams. Recipes and home cooking talk about salt in teaspoons and grams. To be exact about the mass: sodium makes up about thirty-nine per cent of the weight of table salt, which is how the back-of-envelope conversion works — three grams of salt hold roughly 1,170 milligrams of sodium, ten grams hold about 3,900.

One teaspoon of table salt weighs about six grams. It contains approximately 2,300 milligrams of sodium. The NIH expert consensus on POTS gives a single range rather than a ladder: sodium intake can be raised to somewhere between 3,000 and 10,000 milligrams a day, and the consensus says plainly that the optimal dose has not been established [2]. Where clinicians sit within that range for a particular patient is a clinical judgement, not a published number [3]. Translated into salt, that is roughly 1.5 to 4.5 teaspoons of table salt, with the higher end reserved for clinical settings. This is also why many sports drinks are close to useless for orthostatic symptoms in POTS: a bottle with 200 milligrams of sodium per serving is formulated for a well person who is sweating, not for someone with a regulated plasma-volume problem.

The preload argument from Chapter 4 holds: more sodium increases the blood volume returning to the heart, a well-calibrated heart compensates through the Frank-Starling mechanism, and hearts with less reserve can run into trouble instead of benefit. What the arithmetic above is for is reading labels correctly when you have that conversation.

> **SAFETY —** Any significant change in sodium intake is a cardiovascular intervention and a clinician-supervised change, not a kitchen experiment. This holds for everyone considering it, not only for people with a listed risk factor. If you have high blood pressure, kidney disease, heart failure risk, or take medication that affects fluid balance, the conversation happens before the first change, not after it.

The morning protocol: hydrate before you are vertical

For most people with POTS, the worst hour of the day is the first hour. Overnight the body loses water through breath and skin, plasma volume settles toward its lowest point, and the moment you swing your legs over the side of the bed you are asking an already reduced circulation to handle gravity without warning.

The simplest intervention that changes this for some people — and it is a small, clinician-sanctioned intervention rather than a cure — is to start rehydrating before you stand up. A glass of water with electrolytes on the bedside table, drunk lying down or sitting up on the edge of the mattress, ten to fifteen minutes before you put your feet on the floor, gives the kidneys a head start. The plasma volume you stand up on is not the volume you went to sleep with.

What to put in that glass, in roughly decreasing order of evidence:

Oral rehydration salts — the formulation developed by the World Health Organization — are the best-studied option. They carry a specific ratio of sodium and glucose that maximises intestinal absorption through the sodium-glucose co-transport pathway in the small intestine; a packet in 500 to 1,000 millilitres of water gives you more retained fluid than the same water without the sugar [4]. Electrolyte powders with a high sodium content, read off the back of the sachet rather than the front of the packaging, are a practical alternative. Salty broth works well for people who cannot stomach something sweet at six in the morning. A home version — water with a small pinch of table salt and a splash of juice for palatability — is a reasonable budget equivalent of ORS for a single morning dose.

The phrase from the rehabilitation literature that I keep coming back to is from a woman who had been using this approach for months before it clicked for her.

"It took me a while to fully understand that increasing my electrolyte and fluid intake would make a difference to my blood pressure. I had to work hard at it."

— *Rehabilitation and Management of Long COVID*, clinician interview

She was not describing doubt about the biology. She was describing the small emotional barrier to taking an unglamorous intervention seriously and being consistent with it. That barrier is real. The boring tools are the ones a body in crisis can actually sustain. They deserve to be taken seriously on that basis alone.

A word on the shower, because the shower is the classically catastrophic morning ritual for untreated POTS (the five-task stack was introduced in Chapter 1). Hot water dilates peripheral vessels, peripheral dilation worsens pooling, and a shower taken while already symptomatic can become the beat when you sit on the bathroom floor waiting for the room to return. An extra dose of electrolytes twenty to thirty minutes before showering, water temperature lukewarm rather than hot, a shower stool for the actual shower, and compression on immediately afterward — those four elements turn the shower from a daily ambush into an ordinary task for a very large share of people who try them.

Distribute sodium across the day, not only at breakfast

A single large morning dose of sodium is better than nothing. A steady distribution across the day is better still.

Plasma volume demand does not end at nine in the morning. If all your sodium arrives at breakfast, by lunchtime the kidneys have already excreted most of it and your circulating volume is drifting back to baseline. What works for many patients is the simplest possible scheduling: add sodium to every meal, put a salted snack between breakfast and lunch, drink an electrolyte beverage in the early afternoon or before any planned activity, and — for people whose symptoms worsen toward evening — have a cup of broth with dinner. The snack is a delivery mechanism for the sodium, and that is the point of the schedule.

Compression: the abdomen carries more weight than the calves

Compression is an underused tool in POTS, partly because the most commonly sold versions address the wrong region.

The intuition most people start with is calf-and-thigh: knee-high socks, full-length stockings. Those work, modestly, and they are better than nothing. The surprise in the literature is that abdominal compression — a binder, high-waisted compression shorts, or a leotard-style garment that covers the torso — has been shown to relieve POTS symptoms more effectively than leg-only compression in head-to-head comparison [5]. The reason is anatomical. The splanchnic bed, the network of veins serving the gut and the abdominal organs, is one of the largest blood reservoirs in the body; at rest it holds roughly a quarter of circulating volume, and in a body with unreliable vessel tone it can hold more. When you stand, that region can act like an expandable basin for blood that should have returned to the heart. Compressing the abdomen keeps that basin smaller.

The practical sequence is useful. If you are choosing a single garment, pick abdominal compression over leg compression. Readers who already wear knee-highs with modest benefit often find that adding an abdominal binder on top produces an additive effect larger than either alone. And for anyone who has tried knee-highs and given up, one more attempt with an abdominal garment is worth making before concluding that compression does not help you.

Compression class matters less than correct wearing. A 20–30 mmHg gradient at the ankle is the standard starting point for POTS in both adult and adolescent guidance [6]. Higher grades are not automatically better; a garment too tight for you is a garment you will stop wearing by Wednesday.

> **SAFETY —** Contraindications to know before you start: peripheral arterial disease, deep vein thrombosis, advanced chronic venous insufficiency with skin breakdown, severe peripheral neuropathy with loss of sensation, and open wounds on the area to be compressed. For abdominal garments, add pregnancy beyond the first trimester, recent major abdominal surgery, active inflammatory bowel disease or significant abdominal pain and bloating, and known pelvic organ prolapse or marked pelvic floor weakness. Any of those is a conversation with a clinician before you shop.

The wearing rule that most changes effectiveness is timing. Compression works best when it is on *before* you stand. If you stand, walk for twenty minutes, notice the pooling, and then put on the compression, the blood has already moved into the region you are trying to compress and you are compressing an already-full basin rather than preventing it from filling. Keep the garments beside the bed. Put them on sitting up or lying down.

A seasonal note for anyone in warm climates: heat and compression are an uncomfortable pairing, and summer adherence tends to drop. If wearing compression all day in July is not realistic, use it for the windows that matter most — the first two hours of the morning, showers, any planned period of upright activity, travel — and let it go for the rest.

Physical counter-manoeuvres: the cheapest tool in this chapter

If you ever need to stand in a queue, chop vegetables, wait on a train platform, or hold a conversation at a reception — situations where sitting is not socially available — there is a set of small body-position tricks that work immediately, cost nothing, and activate the muscle pump without requiring a walk.

The most useful ones, named individually so you can practise them:

Leg crossing — standing with one leg crossed in front of the other and squeezing both legs toward the midline reduces venous pooling in the lower body measurably.

Gluteal and thigh squeezing — isometric contraction of large muscle groups in the legs and buttocks engages the calf-and-thigh pump even while you are standing still.

Weight shifting — the smallest version of walking in place, rocking subtly from one foot to the other, is still better than holding a completely static posture.

One foot on a step — in a kitchen, a low stool or oven door makes an excellent occasional foot rest; on a train, the lip of a suitcase does the same. Asymmetric loading breaks up pure static standing.

Forward lean with support — resting forearms on a counter or the back of a chair and tipping forward slightly reduces orthostatic loading on the circulation without requiring you to sit.

All of these work best as prevention rather than rescue. Use them before symptoms arrive, not after. The physiology does not wait for a warning.

Two further tools belong in this paragraph even though they are not strictly manoeuvres. A shower stool is a medical device that removes orthostatic load and heat stress from the single most reliably difficult daily ritual in POTS; a large share of patients describe it as among the more meaningful changes they have made to daily life, and the social meaning of using one is worth letting go of. Raising the head of the bed by ten to fifteen centimetres — by placing blocks under the bed frame at the head end, not by sleeping propped up on extra pillows — can reduce overnight urinary output and help preserve plasma volume into the morning, and is a well-established conservative measure in the guidelines [6]. Neither is necessary for everyone. Both are worth knowing about.

The post-lunch dip: meals as an orthostatic stressor

A pattern patients recognise without always having the name for it: mid-afternoon is worse than mid-morning, and the difference tracks lunch.

After a meal, blood redistributes toward the gut to support digestion. In a well-regulated circulation this redistribution is invisible. In a POTS body, it adds an orthostatic tax on top of the one you are already paying [7]. A heavy meal rich in fast-digesting carbohydrates amplifies the effect, because the carbohydrate load triggers an insulin release, glucose drops, and the sympathetic nervous system fires a compensatory response on top of the digestive blood shift. The afternoon slump you feel has a hemodynamic fingerprint, and framing it as a discipline problem or a sugar-craving problem tends to miss the intervention.

What tends to help:

Smaller portions more often. Four small meals tolerate better than two large ones.

More protein and fat relative to simple carbohydrates.

An electrolyte beverage with or just before the meal — the same tool used for mornings, adapted for the post-lunch window.

A brief horizontal or reclined rest after lunch if circumstances allow. Thirty minutes reclining supports venous return at the exact moment venous return is under the highest demand, and the alternative — pushing through the dip on willpower — tends to produce a longer, worse afternoon.

Caffeine sits in its own category. For some patients it worsens symptoms by amplifying tachycardia. For others it helps, modestly, through mild vasoconstriction. The only reliable way to know which group you belong to is to remove it for two weeks as a controlled experiment, observe, and then decide. Broad recommendations on this point are noise.

Medications: matching the mechanism to the pattern

This book is not a manual of pharmacotherapy, and the decision to prescribe any of the medications below belongs to a clinician who knows your situation in detail. What a reader can usefully bring to that conversation is a clear sense of what the available tools do, so that "what are our options?" is a question you can ask with specificity rather than a request to be handed a pill bottle and sent home.

The four medications most often discussed in POTS, each with its target subtype:

Midodrine. A peripheral α1-adrenergic agonist; it constricts arterioles and venules, reducing pooling. It is typically started at 2.5 or 5 mg and can be dosed three times daily during upright hours. Dosing is not given at bedtime because supine hypertension is the characteristic adverse effect. It tends to help most in neuropathic POTS with low blood pressure, where the underlying problem is vessel tone failing to tighten on standing. It does relatively little for people whose primary pattern is high heart rate with normal or elevated blood pressure.

Fludrocortisone. A synthetic mineralocorticoid. It acts on the kidneys to increase sodium and water retention, expanding plasma volume. Typical starting dose is 0.1 mg once daily. It tends to be most useful in hypovolemic POTS, where circulating volume runs low at baseline. Potassium needs to be monitored; low potassium is the characteristic metabolic adverse effect. It is a cardiovascular intervention in the same sense as high-dose salt, and the same clinical oversight applies.

Beta-blockers. A family of medications — propranolol, metoprolol, and nadolol are common choices — that slow the heart by blocking the effect of sympathetic activation at the β1 receptor on the sinus node. Low doses often outperform high doses in POTS. They help when tachycardia is the dominant burden, and they can lower blood pressure as a side effect, which makes them a more complicated choice for people whose blood pressure is already low on standing.

Ivabradine. An If-channel blocker that slows the sinus node directly, without affecting vessel tone or blood pressure. Typical dosing is 2.5 to 7.5 mg twice daily. It is often the preferred heart-rate-lowering agent in POTS patients with low or normal blood pressure, because beta-blockers can drop their numbers further while ivabradine usually does not [8]. One caveat worth raising in the clinician conversation: ivabradine has been associated with a modestly increased risk of atrial fibrillation, and it is typically avoided in patients with an already low resting heart rate. Clinical experience in non-heart-failure POTS populations suggests it is well tolerated in otherwise healthy patients with resting rates in the sixties [9].

A smaller group of agents — the **central α2-agonists clonidine and guanfacine** — is sometimes used in hyperadrenergic POTS to dampen central sympathetic outflow. They are not first-line tools; they come up in cases where the dominant feature is the sympathetic surge itself rather than pooling or volume. A small case series in 2023 reported that the same medication family, at lower doses, improved attention and working memory in a twelve-patient Long-COVID cognitive-symptom cohort — a separate indication from POTS, grounded in the same receptors acting in a different part of the brain, and a finding that needs larger trials before it becomes a recommendation.

The organising principle of this paragraph is the same as the organising principle of Chapter 4: match the mechanism, not the symptom. A high pulse at low blood pressure and a high pulse at normal blood pressure are treated with different tools, even though the complaint looks identical on paper. Knowing your subtype — from your NASA Lean Test numbers, from symptom pattern, sometimes from more detailed testing — is what makes the medication conversation efficient. A prescriber asked "what are our options for *neuropathic* POTS with orthostatic hypotension" has a much cleaner decision surface than a prescriber asked "I feel awful when I stand, what can you give me?"

Reconditioning in a POTS body with PEM

Chapter 4 covers why the classical Levine reconditioning protocol — a structured, graded exercise programme originally developed to retrain the cardiovascular system in POTS, which begins with recumbent exercise and increases the load on a fixed weekly schedule — was not built for the PEM-positive body and what a safer starting shape looks like. The short version, for this chapter's purposes, is that the underlying logic holds — deconditioning does worsen POTS, and movement does help —

but the weekly-increase schedule is what fails in the presence of a 48-to-72-hour crash window.

If you and your clinician decide a movement programme is appropriate, start horizontal and start small. Recumbent bike, rowing machine, floor-based exercises done lying down, swimming in a cool pool. Build the muscle pump in the legs, glutes, and trunk before asking the body to hold weight upright for sustained periods. Minutes added, not ten-minute jumps. Progression happens after the body has signalled it can sustain the current level, not because a week counted forward on the calendar.

One quiet success story in the rehabilitation literature describes a young patient with post-viral POTS who started with daily floor-based supine exercises and three-minute sessions on a recumbent bike. Over six months she moved in stages from lying to kneeling to seated to standing strength work, and extended the bike to thirty minutes. Pacing, fluids, light meals rather than large ones, and compression were in place the whole time. She did not need medication to reach a meaningfully better baseline [10]. Her trajectory is one working example of what "horizontal-first, patient with progression" looks like when the pieces are assembled correctly; it does not predict the same arc for everyone, and individual variation in post-viral POTS is large.

Putting the morning together as one system

The tools above work better in combination than individually. A morning protocol that pulls the main levers at once — pre-stand electrolytes, compression donned lying down, a small breakfast with protein and a modest sodium dose, and an unhurried first half-hour — is one system, not three or four separate interventions. For a large share of people with POTS, the morning is the day's hardest hour; a system that handles the morning changes the shape of everything else.

Two warnings that matter more than they sound like they should.

Do not change five things at once. Pick one. Use it for two weeks. Note what, if anything, changed. Then pick a second. When you change five things together and feel better, you do not know what helped. That knowledge matters, because the illness evolves: seasons change, menstrual cycles change, infections change, and you need to know which lever you can pull when something slips. Pacing introductions one at a time is how you build an accurate map of your own response.

None of this corrects the underlying regulation; it gives your circulation better conditions while the upstream work — specialist evaluation, targeted medication, and the part of this book that belongs to the pacing chapter next — is happening. Tools that let you live while the deeper work proceeds are the scaffolding the rest of the treatment stands on, and installing them well is worth the time it takes.

Catherine did not decide on a five-step plan that morning. She decided on one change. She put a glass of water with a quarter teaspoon of salt and a squeeze of lemon on the bedside table the night after she finished this chapter, set her phone alarm for fifteen minutes before she usually got up, drank the glass sitting on the edge of the bed, and waited. Then she stood.

Her pulse at minute five that morning was eighty-seven, not ninety-six. It is one data point. She wrote it down anyway, under the earlier numbers, on the same spreadsheet, ready for the next appointment with Dan. The conversation she had been unable to have for six months was starting to find its language.

Optional self-observation prompt

Run a single, low-stakes version of the morning hydration experiment, if you have no medical contraindications to a small one-off salt dose.

Dissolve roughly a quarter teaspoon of ordinary table salt — about 575 milligrams of sodium — in 500 millilitres of water, with a little juice or cordial for palatability. A commercial electrolyte sachet rated at 300 to 500 milligrams of sodium per serving is a reasonable alternative. Put the glass on the bedside table tonight. In the morning, drink it sitting or lying on the bed, wait ten minutes, and then stand. Note, with honesty, how the first five minutes feel compared with a normal morning.

Treat this as a single, low-dose check on whether morning volume expansion moves something for you, rather than as the start of a protocol. A change worth chasing shows up fairly clearly. No change is also useful data — the morning problem may sit more in the vessel-tone or the sympathetic-drive side of the system than in volume, and the tools to address those sit in the compression, counter-manoeuvre, and medication paragraphs above.

> **SAFETY —** Do not scale this up on your own. The safe dose for a single trial differs from the safe dose for daily use, especially if you have high blood pressure, kidney or heart conditions, or are on medication that affects fluid balance. If something useful happens tomorrow morning, bring the observation to your next clinical appointment and ask what the supported version of this looks like for your body.

One morning's worth of data is the first sentence of a conversation that has been waiting for a shared vocabulary — the start of a plan, not yet the plan itself.

Sources

1. Vernino S, Bourne KM, Stiles LE, Grubb BP, Fedorowski A, Stewart JM, et al. 2021. *Postural orthostatic tachycardia syndrome (POTS): State of the science and clinical care from a 2019 National Institutes of Health Expert Consensus Meeting — Part 1.* Autonomic Neuroscience, 235:102828. DOI: 10.1016/j.autneu.2021.102828 Intravenous saline produces rapid, short-lasting symptomatic relief through plasma-volume expansion.
2. Vernino S, et al., 2021, as above. §13: "Sodium intake can be increased to 3 to 10 g daily using ordinary table salt (1 tsp. is approximately 2.3 g sodium), salt tablets or electrolyte solutions. DOI: 10.1016/j.autneu.2021.102828 Further research is needed to establish the true therapeutic benefit of salt loading and the optimal sodium intake."
3. Hitch D, Wrench J, eds. 2026. *The Rehabilitation and Management of Long COVID: A Handbook for Clinical Practice.* London: Routledge. DOI: 10.4324/9781003528104 Clinical practice on volume loading under supervision. **And:** Garland EM, Gamboa A, Nwazue VC, et al. 2021. *Effect of High Dietary Sodium Intake in Patients With Postural Tachycardia Syndrome.* Journal of the American College of Cardiology, 77(17): 2174–2184. DOI: 10.1016/j.jacc.2021.03.005 Crossover trial: six days at 300 mEq sodium daily — about 6,900 mg — reduced upright heart rate and increased plasma volume against a low-sodium arm.
4. World Health Organization. *Oral rehydration salts: production of the new ORS,* 2006. Sodium-glucose ratio optimised for co-transport absorption in the small intestine. https://www.who.int/publications/i/item/WHO-FCH-CAH-06.1
5. Bourne, K.M., et al., 2021. *Compression Garment Reduces Orthostatic Tachycardia and Symptoms in Patients With Postural Orthostatic*

Tachycardia Syndrome. Journal of the American College of Cardiology, 77(3): 285–296. DOI: 10.1016/j.jacc.2020.11.040 Abdominal compression produces larger reductions in standing tachycardia and symptoms than lower-body compression alone.

6. Sheldon, R.S., Grubb, B.P., Olshansky, B., et al., 2015. *2015 Heart Rhythm Society expert consensus statement on the diagnosis and treatment of postural tachycardia syndrome, inappropriate sinus tachycardia, and vasovagal syncope.* Heart Rhythm, 12(6): e41–63. DOI: 10.1016/j.hrthm.2015.03.029 **And:** El-Sayed, H., Hainsworth, R., 1996. *Salt supplement increases plasma volume and orthostatic tolerance in patients with unexplained syncope.* Heart, 75(2): 134–140. DOI: 10.1136/hrt.75.2.134 Head-up bed tilt and conservative volume measures as components of non-pharmacologic POTS management.
7. Tani H, Singer W, McPhee BR, Opfer-Gehrking TL, Haruma K, Kajiyama G, Low PA. 2000. *Splanchnic-mesenteric capacitance bed in the postural tachycardia syndrome (POTS).* Autonomic Neuroscience: Basic and Clinical, 86(1–2): 107–113. DOI: 10.1016/S1566-0702(00)00205-8 Excess splanchnic pooling at rest in POTS, and postprandial worsening of orthostatic symptoms.
8. Sheldon, R.S., et al., 2015. *2015 Heart Rhythm Society expert consensus statement on the diagnosis and treatment of postural tachycardia syndrome, inappropriate sinus tachycardia, and vasovagal syncope.* Heart Rhythm, 12(6): e41–e63. DOI: 10.1016/j.hrthm.2015.03.029 Ivabradine: sinus-rate reduction without blood-pressure effect; clinical preference over beta-blockers in low- or normal-BP POTS.
9. Hitch D, Wrench J, eds. 2026. *The Rehabilitation and Management of Long COVID: A Handbook for Clinical Practice.* London: Routledge. DOI: 10.4324/9781003528104 Clinical experience with

ivabradine tolerability in otherwise healthy patients with resting heart rates in the 60–69 range.

10. Hitch D, Wrench J, eds. 2026. *The Rehabilitation and Management of Long COVID: A Handbook for Clinical Practice.* London: Routledge. DOI: 10.4324/9781003528104 Case report chapter on post-viral POTS rehabilitation without pharmacological intervention.

Chapter 6. Pacing: the science of not overdoing it

Catherine had a good morning, and the good morning was the problem.

Two weeks after she started the pre-stand glass of water and the quarter teaspoon of salt, she woke up on a Thursday feeling, for the first time in six months, a little more like a person. Her head was clearer. Her eyes focused on the first try. She walked into the kitchen, noticed that the laundry basket by the washing machine had reached the point of rebellion, and thought: *today, I can get ahead*.

By four in the afternoon she had run one load of laundry, answered eleven emails, made both children lunch, taken a short walk around the block, and stood at the kitchen counter chopping vegetables for a dinner she actually wanted to eat. She sat down on the sofa at five, pleased with herself, and felt mildly tired.

On Saturday she could not get out of bed.

She wrote this in her spreadsheet, under the morning heart rates, in a column she had added the week before and labelled simply *what I did*: Thursday's list, then Friday's short entry (*mostly okay, went to school pickup*), then Saturday (*couldn't*). She stared at the three rows for a long time. The Thursday column showed a normal-ish day for a well person. The Saturday column showed a version of her illness she had been in three or four times before but had never put in writing next to its cause. It was the same pattern each time. A good day, a reasonable day, a day that was gone.

This chapter is about that pattern. It has a name in the patient community — *boom-and-bust* — and the literature calls its delayed core *post-exertional malaise*. The work of this chapter is to describe the biology of why the bill arrives two days after the activity, to explain what it means to live inside that delay, and to lay out the tools people use to stop spending through the ceiling without noticing it. Pacing is the name for those tools as a system. The version of pacing that actually works is more technical than most patients hear on a first pass, and it is also the most-cited self-management intervention in the patient literature on what helps people recover [1].

The dangerous day is the good one

The days people with post-viral illness most need to be careful on are not the days they feel worst. They are the days they feel unexpectedly better. On those days the baseline rises, the internal message is “*finally!*”, and the plan written with the day’s new energy is the plan that books the crash approximately 48 hours later. You do not feel the ceiling coming. That is the whole problem.

If you asked Catherine on Thursday evening whether she had overdone it, she would have said no. Nothing she had done that day was unreasonable on its own. The laundry was ordinary. Her walk was short. Cooking was, on any well day in her life before, a relaxing thing. None of those pieces knew about each other, and none of them knew about the internal budget they were drawing against. By the time the crash arrived on Saturday, the acts that caused it were thirty-six to sixty hours behind her, and she could not trace the line.

The delay is the puzzle. In a well body, fatigue lives close to its cause: you feel tired while you are running, you rest the same evening, and the next morning you are roughly the same person you were. In a post-viral body with PEM, effort and consequence come uncoupled in time. What you feel in the moment is a poor reading on how much you have spent, and the uncoupling itself is a measurable feature of the condition — which is why pacing has to be treated as a technical discipline, planned in advance rather than judged by how you feel in the moment.

Why the bill comes late

PEM — the delayed post-exertional worsening introduced in Chapter 1 — is the mechanism pacing is organized around. A shutdown is the right word. The mechanism is plural rather than singular, and several systems drag the tab.

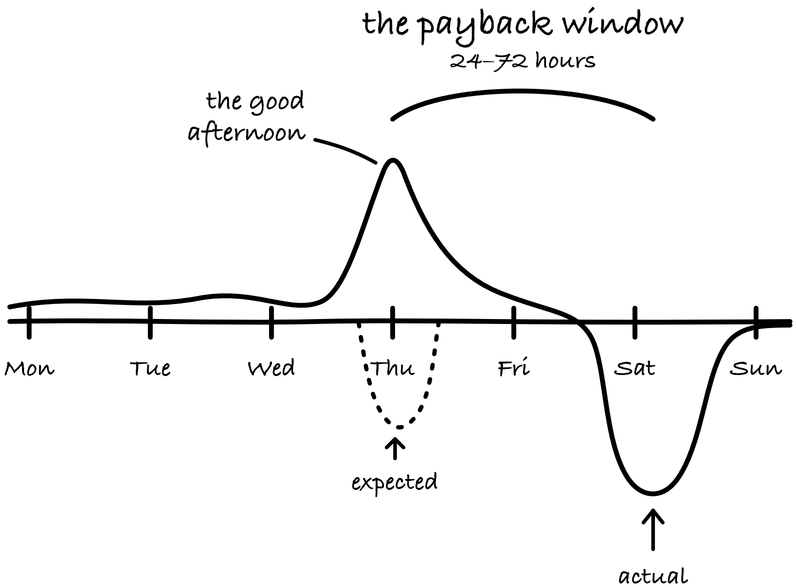
The first is energy production. In Chapter 1 we looked at the evidence that the pyruvate dehydrogenase complex — the enzyme bridge that lets the mitochondria use glucose aerobically — is downregulated in ME/CFS [3]. With that pathway restricted, the body runs more of its metabolism through the anaerobic branch, which yields roughly two molecules of ATP per glucose against the thirty or so that an intact aerobic path produces. Fuel depletes faster. Recovery takes longer. During the activity itself the sympathetic nervous system masks the deficit — adrenaline makes you feel capable — so the internal reading on remaining energy is unreliable in exactly the moment you would use it to decide whether to keep going.

The second is replenishment. After exertion, the body has to rebuild its ATP stores, and that requires adequate purine recycling through the pentose phosphate pathway. In healthy tissue the turnaround is hours. In PEM-prone tissue it runs into days, because the limiting enzymes are themselves operating under a dysregulated cellular environment, and purines that leave the cell through degradation are not quickly replaced [4]. An overdraft on a well day takes one night to clear; an overdraft on a PEM-prone body takes three or four days and compounds if you do not let it settle.

The third is cerebral blood flow. The van Campen–Rowe–Visser tilt-table series — covered in Chapter 2 — quantified a roughly quarter-sized drop in cerebral perfusion on standing in this population [5], and a 2025 SPECT imaging study found the abnormality was already present supine in about 61 percent of POTS patients [6]. This is the mechanism behind the thing patients describe constantly and clinicians rarely take at its word: that reading a difficult email from an upright chair costs about the same as climbing a short flight of stairs. The underperfused prefrontal cortex is the organ doing the reading, and the cost is real.

There are other contributors in the research literature — immune dysregulation, autonomic instability, microvascular changes, mast-cell activity — and the relative weight between them probably differs from one patient to another. The point for this chapter is the opposite of reductive. PEM is multifactorial: reducing it to a single-organ or single-pathway label tends to produce treatment plans that work only for the subset of patients whose physiology happens to be led by that pathway, and misses the rest. The failure modes of single-system models are what make pacing — which is system-agnostic — the most transferable intervention across the IACC family of conditions.

The practical conclusion is blunt. By the time you feel tired, the spending has already happened. In-the-moment sensations are not the dial you plan from. The real unit of planning runs in forty-eight to seventy-two hour windows [2] rather than in single hours or single days. I call this the **payback window**: the interval in which the cost of an effort actually arrives. It is one of the handful of terms this book fixes to a single meaning, alongside the orthostatic tax and hours of upright activity, and the rest of the book uses it in that sense. The right question to ask about an activity is *can I do this today and still do tomorrow and the day after*.

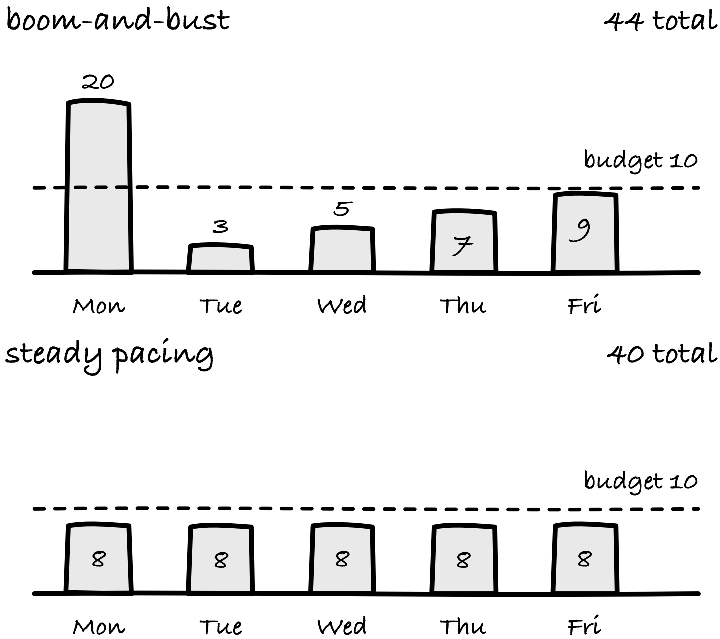


The effort and the price for it sit days apart.

The math of boom-and-bust

Pacing looks like a restriction until you put numbers on it.

Imagine a realistic safe budget of ten units of activity per day. On the good morning, feeling capable, you spend twenty. You are proud of yourself. You have answered the emails, taken the shower standing up, and done a short walk. Over the following four days the overdraft charges itself back: Tuesday you manage three units, Wednesday five, Thursday seven, Friday nine. The five-day total is forty-four units of actual activity, and most of those four days are spent horizontal, tired, low in mood, and grieving for the capable person who showed up on Monday.



Almost the same total. Four very different days.

The paced version is less dramatic. Eight units every day, hedged well under the ceiling. That week's total is forty. The following week, if nothing changes, is forty again, and the week after that a little more — because a system that is not repeatedly crashing is a system that can slowly build tolerance. Over two weeks the paced line passes the boom line for cumulative activity. Over a month it passes by a wide enough margin that the comparison stops being close.

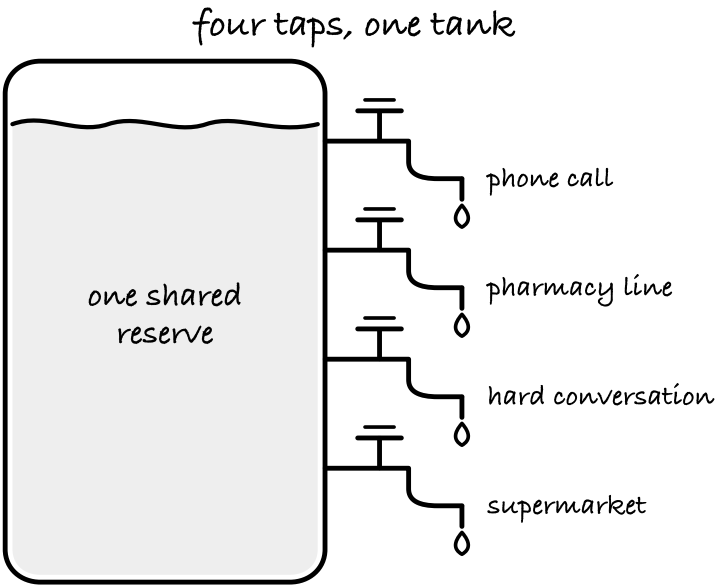
A laptop with a dying battery and a broken fan makes it through more of a workday at fifty percent brightness than at full — not because half is better than whole in principle, but because the machine cannot complete a working day at full without shutting down. The boom-and-bust pattern is the shutdown. Pacing is the setting the machine can actually hold.

Four kinds of spending, not one

Pacing fails more often from accounting errors than from willpower failures. The most common accounting error is counting only physical activity.

A day in which you "did nothing" — meaning you did not leave the house, did not do housework, did not exercise — can still spend the budget down through channels that people underestimate because they do not feel like effort. I have stopped using the word *nothing* when I'm working with patients on a diary, and the reason is that every day written up honestly has at least four categories of spending on it. Tracking only one of them is the most common reason a pacing plan "doesn't work."

Physical is the obvious one. Walking, housework, standing at the kitchen counter, climbing stairs, the shower, carrying groceries. Cognitive is reading, email, phone calls, decisions, screens, conversation that requires attention. Orthostatic is any time spent upright: sitting counts here too, because a non-reclining posture still asks the vascular system to move blood against gravity, and for a person with meaningful orthostatic intolerance, a two-hour upright sit is not a rest. Emotional and sensory is conflict, medical appointments, grief, noise, bright light, and crowded or over-stimulating environments.



One tank. Four ways to empty it.

If cerebral blood flow is lower in an upright posture in your body, reading a complex document while seated at a desk is not recovery time. If a conversation with a difficult family member triggers a sympathetic surge, the twenty minutes on the phone may cost more than the laundry did. A diary that writes down *read for an hour* and a diary that writes down *read sitting for an hour, on day after argument with Dan* are two different accounting systems, and only the second one can explain a Saturday crash.

A small practical change that makes the diary usable: record body position alongside activity. Thirty minutes reading supine on the sofa and thirty minutes reading at the kitchen table are not the same unit of cost. The position is part of the price.

Heart rate as a fire alarm, not a coach

Using a heart-rate monitor for pacing differs from using one for training. The use case is inverted: in training, the monitor marks a target to reach; in pacing, it marks a ceiling you are trying not to break. The device functions as a fire alarm, not a coach.

For people with ME/CFS and PEM-prone Long COVID, the Workwell Foundation — a US research organisation that has done more than any other group to quantify the cardiopulmonary physiology of PEM through two-day CPET — has described a practical, individualised heart-rate ceiling for daily pacing: roughly your resting heart rate plus fifteen beats per minute [7]. For many patients with a resting heart rate in the sixties to low seventies, that places the ceiling in the low- to mid-eighties. If that sounds offensively low, it is. It is supposed to. The number marks a protective threshold to stay under during routine activities — a cap, not a target, and not a workout zone.

It is also an approximation. The gold standard for establishing a true anaerobic threshold is the two-day cardiopulmonary exercise test we covered in Chapter 1 — the protocol that exposes the day-two drop in workload at the anaerobic threshold that Snell, Stevens, and colleagues first documented in this population. A small number of specialist clinics offer it, most patients will not have access, and for those who do not, the resting-plus-fifteen heuristic is the working substitute. In severely affected patients the real anaerobic threshold sits even lower than the heuristic predicts, and in those cases the ceiling has to be revised downward based on symptom tracking rather than on a formula.

The goal is to stay below the anaerobic threshold, because crossing it — sustained, not momentary — is the event that shifts metabolism onto the inefficient pathway and triggers the cascade that becomes PEM forty-eight hours later. In a healthy person that threshold sits high enough that ordinary living does not approach it. In Long COVID it can sit very low: making dinner standing up, or a stressful five-minute phone call, can cross it.

If your heart rate jumps to 100 or 105 beats per minute simply from standing, a heart-rate ceiling set at 85 does not give you usable information in the way it does for someone whose standing rate is 75. In that case the tool shifts from *do not cross this line* to *break everything into pieces short enough that the line isn't relevant*. Workwell's own protocol for severe PEM uses activity blocks as short as two minutes, because the cardiovascular system destabilises on that timescale [7]. A person with that physiology is pacing in seconds, not in hours, and the chapter ahead on micropacing is where this picks up.

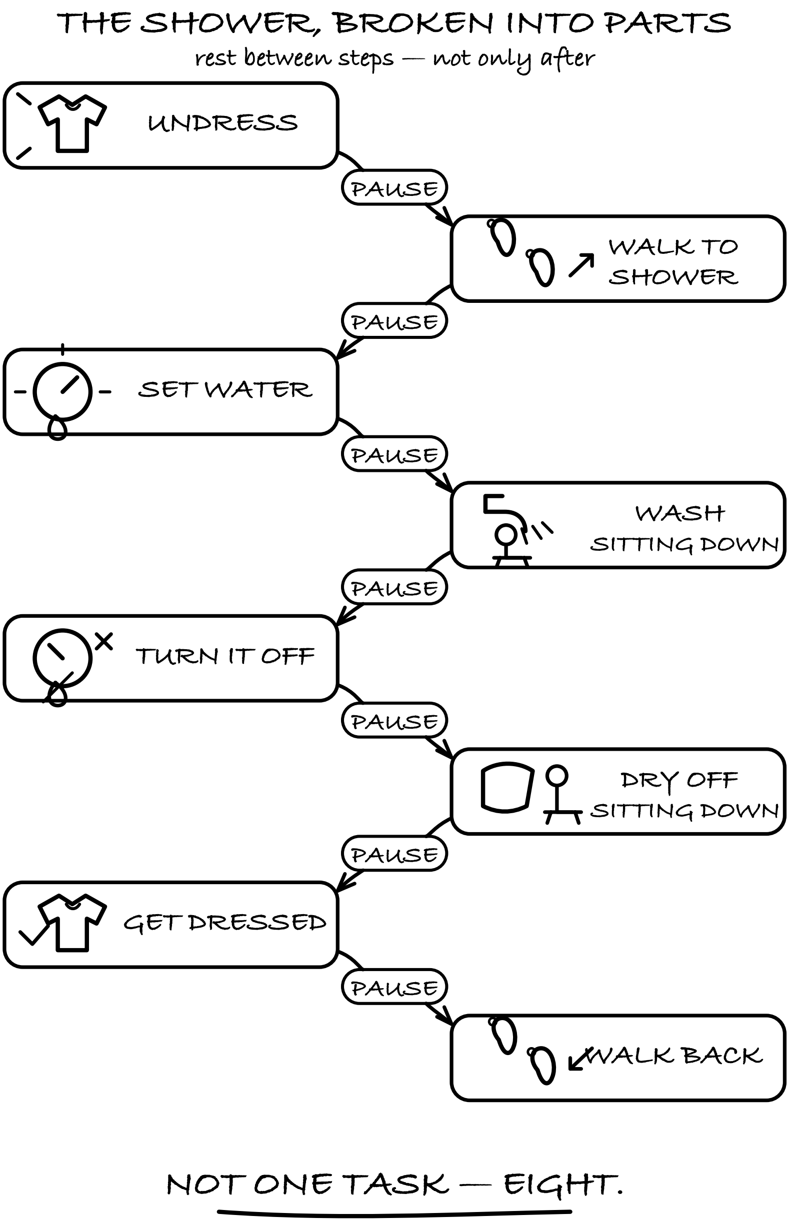
A final note on the monitor. Do not glance at it only when you are feeling it. The utility is that it catches the rise you cannot yet feel. Set an audible alarm on your wearable at the ceiling — most of the consumer devices support this — and when it beeps, you sit down, whether or not you currently have the sensation of needing to. The sensation follows the number by minutes, and by the time it arrives, the spending has already happened.

Micropacing: when even a little is a lot

For the most severely affected patients, the question is not *how do I add a walk* and not even *how do I sit at a desk for twenty minutes*. The question is *how do I get through my morning wash without the crash that finishes the afternoon*.

The answer is to break every task — including tasks a well person would not call a task — into segments of ten to thirty seconds of effort with one to several minutes of full recumbent rest between them. This is micropacing, and it is the pacing literature's most counterintuitive recommendation, because it asks you to rest before you are tired.

A worked example. The shower, which we discussed in Chapter 1 as the classic autonomic trap, is a common micropacing candidate for severely affected patients. One workable choreography: sit on a shower stool, wash your hair with your eyes closed and the water lukewarm; turn the water off; lie down on a towel placed on the bathroom floor for two minutes; get back up, wash your body; turn the water off; sit until the vertical rise in heart rate resolves; towel yourself dry sitting down; leave the bathroom in a robe; and finish dressing lying down on the bed. An ordinary daily task for a well person has become an eight-segment protocol. For a patient at a low functional baseline, the eight-segment shower is the version that does not cost the afternoon; any simpler version does.



Not one task. Eight, with somewhere to stop between them.

The same logic works for email (read one message, close your eyes for two minutes, draft the reply, close the laptop), cooking (slice vegetables seated, lie down, start the pan, lie down, add ingredients, lie down), and the slow work of correspondence (one exchange per session with a real rest in between). None of this is a permanent prescription. It is the version of a day that does not trigger PEM, which is what allows the underlying trajectory to move at all.

The load-bearing distinction is when the rest happens. Resting after you feel overdone falls outside what micropacing does; the mechanism works because the rest arrives before the spike, on a schedule, while the current segment still feels fine. By the time the body signals *now*, the spending has happened and the forty-eight-hour countdown has started. The practical habit is a timer — physical, on a phone, on a watch — that interrupts you at the chosen interval whether or not you feel ready to stop.

Pacing is not a surrender of the recovery project

Most patients' difficulty with pacing lives on the cultural side of the practice, not the mechanical side.

We live inside a set of assumptions about what a productive day looks like, and those assumptions were written for well bodies. A ranking of daily achievement that places *did the laundry, worked, cooked, exercised* above *ate twice, washed sitting, did not crash* is a ranking that no ELC patient's body can meet in the phase they are in, and measuring oneself against it produces a predictable emotional outcome on top of the physical one.

I want to be careful here. Pacing is not an instruction to do nothing; its goal is to stabilise the system at a level it can sustain, so that slow improvements — the kind that are measurable in months rather than weeks — can actually accumulate. A repeatedly crashing system cannot repair. A system kept below its threshold has the conditions for repair, and in a patient who has had no better day in a year, the first sign of recovery is often the simple fact that the crashes stop. Stabilisation of that floor counts as real clinical progress, even when nothing on the calendar has visibly improved.

A meta-analysis of fourteen activity-pacing studies in chronic fatigue syndrome reported a moderate improvement in fatigue against no treatment or usual care — a Hedges' g of −0.52, with a confidence interval running from −0.73 to −0.32 [8]. A separate scoping review of the same literature is more equivocal: of the studies it gathered, eleven reported benefit, four no effect, and two a worse outcome than the control condition, and its authors conclude the evidence base is still too thin to set practice from [8]. In clinical research language, that is a useful effect with an excellent safety profile. NICE's 2021 guideline on ME/CFS (NG206) names pacing as the only first-line activity-management recommendation for the condition, and its formulation is worth citing exactly: pacing is recommended *because it does not carry the risk profile of graded exposure approaches in this population*, and other activity-management protocols carry evidence of harm [9].

Medinger's long-term patient survey published through the LC Handbook found the same pattern from the other direction. Among patients who reported significant improvement or recovery, the self-identified contributors were, in order: pacing (58 percent), time (53 percent), rest (47 percent), and dietary changes (32 percent) [10]. Pacing leads every list like this in every source I know. Across the self-management literature for this family of conditions, it is the single most replicated finding we have.

Gez Medinger's clinical description of what pacing actually is stays the best short version I have seen:

"Pacing is a strategy where you stay within your 'energy envelope' — the amount of activity you can do without triggering post-exertional malaise (PEM). This means learning to recognise your limits and stopping activity before you crash, even if you feel you could do more."

— Gez Medinger & Danny Altmann, *The Long COVID Handbook*

The phrase *even if you feel you could do more* is the whole technical content of the practice. The in-the-moment sensation is not a trustworthy reading, for the reasons described in the mechanism section above. Pacing asks you to stop while the internal voice is still asking for the extra five minutes. Patients who report having cracked it describe this step — learning to stop early, using a timer rather than a feeling — as the hardest behavioural change of the illness.

Why "just exercise more" is a dangerous sentence

I am going to write this section carefully, because I want to be accurate about two things at once.

Exercise helps many people, including many people with many chronic illnesses. A useful pacing chapter cannot end with the impression that movement is the enemy. For patients without PEM — including many with pure POTS who went through the Levine protocol with benefit — graded movement is part of what gets them back. We discussed that in Chapter 4, and it remains true.

What is not true is the extension of that rule to a population whose illness is defined by a delayed crash. And that extension has a documented history, which is worth telling once, because it still shapes the advice ELC patients receive.

In 2011 the PACE trial — at the time the largest randomised trial of treatments for ME/CFS — reported that graded exercise therapy and cognitive behavioural therapy produced modest benefits over adaptive pacing and standard medical care [11]. The headline circulated widely. The trial's methodology carried problems that only emerged through years of patient-led scrutiny and independent reanalysis: outcome definitions were revised after enrollment in a way that made recovery easier to achieve on paper, the primary outcomes depended on unblinded subjective questionnaires, and the trial's own objective measure — the six-minute walk distance — ended at an average of 379 metres in the graded-exercise arm, a distance characteristic of a person who is still functionally impaired [12].

It is important to be precise about the critique. PACE was a large, conscientiously run clinical trial by the standards of its era. The problems were methodological — the post-hoc adjustment to outcome thresholds, the reliance on subjective measures under unblinded conditions — rather than motivational. Methodological errors in medicine produce the same clinical consequences as malicious ones, though: programmes of graded exertion were prescribed to a patient population whose central diagnostic feature is a delayed crash from exertion, and the harm is documented in the patient literature.

Patients did the work of correcting this. They filed freedom-of-information requests, surfaced the raw data, funded independent reanalyses, and built a body of evidence that made the original conclusions unsustainable. In 2021 NICE withdrew its recommendation for graded exercise therapy in ME/CFS and issued NG206 in its current form, which centres pacing and explicitly cautions against fixed-schedule exertion protocols for patients with PEM [9]. That guideline change did not happen because the medical system independently reached the correct conclusion. It happened because patients kept their own records and held them up to the light.

This is why the question to bring to any rehabilitation offer that arrives with a graph of planned weekly progress is: *is this plan fixed, or does it adjust to delayed worsening in my symptoms*. If the answer is that the plan advances on a calendar, regardless of a symptom response that arrives two days later, the plan is not designed for a PEM-prone body. That one sentence, asked early, has saved patients from months of regression. If the answer is that the plan is symptom-responsive — that an advance is conditional on a stable forty-eight-to-seventy-two-hour window after the current level — you are being offered something that fits the physiology. The distinction is worth knowing before the first session.

A line of research by Welltory researchers has examined exactly this axis in an observational, self-reported user cohort (not a clinical trial): which people's wearable data show a delayed heart-rate-variability drop in the two to three days after activity, which do not, and whether the observed patterns could inform hypotheses for individualized pacing rather than for an age-based formula. This describes exploratory research; it is not a feature of the current Welltory app and has not been reviewed or authorized by the FDA for any diagnostic or monitoring purpose. The Welltory app does not diagnose, monitor, or treat any disease. This reference describes a scientific investigation, not available product functionality. That work is one reason this book exists in the form it does.

The repeatability test

Most days of pacing do not require you to remember any of the above. The test that replaces the calculus is a single sentence.

An activity fits inside your budget if you can do it today and also feel roughly the same tomorrow and the day after. If you went for the walk and today was fine but yesterday's version of tomorrow turned out bad, the walk was over budget. If you went for the walk and the following seventy-two hours looked the same as they did without the walk, the walk was inside budget.

The test asks whether you can repeat the activity without a delayed consequence, not merely whether you can survive it once. Repeatability is the operational definition of *within the envelope*. When you are trying to expand what you can do — and you should be trying, when the body will carry it — the expansion is measured in additional minutes, not additional tens of minutes, and every advance is held for two clean weeks before the next advance. A five-percent weekly increase, sustained without a crash, is faster over six months than a thirty-percent weekly increase that ends in a regression every third week.

What to do over the next two weeks

For the coming fortnight, keep three pieces of data and nothing else.

Every morning, before you stand, record your heart rate on the wearable you already use, and rate how you feel on a one-to-ten scale. Do not compare to anyone else's scale; compare to your own entry yesterday. Through the day, write down blocks of activity, each with a position of the body — specifying *walked fifteen minutes on the flat* rather than *walked*, *twenty minutes seated at laptop* rather than *worked*, *forty minutes on the phone, seated, difficult conversation* rather than *phoned Dan's sister*. At twenty-four and forty-eight hours after each day's activity, note the lagged effect: worse, same, better; any fog, heaviness, unrefreshing sleep, new symptoms.

At the end of the two weeks, circle the days that did not produce a delayed worsening. That list is your current real baseline. Take roughly half of what the highest of those days looked like, and call it your starting point — not your target. It will feel preposterously small on paper. That is the point. A starting point below the ceiling is a starting point you can hold for long enough to build from.

One more habit worth installing early: before you raise the ceiling, hold the current level for at least two weeks of clean days. Readers of this book with a decade of ME/CFS experience will tell you that two weeks is the short version of the rule, and that four is safer for meaningful step-changes. The formula I use in clinical conversations is *wait until the* *stability feels almost boring*. If the last two weeks have felt boring at the current level, you are ready to add a minute or a block. If any of the last two weeks has carried a lagged symptom day, you are not.

Catherine did not build a perfect pacing system the Saturday she stayed in bed. She did a smaller thing. On Monday morning, still recovering, she added two columns to the spreadsheet on the kitchen table — one for total minutes upright, one for how she felt two days later on a one-to-ten scale — and set an alarm on her watch for a heart rate of eighty-seven. Her resting rate was seventy-two. Seventy-two plus fifteen gave her a ceiling she could see on a screen.

The first week of that ceiling, on paper, was embarrassing. She paused the laundry twice. In the middle of cooking dinner, she sat down on a stool. The emails she had planned to clear that evening stayed unanswered. By her old measurements of productivity, the week looked poor. She also reached Saturday at the same baseline she had woken up on that Monday, which had not happened in four months.

The second data point was in the spreadsheet. Her first data point had been a heart rate at minute five, after electrolytes. This one was a week without a crash. Two rows in a kitchen-table spreadsheet, fifteen months into an illness that had no clean name when it started, were becoming the file she was going to bring to the conversation with Dan when she was ready to have it.

Optional self-observation prompt

Spend tomorrow as your two-week diary's first entry, as honestly as you can write it.

Before you get out of bed, write a resting heart rate and a one-to-ten state. Through the day, log blocks of activity with position of the body.

At bedtime, write the number for the day as a whole and the forty-eight-hour check on the day before yesterday. Do not change any behaviour yet. This is a measurement week, not an intervention week.

If you already know you have PEM, set a heart-rate alarm on your wearable at your resting rate plus fifteen beats per minute before you start the day. Sit down when it beeps. You can disagree with the number later, after the data comes in; disagreement based on two weeks of your own recorded body is a different argument from disagreement based on how the morning feels.

One thing not to do. Do not start a movement programme, do not add or subtract a medication, and do not change any of the Chapter 5 tools you have been using. This chapter is about reading the system you already have. The adjustments come next, and they are easier when the picture they are adjusting is on paper.

Sources

1. Medinger G, Altmann D, 2022. *The Long COVID Handbook*. London: Hutchinson Heinemann. Patient-reported recovery factor data identifying pacing as the most-cited self-management contributor across a large patient sample.
2. National Institute for Health and Care Excellence, 2021. *Myalgic encephalomyelitis (or encephalopathy)/chronic fatigue syndrome: diagnosis and management* (NG206). Post-exertional malaise as a core diagnostic feature; characteristic delayed onset, typically between 12 and 72 hours after exertion (the often-cited 12–48 hour window is the most frequent but not the only pattern), duration from days to weeks. https://www.nice.org.uk/guidance/ng206
3. Fluge, Ø., Mella, O., Bruland, O., et al., 2016. *Metabolic profiling indicates impaired pyruvate dehydrogenase function in myalgic en-*

cephalopathy/chronic fatigue syndrome. JCI Insight, 1(21): e89376. DOI: 10.1172/jci.insight.89376 Downregulated PDH activity and relative shift toward anaerobic metabolism, with reduced ATP yield per glucose.

4. Mahoney, D.E., et al., 2018. *Understanding D-Ribose and Mitochondrial Function.* Advances in Bioscience and Clinical Medicine, 6(1): 1–5; Teitelbaum, J.E., et al., 2006. *The use of D-ribose in chronic fatigue syndrome and fibromyalgia: a pilot study.* Journal of Alternative and Complementary Medicine, 12(9): 857–862. DOI: 10.7575/aiac.abcmed.v.6n.1p.1 Limited pentose-phosphate-pathway flux creating a bottleneck in purine recycling and ATP replenishment after exertion.
5. van Campen, C.L.M.C., Verheugt, F.W.A., Rowe, P.C., Visser, F.C., 2020. *Cerebral blood flow is reduced in ME/CFS during head-up tilt testing even in the absence of hypotension or tachycardia.* Clinical Neurophysiology Practice, 5: 50–58. DOI: 10.1016/j.cnp.2020.01.003 Extracranial Doppler quantification of 26% cerebral blood-flow reduction in ME/CFS on standing against ∼7% in healthy controls.
6. Seeley M-C, O'Brien H, Wilson G, Coat C, Smith T, Hickson K, Casse R, Page AJ, Gallagher C, Lau DH. 2025. *Novel brain SPECT imaging unravels abnormal cerebral perfusion in patients with postural orthostatic tachycardia syndrome and cognitive dysfunction.* Scientific Reports, 15(1):3487. DOI: 10.1038/s41598-025-87748-4 Fifty-six participants; abnormal cerebral blood flow in 61%, present even while supine, with the lowest mean z-scores in lateral prefrontal and sensorimotor cortices.
7. Stevens, S., Snell, C., et al. Workwell Foundation clinical materials on cardiopulmonary exercise testing and pacing in ME/CFS. Practical pacing heart-rate ceiling described as resting heart rate

plus approximately 15 beats per minute for most deconditioned PEM-prone patients; severely affected patients may require a lower ceiling and activity blocks as short as ~2 minutes. https://workwellfoundation.org

8. Casson S, Jones MD, Cassar J, Kwai N, Lloyd AR, Barry BK, Sandler CX. 2023. *The effectiveness of activity pacing interventions for people with chronic fatigue syndrome: a systematic review and meta-analysis.* Disability and Rehabilitation, 45(23): 3788–3802. DOI: 10.1080/09638288.2022.2135776 Fourteen studies; fatigue Hedges' g −0.52 (95% CI −0.73 to −0.32) against no treatment or usual care. **And:** Sanal-Hayes NEM, Mclaughlin M, Hayes LD, et al. 2023. *A scoping review of 'Pacing' for management of Myalgic Encephalomyelitis/Chronic Fatigue Syndrome (ME/CFS): lessons learned for the long COVID pandemic.* Journal of Translational Medicine, 21(1): 720. DOI: 10.1186/s12967-023-04587-5 Eleven studies reported benefit, four no effect, two a detrimental effect; the authors judge the literature insufficient to inform treatment practice.
9. National Institute for Health and Care Excellence, 2021. *NG206*, as above. Pacing recommended as first-line activity-management for ME/CFS; graded exercise therapy de-recommended for the condition. https://www.nice.org.uk/guidance/ng206
10. Medinger G, Altmann D, 2022. *The Long COVID Handbook*, Chapter 11 patient recovery survey. Self-identified contributors among substantially improved or recovered patients: pacing (58%), time (53%), rest (47%), dietary changes (32%).
11. White, P.D., et al., 2011. *Comparison of adaptive pacing therapy, cognitive behaviour therapy, graded exercise therapy, and specialist medical care for chronic fatigue syndrome (PACE): a ran-*

domised trial. The Lancet, 377(9768): 823–836. DOI: 10.1016/S0140-6736(11)60096-2

12. Wilshire, C.E., et al., 2018. *Rethinking the treatment of chronic fatigue syndrome — a reanalysis and evaluation of findings from a recent major trial of graded exercise and CBT.* BMC Psychology, 6: 6. DOI: 10.1186/s40359-018-0218-3 Independent reanalysis of PACE data showing that original recovery and improvement claims were not supported under the trial's pre-specified outcome definitions.

Chapter 7. Food as a tool, not a diet

There is no "Long COVID diet." If one existed, the patient internet would have found it by now, fought over it in three languages, and built a spreadsheet.

What exists instead is a set of levers, each matched to a specific job. Salt and water, which we covered in Chapters 4 and 5, is a blood-volume intervention for orthostatic intolerance, not a diet. A low-histamine elimination is a short diagnostic probe for mast-cell involvement, not a wellness trend. Meal structure is autonomic damage control, not clean eating. And most supplements, at the dose ranges sold to people with chronic illness, fall somewhere between targeted repair and expensive urine.

The wrong question here is "what should I eat?" — as if one perfect answer is waiting to be found. The better question is: *what job am I asking this food to do?*

If standing is the main problem, the job is blood volume, and the levers are in Chapters 4 and 5. If meals trigger flushing, itching, headaches, gut chaos, or the classic "why did yesterday's chicken wreck me?", the job is histamine tolerance. If you get shaky, sweaty, wired, sleepy, or tachycardic after carb-heavy meals, the job is glucose stability. If none of those is loudly your picture, the job is having a sane default that doesn't pick fights with the rest of your physiology — and that's what the Mediterranean pattern is for.

Food is chemistry that runs on a schedule. The moral framing — clean eating, pure eating, sinful eating — is the wrong frame for a body with dysautonomia. Think tool, not virtue.

Histamine: when yesterday's leftovers hit harder than the workout

Let's start with the most practical experiment in this chapter.

We touched this paper briefly in Chapter 3, where it sat in the mast-cell layer of the five-diagnoses frame; in this chapter the same symptom-count arithmetic does a different job — it is the reason to consider a short histamine probe at all.

In 2021, a team led by Leonard Weinstock published a study in the *International Journal of Infectious Diseases* comparing 136 Long COVID patients, 136 controls, and 80 patients with diagnosed mast cell activation syndrome (MCAS) [1]. They used a standard symptom instrument — the Mast Cell Mediator Release Syndrome score — to ask whether the pattern of complaints in Long COVID looked like the pattern in MCAS. What they found was that the Long COVID group's mean symptom count had risen from 11.6 before their infection to 22.0 after, almost exactly matching the MCAS group's 20.4. Severity scores told the same story: 21.1 in Long COVID versus 20.1 in untreated MCAS, statistically indistinguishable. The authors were careful not to claim that Long COVID patients *have* MCAS — formal MCAS diagnosis requires mediator-level laboratory testing that wasn't done here. What they were willing to say was that mast cell activation symptoms were increased in Long COVID, and they mimicked the severity and symptom profile reported by people who have been formally diagnosed with the condition.

What the paper supports, in plain terms, is that mast-cell-type complaints show up often enough in the Long COVID population that dismissing them has a real cost — which is a narrower and more defensible claim than "two-thirds of Long COVID is MCAS."

Histamine is not just the thing that makes you sneeze in pollen season. Mast cells — immune cells that participate in allergic and inflammatory responses — store histamine and release it, along with a range of other mediators, when they're activated. Some foods contain histamine directly, at concentrations that depend heavily on how long the food has been sitting. Others trigger mast cells to release their own. When the rate of histamine production exceeds the rate at which your body can break it down — primarily through the enzyme diamine oxidase, or DAO, in the gut lining — the buildup can feel like a recognizable mess. Rapid heart rate. Flushing. GI upset. Headaches. Poor sleep. A vague internal sense that Tuesday is allergic to you.

The detail many people miss is that freshness matters chemically, not just aesthetically. Histamine levels rise as protein-containing food sits — in the fridge, in the pan, on the counter. That's why someone can say, with complete sincerity, "chicken is fine the day I cook it, but day-two chicken gives me a migraine." That sounds like magical thinking until you read the chemistry.

Wine is a four-way attack in a single glass: the drink itself contains histamine, it activates mast cells directly, it dilates peripheral vessels, and it fragments sleep. Fermented foods sit in a strange place in the literature — a microbiome booster in one kind of article, a histamine disaster in another. For someone with mast-cell sensitivity, both can be true at once: fermentation may genuinely help the microbiome and still wreck a histamine-sensitive person's evening.

A low-histamine trial is a short-term diagnostic probe, not a way of life. Its purpose is to test whether histamine is part of your picture, quickly, and then get you back to a broader diet.

Here's how to run it cleanly over ten to fourteen days. Keep everything else stable: your caffeine, your antihistamines if you're on them, your hydration and sodium routine, your supplements. Change only the food. Remove the obvious high-histamine group — aged cheeses, fermented foods, alcohol, processed and cured meats, smoked or canned fish, vinegar-heavy dishes, and leftovers that sat in the fridge overnight. Cook fresh. If you batch-cook, portion and freeze the extras immediately so they never get a chance to mature in the fridge. Track the same few symptoms every day: post-meal tachycardia, flushing or itching, gut chaos, sleep quality, brain fog.

Then, at the end of the window, reintroduce one category at a time — say, aged cheese on day one, a glass of wine on day three, fermented vegetables on day five. Watch the same tracked symptoms for the same flares.

If symptoms settle clearly on the clean protocol and come back with specific reintroductions, you've found a real piece of your pattern — *histamine is part of my picture,* in the modest sense, rather than *I have solved Long COVID.* That is enough to be useful.

And it may only be a piece. If your flushing and GI chaos improve but standing is still hard, the experiment hasn't failed — it's told you that mast-cell mediators are one mechanism among several, and that your orthostatic intolerance (Chapter 4) has a different engine than your post-meal itch.

One last warning. Do not stack restriction on restriction until your diet is five bland foods and a mountain of fear. Low-histamine is a short diagnostic test, never a lifelong identity.

The 2024 open-label signal for an H1 + H2 antihistamine combination (fexofenadine and famotidine at standard doses), and the STIMULATE-ICP readout that frames it, are covered in Chapter 3 (see Ch03 ref 10); the short version is that the mechanism is plausible, the observational data is hopeful, and a placebo-controlled trial is imminent. This one is clinician-prescribed territory, not a self-directed experiment like the elimination diet.

Glucose: when the "panic attack" starts at breakfast

One of the more confusing mismatches in Long COVID and POTS is that post-meal hypoglycemia can feel exactly like a panic attack.

Shaky hands. Sweating. A heart that starts thudding without warning. A sudden, unprovoked sense that something is about to go wrong. If this happens reliably after a carb-heavy meal, after a missed meal, or in the middle of the night, the first suspect is not your psyche. The first suspect is your counter-regulatory hormonal response to a falling glucose level.

Here's the mechanism. A meal with fast carbohydrates pushes blood sugar up quickly, which provokes a brisk insulin release, which pulls the glucose back down — sometimes below where it started. In a healthy, untaxed body, that dip is buffered smoothly. In a dysautonomic body, the dip triggers a counter-regulatory surge: glucagon, cortisol, and epinephrine from the adrenal medulla, which is the one place in this book where *adrenaline* is the right word. The epinephrine tells the liver to dump stored glucose back into circulation. It also produces the tremor, the sweating, the racing heart, the internal buzzing, and the sense of dread.

The body doesn't send a polite notification that its glucose is low. It sends its emergency hormone. And that hormone feels the same whether the trigger is a low blood sugar, a threat in the environment, or a random misfire — which is why someone can leave an afternoon "panic attack" convinced that their anxiety is getting worse, when the real event was a glucose crash after a breakfast of coffee and toast.

POTS makes this worse in a specific way. After any meal, blood preferentially shifts to the gut to handle digestion; this is normal splanchnic physiology, and in a dysautonomic body it stacks directly on top of the orthostatic shortage we spent Chapter 4 describing. Post-meal tachycardia, post-meal dizziness, and the classic "I need to lie down after lunch" feeling are hemodynamic before they are psychological. Combine that with a glucose spike-and-drop and you get what one patient I know calls the after-lunch existential crisis, which is funnier than it deserves to be.

A two-week experiment often resolves the question. The rule is simple: never eat naked carbohydrates. Pair any starch or fruit with protein, fat, or fiber. Change the order of eating — protein and vegetables first, then the carbohydrate part — which in controlled studies visibly blunts the post-meal glucose excursion [2]. Retire the coffee-and-toast breakfast in favor of something with an anchor: eggs, Greek yogurt, cottage cheese, nuts, smoked fish. Make lunch smaller, especially if your afternoons tend to fall off a cliff.

Track what happens thirty minutes, ninety minutes, and three hours after meals. Note your pulse, whether you get shaky, whether you get drowsy, and whether standing becomes harder. Three hours is the important data point — that's where the late dip tends to land, and it's the interval that gets missed by people who only check how they feel immediately after eating.

If you can borrow a basic glucometer for a week, use it in the exact moments when "panic" hits. Some patients discover that their anxiety has a startlingly specific trigger, and the trigger is a 70 mg/dL reading forty-five minutes after a croissant.

Catherine, who by this point had her HR ceiling and her pacing diary from Chapter 6, had her own version of this. On a Wednesday afternoon in early October, after a lunch of toast, an apple, and a second coffee, she felt the specific kind of awful she had started calling the "2 PM cliff" — shaky, heart pounding, a rush of dread that felt like panic. Her wearable record showed a heart rate spike that didn't match her pacing-limit math; she had been sitting at her kitchen table, not exerting. The record could show the spike but not its cause. The shape of the event didn't fit anxiety and didn't fit exertion, and the timing — about an hour after a lunch of fast carbohydrate — pointed at a post-meal glucose swing as the thing worth testing. She spent the next four mornings pairing her breakfast carbs with two eggs and a handful of walnuts, and the 2 PM cliff flattened. She didn't start a formal two-week trial; she filed the protein-first rule on the same shelf as her hydration protocol and her HR ceiling, as one more tool she knew how to reach for. That is the right pace for adding levers.

Gut and serotonin: what we know and what we don't know yet

This is the mechanism-heavy section of the chapter, and I'm going to be careful with the language, because the gut-serotonin-brain story has been oversold in popular writing in ways that make the actual science harder to talk about.

The well-established part first. Roughly ninety to ninety-five percent of the serotonin in your body is produced in the gut, specifically in the enterochromaffin cells of the intestinal lining [3]. Most of its job there is local. It regulates peristalsis — the rhythmic squeeze that moves food through the GI tract. It modulates intestinal secretion. It tunes visceral sensitivity, which is why serotonin-pathway drugs are sometimes effective in irritable bowel syndrome.

Here is where popular writing gets sloppy. *Gut serotonin* and *brain serotonin* are two functionally separate systems. Serotonin itself cannot cross the blood-brain barrier, so the serotonin your gut produces does not travel to your brain and affect your mood directly. The brain has its own serotonin, synthesized locally by neurons in the raphe nuclei, operating on its own schedule. Writing that treats "low gut serotonin" and "depression" as the same phenomenon is conflating two separate organs.

What does exist is indirect signaling. Gut serotonin can act on 5-HT3 receptors located on the endings of afferent fibers of the vagus nerve [4]. That is one mechanism through which the gut passes information upward — not by shipping serotonin molecules to the brain, but by firing vagal sensory fibers that carry the signal. It's worth being careful even here: the vagus nerve's own primary neurotransmitter is acetylcholine, not serotonin. The picture is "serotonin as one of several signals the gut uses to talk to the vagus," not "serotonin is the language of the gut-brain axis."

In 2023, a team at the University of Pennsylvania published a paper in *Cell* proposing a specific mechanism by which this matters in Long COVID [5]. The model went like this: viral fragments persist in the gut → the immune system sustains an interferon response → interferons interfere with the absorption of tryptophan, the amino acid precursor for serotonin → peripheral serotonin falls → vagal signaling weakens → hippocampal function, and memory along with it, is impaired. The paper was widely reported as an explanation for brain fog.

A few things are worth knowing about this. Brain serotonin levels in the study were unchanged — the authors themselves note that the mechanism is peripheral. The animal-model evidence was more convincing than the human data; in humans, the finding was strong in one of the study's cohorts (in Ireland), absent in a second cohort, and weak in a third. And in 2024, a separate group published a study in *Infection* reporting that they could not find reduced serum serotonin in their own Long COVID cohort, and that serotonin levels did not correlate with the severity of fatigue or depressive symptoms in the patients they examined [6]. A methodological commentary published the same year raised a specific concern about the original work: the centrifugation protocol used to prepare the plasma samples produces plasma that is almost entirely depleted of platelets, and platelets are the body's main reservoir of serotonin [7]. The concern is that the reduction observed may reflect the sample preparation rather than the disease.

Where that leaves us: the peripheral-serotonin hypothesis is a serious idea with open methodological questions and partial replication. It is worth knowing about. It is not yet a settled mechanism, and any supplement or dietary intervention pitched on the premise that it *is* settled is running ahead of the evidence.

A parallel piece of biochemistry is worth holding next to this. Gut serotonin and brain serotonin are synthesized in separate places — the gut in its enterochromaffin cells, the brain in its raphe neurons — but they draw on the same dietary tryptophan pool, so a body that has been routing a lot of tryptophan into gut synthesis has less available to the brain's own machinery. That is part of the logic behind why some clinicians suggest an SSRI can be useful in post-COVID cognitive symptoms: not because the gut makes the brain's serotonin, but because raising the efficiency of central serotonin signaling is one of the few pharmacological levers that does not require solving the input-supply problem first. Tryptophan itself is the raw material for serotonin, but it is also the raw material for a completely different metabolic route called the kynurenine pathway. It is tempting to describe this as "normally the body builds serotonin; during inflammation it diverts tryptophan to kynurenine instead." That framing is misleading. Under perfectly healthy baseline conditions, most tryptophan already flows down the kynurenine pathway — the body's need for serotonin, at the molecular scale, is relatively small, and a serotonin molecule is a signaling mediator, not a bulk-consumed nutrient. What inflammation does is shift the balance slightly further toward kynurenine. Some of the kynurenine pathway's downstream metabolites, notably quinolinic acid, have been associated with fatigue and cognitive symptoms in animal models. The shift is the story here, not the pathway itself.

One more piece of the puzzle, with a twist. *Faecalibacterium prausnitzii* is a gut bacterium whose main claim to notice is that it produces butyrate, a short-chain fatty acid that feeds the cells of the intestinal lining and has local anti-inflammatory effects — locally, on the gut wall, not as a systemic immune-calming signal. In 2022, Yeoh and colleagues published a prospective study of 106 patients in *Gut* showing that reduced levels of *F. prausnitzii* and *Bifidobacterium pseudocatenulatum* at acute-phase had the strongest inverse correlation with persistent symptoms at six months — the lower these organisms were at the start, the more likely a person was still symptomatic half a year later [8]. The twist is that *F. prausnitzii* is an extreme anaerobe, exquisitely sensitive to oxygen and particularly vulnerable to antibiotics and heavy non-steroidal anti-inflammatory use. Both of those were given widely in the acute phase of COVID, often empirically, often without a confirmed bacterial infection to justify them. A meaningful fraction of the gut-microbial disruption attributed to Long COVID may in fact be an artifact of the acute-phase treatment rather than a direct consequence of the virus itself.

While we're here, a related terminology note. The phrase "leaky gut" is doing more work in the popular press than the anatomy supports. There is no hole. What the research describes is increased intestinal permeability — a change in the way the tight junctions between gut epithelial cells regulate what passes through, including the translocation of lipopolysaccharide (LPS), a molecular fragment of the outer membrane of certain bacteria, into the circulation. LPS then engages the innate immune system and contributes to low-grade chronic inflammation. This is a real phenomenon. "Leaky gut" is a useful shorthand only if you remember it refers to altered junctional permeability, not to anything perforated.

The practical part of all this is short. Make sure your diet has adequate tryptophan — it shows up in turkey, fish, eggs, dairy, and nuts. Serotonin is not built out of nothing. If your diet has been hollowed out by a stacked elimination protocol, a body already working on reduced reserves will not have the raw material to build what it needs.

Mediterranean pattern: the sane default

If you don't have a loud histamine pattern, you don't have obvious glucose swings, and you simply need a default — a way of eating that won't pick fights with your cardiovascular system or your gut — the Mediterranean pattern earns its reputation.

The point is not that olive oil is magical or tomatoes are a religion; the pattern earns its keep by being boringly stable — minimally processed foods, plenty of plants, ample fiber, unprocessed fats, moderate animal protein, and a shortage of the glucose bombs that are the first thing to cause trouble. In the general adult population, this pattern reduces cardiovascular disease; the PREDIMED trial, published in its revised form in the *New England Journal of Medicine* in 2018, remains the largest randomized evidence on that point [9]. That is not a Long COVID claim; it is a cardiovascular and inflammatory-systems claim. For a body already struggling with both, "doesn't make either worse" is a useful property.

The pattern has to be adapted to the specific body eating it. When histamine is a problem, a Mediterranean plate looks like freshly cooked fish or chicken with olive oil, potatoes, oats, and fresh vegetables — minus the aged cheeses, the wine, the vinegar, and the week-old leftovers. When irritable bowel syndrome is your loudest symptom, a giant bowl of raw vegetables may be the worst possible lunch, regardless of how saintly it looks on social media. For people whose long-term post-COVID smell and taste changes have wrecked their appetite, the goal shifts — anti-inflammatory perfection matters less than simply getting enough calories and enough protein, so that the body has the raw material to do repair work at all. Undereating while chronically ill is one of the quieter ways a person can undermine their own recovery.

This is where diet culture becomes a real hazard, not a mild one. Low-histamine stacked on low-FODMAP stacked on gluten-free stacked on dairy-free stacked on intermittent fasting looks cautious on paper and acts, in a body already running on reduced energy-production capacity, as a path to caloric and micronutrient insufficiency that can produce a measurable deterioration. When every suspect food is eliminated simultaneously, you have lost the ability to learn anything about any of them, and you have likely lost a third of the calories and protein your recovery depends on.

A cultural note worth saying out loud: the old idea that you should fast when you're sick — lose your appetite, skip meals, let the body "focus on healing" — is not supported by current evidence. Sickness behavior, the cluster of low appetite, social withdrawal, and sleep that appears during acute infection, is best understood in evolutionary terms as a strategy that reduces the chance of infecting others (an immobile, withdrawn sick person spreads less disease than an active one), not a strategy that optimizes individual recovery. The immune response is metabolically expensive, protein-intensive, and benefits from nutrition rather than abstention. For a condition as long-lasting as Long COVID, the "fast and see" approach is a recipe for a slow hollowing-out of reserve.

A giant, dressing-free, salt-free salad can be one of the worst possible lunches for someone with POTS. No sodium to support plasma volume. Low caloric density in a body that tires quickly. A large fiber bolus that pulls blood to the gut for hours. It looks impeccable on paper and functions poorly in practice. Pattern matters more than purity here.

Supplements: targeted repair, not magic beans

Here is the myth to put down first. *Supplements will cure you.*

Real deficiencies matter. A low vitamin B12 level can produce what looks exactly like small-fiber neuropathy, brain fog, and a mild depression — correcting it is not placebo, it's repair, and the difference it makes can be dramatic. Iron deficiency can masquerade as chronic fatigue with a completely different treatment. Low vitamin D shows up on basic labs for a reason. A supplement that fixes a measured deficiency is a tool with a specific job.

Most supplements, bought by most people without a measured deficiency, do one of two things. They do nothing. Or they cause side effects. This is the part the marketing doesn't mention.

Before buying anything expensive, the single most useful investment is a basic lab panel. Complete blood count. Ferritin. Vitamin B12. Vitamin D. A basic metabolic panel. A blood test that tells you *not* to buy five new bottles is one of the better investments available in this space.

The short map below is a rough guide rather than a prescription — a summary of what the evidence says about each of the commonly discussed items in the Long COVID and ME/CFS conversation. Discuss any of these with a clinician who knows your full picture before starting, especially if you are on medications; interactions between supplements and prescription drugs are real, and some of them matter.

Coenzyme Q10 (CoQ10) supports mitochondrial electron transport. The evidence is mixed. Several small trials in ME/CFS have reported improvement in fatigue; one trial specifically in Long COVID found no clear signal. In patient-reported data, a combination of CoQ10 with alpha-lipoic acid showed resolution of fatigue in about half of Long COVID patients in an open-label study [10]. Two further placebo-controlled trials have since read out without a clear benefit signal; a larger trial looking at blood-vessel function is due to report in 2028, and the confirmatory picture stays thin until it does. Clinical benefit, where it appears, typically requires eight to twelve weeks of consistent dosing at two hundred to four hundred milligrams per day. Take it with a meal that contains fat — CoQ10 is fat-soluble, and absorption without fat is poor.

D-ribose is a sugar that participates in the synthesis of adenosine triphosphate (ATP) and the nucleotide pool the body uses to run its energy machinery. A small pilot study by Teitelbaum and colleagues in 2006 reported subjective improvement in ME/CFS and fibromyalgia at a dose of five grams three times daily. No large controlled trial has been conducted. One practical note: D-ribose can lower blood glucose, which is useful to know if your glucose is already unstable. Combining D-ribose with the previous section's glucose discussion is worth thinking through before starting.

Niacin (vitamin B3) theoretically raises cellular NAD+, the cofactor that mitochondrial energy production depends on. When NAD+ availability is low, cells shift to less efficient energy pathways — basic function continues, but load tolerance is reduced, and fatigue accumulates faster. The strongest signal for niacin in Long COVID comes from patient-led survey data, not randomized trials [11]. Reported doses range from fifty to five hundred milligrams daily. Start at the low end. Niacin produces a characteristic "flush" — skin warmth, redness, sometimes a burning sensation — that is harmless but unpleasant; the no-flush forms have been less studied in this context.

Magnesium glycinate is the most commonly recommended form for the "tired but wired" pattern and for muscle tension. Typical doses are two hundred to four hundred milligrams of elemental magnesium per day. No Long COVID-specific trials exist, but the safety profile is excellent and magnesium deficiency is common enough to make this a reasonable default. Read the label carefully — the weight listed is often the weight of the whole magnesium compound, which is not the same number as the elemental magnesium content.

Vitamin D3 is worth replacing if your level is low. Chronic illness plus reduced time outside predictably depletes it. A standard maintenance dose is one thousand to two thousand international units daily; a clinician can prescribe higher doses for a defined period if you're frankly deficient. Megadosing when your level is already normal does not produce an extra benefit and can cause toxicity.

Vitamin B12 matters when your level is low or borderline, particularly if you're seeing neuropathic symptoms or brain fog. The sublingual methylcobalamin form at one thousand micrograms per day absorbs better than tablets; for significant deficiency, a clinician may recommend injection.

Low-dose naltrexone (LDN) is one of the most actively discussed experimental treatments in Long COVID and ME/CFS. Naltrexone at standard doses (fifty milligrams and up) is used to treat opioid use disorder; at doses below five milligrams it appears to act through a different mechanism, modulating microglial activity in the central nervous system and dampening pro-inflammatory signaling, including 2025 work showing partial restoration of TRPM3 ion-channel function in natural killer cells from Long COVID patients [12]. As of May 2026, no Long COVID RCT has yet reported full results, but several are close, and 2024–2025 systematic reviews of the pre-post observational data continue to show a moderate fatigue benefit with roughly half of patients meeting responder criteria by twelve weeks [13]. A double-blind Phase II trial in British Columbia (n=160, titrated to 4.5 mg) finished participant activities in early 2026, with results expected later this year. The LIFT trial at Brigham and Women's Hospital is testing LDN alone and combined with pyridostigmine in ME/CFS with orthostatic intolerance. The NIH's pivotal pediatric LDN trial under RECOVER-TLC (1,300 participants, ages 6–25) was delayed; enrollment is now expected to begin in summer 2026. LDN is a prescription medication; it cannot be used alongside any opioid medication, and its most common side effects are insomnia and vivid or unsettling dreams — worth considering carefully if your sleep is already disturbed. Nausea and headache are also possible, especially in the first weeks. Most patients begin at a very low dose — 0.5 to 1 milligram — and titrate slowly.

Urolithin A is a natural metabolite produced in the gut from the polyphenols found in pomegranates, walnuts, and berries, in people whose gut microbiome includes the specific bacteria that perform the conversion. Only about thirty to forty percent of people have that capacity, which is why urolithin A has become available as a direct supplement. Its proposed mechanism is the activation of mitophagy — the cellular process of clearing damaged mitochondria. A randomized trial in middle-aged healthy adults showed improved muscle strength and mitochondrial efficiency [14]. No trial has been completed in Long COVID or ME/CFS. Extrapolating from healthy-adult results to a population with documented mitochondrial and immune changes is not justified at this point. If you want to spend money on this one, wait for the clinical data.

One category earns a specific heads-up on top of the general rule. Supplements sold against the post-COVID microclot hypothesis — nattokinase is the most common — have no human Long COVID trials behind them, and their fibrin-breaking action can interact meaningfully with blood-thinning medications. In-vitro signal in cell cultures is not the same as evidence in people, and community enthusiasm around this one is running ahead of the data.

A few practical rules for the whole category. Change one supplement at a time and wait two weeks before adding another, or you will learn nothing from the exercise. Keep a short log — what you started, when, at what dose, and what you noticed. Re-measure labs where it makes sense.

> **SAFETY —** A clinician who knows your medications is the right person to consult before adding any of these. Interactions and contraindications are real, and some are not obvious.

N-of-1: treat food like science

Your body is a noisy system. To hear what it is saying, you need a method.

The "N-of-1" experiment is an underused tool in this space. The name is biostatistical shorthand for a study with a single participant — you — who is both subject and investigator. The rules are simple and discipline matters more than ambition.

Pick one tool and one target symptom. Not "health" in general. Not "feeling better." Something you can measure, even crudely: your standing pulse on the first morning after waking, your afternoon brain fog at 2 PM, your post-meal heart rate, the hour at which you hit a wall. Measure a three-day baseline before you change anything. Run the intervention long enough for it to show a signal — sodium and fluid work in days, a glucose-pairing rule takes a week or two, a low-histamine probe needs ten to fourteen days, a CoQ10 trial takes two to three months. Keep everything else you do stable during the test window, including your caffeine, your sleep schedule, and any other supplements you're on. Predefine what success looks like before you start: my standing heart rate drops by ten beats per minute; my 2 PM crash gets visibly shorter; the post-meal fog goes from ninety minutes to thirty. You do not need an app for this. A phone note is enough. Write down what you ate, what you took, and what you noticed.

If you treat food changes as experiments rather than as identity commitments, you stop being hostage to the labels — "paleo person," "low-carb person," any of them. What you are, for the duration of the experiment, is a person running a specific test on a specific system, and the only question that matters is whether the numbers, and the way you feel, agree that it helped.

Food will not cure Long COVID. It can, at best, give you back standing time, a calmer morning, and a night of sleep that doesn't start at 3 AM. That is not nothing — that is three levers you didn't have yesterday. The frame here is debugging a system, with food as one of its inputs, rather than eating your way out of an illness.

The delay between a food and its effect is part of what makes this hard. A histamine reaction can land hours after the meal. A glucose crash can land ninety minutes out. An inflammatory reaction to a food you shouldn't be eating can take a day to surface. The pattern of the delay is what you're looking for, not an exact hour. Write down the timing of the meal, write down the timing of the symptom, and let the pattern accumulate over days. If the delay could be pinned to the hour reliably — as Chapter 9 says when this diary question returns — medicine would have solved this illness already. The delay is a central feature of the problem rather than a nuisance around the edges.

Optional self-observation prompt

Pick one of three experiments. Not all three. One.

Option one, the glucose test. For the next three days, never eat a carbohydrate without pairing it with protein, fat, or fiber. Change the order of your meal: protein and vegetables first, then the starch. Track your pulse and your energy ninety minutes after each meal. If the 2 PM cliff flattens, you have found a lever.

Option two, the histamine probe. Look at what you ate for dinner last night — were aged cheeses, cured meats, wine, fermented foods, or day-old leftovers on the plate? If yes, try one day without any of those. Compare how the next morning feels.

Option three, the lab order. Before you spend money on any supplement, get a basic panel — complete blood count, ferritin, vitamin B12, vitamin D, a basic metabolic panel. Bring the results to your clinician. A test that tells you what *not* to buy is cheaper than six months of bottles.

One experiment. One variable. Two weeks of observation. The frame here is running a study in which you are both subject and investigator — which is a different activity from the moral performance of eating healthy.

Sources

1. Weinstock LB, Brook JB, Walters AS, Goris A, Afrin LB, Molderings GJ. 2021. "Mast cell activation symptoms are prevalent in Long-COVID." *International Journal of Infectious Diseases* 112:217–226. DOI: 10.1016/j.ijid.2021.09.043 Mean mast cell symptom count rose from 11.6 pre-COVID to 22.0 post-COVID in 136 Long COVID patients, statistically indistinguishable from untreated MCAS patients (mean 20.4). Mean severity score 21.1 vs 20.1 in MCAS. Authors explicitly note the finding is not equivalent to a formal MCAS diagnosis, which requires mediator-level laboratory testing.
2. Shukla AP, Iliescu RG, Thomas CE, Aronne LJ. 2015. "Food order has a significant impact on postprandial glucose and insulin levels." *Diabetes Care* 38(7):e98–e99; and Zeevi D et al. 2015. "Personalized nutrition by prediction of glycemic responses." *Cell* 163(5):1079–1094. DOI: 10.2337/dc15-0429 Eating protein and vegetables before carbohydrate reduces post-meal glucose excursion and insulin response.
3. Yano JM, Yu K, Donaldson GP, Shastri GG, Ann P, Ma L, Nagler CR, Ismagilov RF, Mazmanian SK, Hsiao EY. 2015. "Indigenous bacteria from the gut microbiota regulate host serotonin biosynthesis." *Cell*

161(2):264–276. DOI: 10.1016/j.cell.2015.02.047 Roughly 90–95% of body serotonin is produced by enterochromaffin cells of the intestinal mucosa.

4. Browning KN, Travagli RA. 2015. "Central nervous system control of gastrointestinal motility and secretion and modulation of gastrointestinal functions." *Comprehensive Physiology* 4(4):1339–1368. DOI: 10.1002/cphy.c130055 Gut-derived serotonin acts on 5-HT3 receptors on vagal afferent terminals; the vagus nerve's own primary neurotransmitter remains acetylcholine.
5. Wong AC, Devason AS, Umana IC, Cox TO, Dohnalová L, Litichevskiy L, Perla J, Lundgren P, Etwebi Z, Izzo LT, Kim J, Tetlak M, Descamps HC, Park SL, Wisser S, McKnight AD, Pardy RD, Kim J, Blank N, Patel S, Thum K, Mason S, Beltra J-C, Michieletto MF, Ngiow SF, Miller BM, Liou MJ, Madhu B, Dmitrieva-Posocco O, Huber AS, Hewins P, Petucci C, Chu CP, Baraniecki-Zwil G, Giron LB, Baxter AE, Greenplate AR, Kearns C, Montone K, Litzky LA, Feldman M, Henao-Mejia J, Striepen B, Ramage H, Jurado KA, Wherry EJ, Abdel-Mohsen M, Meyer NJ, Kohli RM, Weiss SR, Collman RG, Bushman FD, Thaiss CA, Levy M. 2023. "Serotonin reduction in post-acute sequelae of viral infection." *Cell* 186(22):4851–4867.e20. DOI: 10.1016/j.cell.2023.09.013 Proposed mechanism: viral persistence in gut → interferon response → reduced tryptophan absorption → reduced peripheral serotonin → reduced vagal signaling → cognitive impairment. Brain serotonin was unchanged. Cohort-level findings were inconsistent — strong in Ireland, absent in UNCOVR, weak in a third cohort.
6. Mathé P, Götz V, Stete K, Walzer D, Hilger H, Pfau S, Hofmann M, Rieg S, Kern WV. 2024. "No reduced serum serotonin levels in patients with post-acute sequelae of COVID-19." *Infection* 53(1):463–466. DOI: 10.1007/s15010-024-02397-5 Independent cohort did

not confirm the peripheral serotonin reduction reported in Wong et al. 2023; no correlation with fatigue or depressive symptom severity.

7. Anderson GM, Cook EH, Blakely RD, Sutcliffe JS, Veenstra-Vander-Weele J. 2024. "Long COVID-19 and peripheral serotonin: a commentary and reconsideration." *Journal of Inflammation Research* 17:2169–2172. Methodological critique: centrifugation at 1,500 × g for 10 minutes produces plasma with fewer than 10,000 platelets per microliter — approximately 4% of typical whole-blood platelet count — and platelets are the body's primary reservoir of serotonin, so the reported reduction may reflect sample preparation rather than pathway change. DOI: 10.2147/JIR.S456000
8. Liu Q, Mak JWY, Su Q, Yeoh YK, Lui GC-Y, Ng SSS, Zhang F, Li AYL, Lu W, Hui DS-C, Chan PKS, Chan FKL, Ng SC. 2022. *Gut* 71(3):544–552. DOI: 10.1136/gutjnl-2021-325989, Ng SC. 2022. "Gut microbiota dynamics in a prospective cohort of patients with post-acute COVID-19 syndrome." *Gut* 71(3):544–552. Prospective study of 106 patients; *F. prausnitzii* and *Bifidobacterium pseudocatenulatum* showed the strongest inverse correlation with persistent symptoms at six months. Acute-phase antibiotic use is discussed as a likely confounder of the microbiome shift.
9. Estruch R, Ros E, Salas-Salvadó J, et al. 2018. "Primary prevention of cardiovascular disease with a Mediterranean diet supplemented with extra-virgin olive oil or nuts." *New England Journal of Medicine* 378(25):e34. DOI: 10.1056/NEJMoa1800389 Revised PREDIMED analysis confirming cardiovascular risk reduction on a Mediterranean dietary pattern.
10. Castro-Marrero J, Sáez-Francàs N, Santillo D, Alegre J. 2017. "Treatment and management of chronic fatigue syndrome/myalgic encephalomyelitis: all roads lead to Rome." *British Journal*

of Pharmacology 174(5):345–369; and Barletta MA, Marino G, Spagnolo B, et al. 2023. “Coenzyme Q10 + alpha-lipoic acid for chronic COVID syndrome.” *Clinical and Experimental Medicine* 23(3):667–678. DOI: 10.1111/bph.13702 Combination therapy showed fatigue resolution in approximately half of treated Long COVID patients in open-label conditions; confirmatory randomized data still pending.

11. Medinger G, Altmann D. 2022. *The Long COVID Handbook*. Patient-led survey of 812 Long COVID patients (November 2020) reported statistically significant symptom improvement with low-histamine diet ($p=0.0001$) and niacin supplementation ($p=0.01$) versus non-intervention control.
12. O’Kelly B, Vidal L, McHugh T, Woo J, Avramovic G, Lambert JS. 2022. “Safety and efficacy of low dose naltrexone in a long covid cohort; an interventional pre-post study.” *Brain, Behavior, & Immunity — Health* 24:100485; and Sasso EM, Eaton-Fitch N, Smith P, Muraki K, Marshall-Gradisnik S. 2025. “Low-dose naltrexone restored TRPM3 ion channel function in natural killer cells from long COVID patients.” *Frontiers in Molecular Biosciences* 12:1582967 (PMC12127304). DOI: 10.1016/j.bbih.2022.100485 The 2022 O’Kelly paper is the canonical pre-post pilot signal in Long COVID; the 2025 Sasso paper is the first ex-vivo demonstration that LDN partially rescues a specific ion-channel deficit (TRPM3 in NK cells) documented in LC patients, providing a more specific candidate mechanism than the older “microglial modulation” framing.
13. Byambasuren O, Atkins T, Baptista S, Glasziou P, Chakraborty S. 2026. “Effect of low-dose naltrexone for long COVID: a systematic review and meta-analysis.” *BMJ Open* 16(7):e111253. DOI: 10.1136/bmjopen-2025-111253 (preprint: medRxiv DOI: 10.1101/2025.09.09.25335451; and Du A, Nguyen ADK. 2025.

"Does Low-Dose Oral Naltrexone Alleviate Symptoms of Long COVID? A Systematic Review and Meta-Analysis." *COVID* (MDPI) 5(12):198. Both 2025 reviews pool four observational pre-post LC cohorts (combined n ≈ 155, dose range 1–6 mg) and converge on a moderate fatigue effect with approximately 50% of patients meeting responder criteria at 12 weeks; both note the absence of completed randomized controlled trial data and identify the Nacul (NCT05430152) and RECOVER-TLC LDN arms as the upcoming definitive readouts.

14. Singh A, D'Amico D, Andreux PA, Fouassier AM, Blanco-Bose W, Evans M, Aebischer P, Auwerx J, Rinsch C. 2022. "Urolithin A improves muscle strength, exercise performance, and biomarkers of mitochondrial health in a randomized trial in middle-aged adults." *Cell Reports Medicine* 3(5):100633. DOI: 10.1016/j.xcrm.2022.100633 Muscle strength improvement of approximately 12% and improvement in mitochondrial biomarkers in healthy adults aged 40–64. No Long COVID trial has been completed. About 30–40% of people have gut microbiomes capable of producing urolithin A endogenously from dietary polyphenols.

Chapter 8. Sleep architecture, not sleep hours

You can spend nine hours in bed and wake up as though you never slept.

What breaks is sleep architecture — the internal structure of the night. Among people with Long COVID, ME/CFS, and the broader cluster of infection-associated chronic conditions, architectural disruption is common enough to deserve its own chapter. The reason is specific: the overnight window is when most of the repair work your body is trying to do actually gets done. When that window fragments, everything downstream of it fragments too.

What happens while you sleep

Sleep is a structured, active process with specific jobs. It has its own moving parts, its own timing, and its own ways of going wrong.

A night of sleep is made of 90–110-minute cycles, each of which passes through several stages. Slow-wave sleep — the deepest stage, dominated by large-amplitude delta-band EEG waves — is when most of the body's physical repair work runs. Growth hormone release peaks. Tissue repair happens at higher rates than in waking hours, including the rebuilding of muscle microdamage and connective tissue. Immune memory is consolidated — the process by which exposures learned during the day are filed into longer-term defenses [11]. And the glymphatic system, a waste-clearance pathway along the perivascular spaces of cerebral blood vessels, runs at its full speed. Nedergaard and Goldman's 2016 overview describes it this way: during slow-wave sleep, cerebrospinal fluid moves through the brain's extracellular spaces and flushes out metabolic byproducts, including amyloid-beta and tau fragments, that accumulate during waking activity [1]. If that flush does not happen, the debris stays.

REM sleep — rapid-eye-movement sleep, the stage in which most vivid dreaming occurs — is doing something different. It is consolidating memory and metabolizing the emotional weight attached to recent experience. Work by Matthew Walker and colleagues has shown that REM sleep selectively reduces the next-day reactivity of the amygdala to emotionally charged memory content, while preserving the factual core of the memory itself [2]. In plain terms: events you lived through during the day get stored overnight in a form that remembers what happened but carries less of the distress. When REM is fragmented or shortened, the content stays; the processing does not.

In Long COVID and ME/CFS, both stages are under attack from multiple directions at once. Inflammatory signals arrive at the wrong times. The sympathetic branch of the autonomic nervous system stays active when it should be quieting. Slow-wave sleep gets fragmented and REM gets truncated. The "overnight updates" do not finish installing.

The result is a subtly misleading arithmetic. Eight hours in bed can contain only two or three hours of genuinely restorative sleep. Your body tracks the restorative minutes — the ones spent on the pillow alone do not add up the same way.

Why standard sleep-hygiene advice often stalls out

"Go to bed at the same time every night." "Put away screens an hour before sleep." "Keep the bedroom cool and dark."

This is reasonable advice. It is also aimed at the wrong problem for many people in this population.

Standard sleep-hygiene rules were built for a healthy nervous system that sleeps badly because of habit or stress. They assume the hardware works and that behavior is the thing to change. In a body with dysautonomia, chronic inflammation, and a sympathetic branch that will not stand down at night, you can produce a textbook-perfect bedroom — cool, dark, quiet, phone charging in the kitchen — and still wake at three in the morning with a pounding heart and a sense of catastrophe you cannot attach to anything.

The hygiene framework cannot reach that mechanism. It was built for a different problem wearing the same clothes.

To pick an intervention that will actually move the needle, the first step is to identify which of several specific mechanisms is breaking your sleep right now.

Four mechanisms that break sleep in Long COVID

Inflammatory signaling. Chronic low-grade inflammation disrupts both circadian rhythms and sleep architecture. The connecting molecules are cytokines — immune signaling proteins — which act not only in the periphery but directly on the brain regions that regulate sleep and wake. Imeri and Opp's 2009 review in *Nature Reviews Neuroscience* lays out the specific pathway: interleukin-1 beta and tumor necrosis factor alpha promote non-REM sleep at low doses and fragment it at sustained elevated doses, acting through receptors in the basal forebrain and the preoptic area [3]. This is why acute infections feel sedating — the immune system is actively producing the "sleep now" signal. The same machinery, kept on chronically, stops working correctly. The signal is always on and therefore informs nothing.

Dysautonomia and nighttime sympathetic surges. Many patients with POTS and autonomic dysfunction describe a stereotyped pattern: flattened by fatigue at nine in the evening, suddenly alert at midnight, wrecked by eight in the morning. The pattern is a mis-scheduled activation of the sympathetic branch. Instead of the expected overnight dominance of the parasympathetic branch — slower heart rate, lower blood pressure, quieter breathing — the sympathetic branch intermittently breaks through. Waking from sleep with a racing heart, a damp shirt, and a sense that something is wrong is one common presentation. The event is physiology arriving with an emotional label attached; the somatic signal comes first, the interpretation after.

Nighttime glucose drops in people with diabetes. An older online claim is that waking at three in the morning with a sense of doom is caused by a glucose *spike*. This was corrected in scientific review, and the reframe is correct: the mechanism is better described as a glucose *drop* that triggers a counter-regulatory hormonal response. In a diabetic body — particularly one on insulin or a sulfonylurea — a falling nighttime glucose provokes glucagon and epinephrine release from the adrenal medulla, and that hormonal surge wakes the sleeper. Blood sugar, tremor, sweating, and a racing heart at three in the morning, in someone with diabetes, can be the nighttime version of the glucose-crash sequence I described in Chapter 7. Outside of diabetes, the same mechanism is speculative. It may contribute in individual cases; the confident first explanation for most nondiabetic patients usually lies elsewhere, and the autonomic surge described above is the more likely candidate. If you are diabetic and these wakings are frequent, talk to your endocrinologist about continuous glucose monitoring overnight before changing anything else.

Histamine as a wake-promoting signal. Histamine is not only the molecule that makes you sneeze in pollen season. In the central nervous system it is one of the principal wake-promoting neurotransmitters, released by neurons in the tuberomammillary nucleus of the hypothalamus that project broadly across cortex. Thakkar's 2011 review in *Sleep Medicine Reviews* describes histamine's role as part of the ascending arousal system — activity is maximal during waking, suppressed in REM, and nearly silent during slow-wave sleep [4]. If your mast-cell activity is elevated in the evening, particularly after a dinner high in aged or fermented foods, that ambient histamine load is competing directly with the sleep system. This is another reason the low-histamine probe from Chapter 7 is often worth running before stacking sleep aids.

If you are not sure which of these four is most active in your case, the order I would suggest is behavioral: start with the autonomic surge and histamine patterns, because both have low-cost probes you can try without prescriptions. Inflammatory contribution is harder to intervene on directly outside of the pacing tools from Chapter 6. Nighttime glucose is worth investigating specifically when it applies.

Breathing as a lever, with red flags

Slow breathing with a lengthened exhale is one of the most studied non-pharmacological interventions on autonomic state. The mechanism is straightforward: exhalation preferentially engages the parasympathetic branch, the braking system to sympathetic activation [5]. There are two variants worth distinguishing, because they do related but different things.

Extended-exhale breathing uses an inhale of about four counts and an exhale of six to eight. The lengthened exhale accents parasympathetic activation. This is the calmer of the two patterns and the better choice close to bedtime.

Resonant breathing uses an inhale and an exhale of about five seconds each — roughly six cycles per minute. At this rate, the oscillation of the breath and the slower oscillation of arterial blood pressure synchronize; the phenomenon is called resonance and sits around 0.1 Hz. The effect is that the baroreceptors — stretch receptors in the carotid sinus and aortic arch that signal pressure changes — get a cleaner, higher-amplitude signal to work with, and heart-rate variability, a measure we will look at more carefully in Chapter 9, increases. Individual resonant frequency varies slightly; the window of 4.5 to 6.5 seconds per inhale and exhale covers most adults.

> **SAFETY —** Here is where the red flags matter, because neither of these patterns is universally safe in the Long COVID population. If you have mast-cell activation symptoms, deep inhalations, cold dry air, or breath-holds can directly irritate the airway and provoke a flare. In severe ME/CFS, even ten minutes of structured breathing can be enough cumulative sensory-motor exertion to produce post-exertional malaise the following day — breath practice counts as exertion in that body. A history of hyperventilation changes the calculation differently: deliberately deep inhalations lower your arterial carbon dioxide, which narrows cerebral blood vessels and produces lightheadedness and tingling that your nervous system misreads as a threat, escalating rather than calming the state you were trying to fix.

The rule for any breathing practice is the 48-hour audit. If it helps at ten in the evening but the next day is worse, the tool is wrong for your current state. If it helps and does not create a tax the following day, keep it.

The gentlest place to start is inhaling through the nose for four counts and exhaling through the nose for six, three minutes at a time, with the lower ribs expanding slightly and the shoulders still. Do not try to breathe *deeply*. Deliberate deep breathing is counterproductive here.

A softer alternative, for people who find any deliberate inhalation uncomfortable, is humming on the exhale. Vibration of the vocal cords stimulates branches of the vagus nerve in the larynx and shifts the autonomic state toward rest without requiring any change in breathing volume. Breathe normally, close the mouth, add a quiet “mmm” on the exhale. It is mechanically minimal and does not provoke hyperventilation.

Practical tools for sleep architecture

The tools in this section are organized by mechanism. Each one targets a specific breakdown point in the system we just described.

Temperature. A falling core body temperature is one of the triggers of the switch into slow-wave sleep. A bedroom around 65–68°F (18–20°C) supports that process [6]. A hot shower right before bed is a bad choice if you have POTS — it dilates peripheral vessels, amplifies orthostatic symptoms, and heats the body at the moment it is supposed to be cooling. A warm, not hot, shower an hour or two before bed is the working compromise.

> **SAFETY —** Ice baths and extreme-cold exposure protocols — including the Wim Hof method, which combines full cold-water immersion with intense breathing — have become popular as wellness practices. In Long COVID, MCAS, and severe dysautonomia, full cold-water immersion is a different animal entirely. The primary cold-shock response is a sharp burst of vasoconstriction, a steep rise in blood pressure, and a tachycardic surge — exactly the kind of autonomic challenge a fragile system handles badly. Cold plunges and mast-cell disease are an unsafe pairing. A cool rinse at the end of a warm shower sits in a different category from a full cold-water immersion. If you have mast-cell reactions or pronounced orthostatic intolerance, the extreme-cold culture is one to sit out.

Head-of-bed elevation. For patients with POTS and other forms of orthostatic intolerance, raising the head of the bed by 10–15 centimeters — about four to six inches, or roughly 3–5 degrees of incline — reduces nighttime urine output, helps maintain plasma volume overnight, and lowers the frequency of nighttime orthostatic episodes (also discussed in Chapter 5 as a conservative measure for plasma volume). Claydon and Hainsworth's 2004 review of non-pharmacological measures for orthostatic intolerance reported a consistent benefit [7]. This is a low-cost, low-risk intervention that you can run as a one-person experiment for a week by propping up the head of the bed frame with a couple of sturdy blocks.

Light. Blue-wavelength light suppresses melatonin release. In a body where circadian signaling is already disrupted, adding late-evening screen light is one more insult to a system that is trying to synchronize. Screen-side blue-light filters help; stopping screens an hour before bed helps more. The argument here is biological — melatonin timing is a chemical event, and its signal can be blunted by the wrong spectrum at the wrong time.

Melatonin, at low doses, as a chronobiotic. In circadian-rhythm disruption, a small dose of melatonin — 0.5 to 1 milligram, 30 to 60 minutes before the desired sleep time — can help re-anchor the rhythm. Note the dose carefully. At 0.5 to 1 milligram, melatonin acts as a chronobiotic, a signal that tells the circadian system what time it is. At 5 to 10 milligrams — the doses most commonly sold over the counter — melatonin produces supraphysiological concentrations that often produce heavy next-day fog and occasionally the opposite of the intended effect.

> **SAFETY —** Note the dose. At 5 to 10 milligrams — the strengths most commonly sold over the counter — melatonin reaches supra-physiological concentrations. A useful rule: if a dose produces a next-day fog heavier than whatever it helped you sleep through, the dose is too high.

One important caveat. Melatonin is immunomodulatory — at higher doses it has measurable effects on T-cell activity and cytokine production — and in autoimmune illness the effect is not always desirable. Given that Long COVID may include an autoimmune component (the autoantibody findings discussed in Chapter 4), melatonin is worth discussing with a clinician who knows your full picture before routine use, even if you have no formal autoimmune diagnosis.

Daytime naps. For people with moderate to severe ME/CFS, a daytime rest is often structurally necessary rather than a sign of poor discipline. The nuance is that a nap longer than 20–30 minutes can erode the nighttime "sleep pressure" — the accumulated drive for overnight sleep — and worsen the quality of the night. If your nights are poor, cap daytime rest at around 20 minutes and try to keep it before three in the afternoon. If daytime rest is the only way you can function at all, take it; just watch how it interacts with the night.

Caffeine and the CYP1A2 caveat. Caffeine has a half-life of 5 to 7 hours in most people — meaning that half the dose is still in circulation that many hours after you drank it. In people with specific variants of the CYP1A2 gene, which encodes the liver enzyme that metabolizes caffeine, the half-life can be considerably longer. Sachse and colleagues' work on the intron-1 polymorphism of CYP1A2 documented this variability [8]. If you drink coffee at noon and sleep badly, this is worth testing. Move all caffeine to before noon for two weeks. If sleep improves, the variable was real.

Catherine, seven months post-COVID and already running on a hydration protocol, a pacing ceiling, and the protein-first breakfast rule from Chapter 7, had been sleeping badly for months. She had chalked it up to the illness generally. In early October, at my suggestion, she kept a three-line sleep log for a week — bedtime and wake time, the number of awakenings and roughly when, and a morning score out of ten. On review, the pattern was not randomly bad. Four of the seven nights included a waking somewhere between 2:30 and 3:45, often with a heart rate on her watch that was visibly higher than her resting baseline. She does not have diabetes, so the glucose-drop mechanism did not fit her case. What she does have is a clear orthostatic history, and her late-night heart-rate elevations looked much more like an overnight autonomic surge than a hypoglycemic event. She tried the single cheapest intervention first. She put two stacked hardcovers under each leg at the head of the bed frame and slept that way for ten nights. The waking pattern flattened without vanishing — two of the ten nights included a waking, and when she woke, she fell back to sleep faster. That is the right size of result for one tool, applied correctly, in the right case.

Sleep and PEM: the overnight buffer

For people whose illness includes post-exertional malaise, sleep does something additional that often goes unnamed. It is the buffer against activity debt.

Many patients find that the quality of the previous night is one of the best predictors of how vulnerable they are to PEM the next day. A bad night lowers the ceiling on what the body can do the following day before tipping into delayed worsening. The relationship is measurable in autonomic and inflammatory markers — observable downstream physiology rather than interpretation. A nervous system that has not completed a full restorative cycle has less headroom for load.

The practical implication is counterintuitive. On the day after a bad night, reduce activity preemptively — before the crash — rather than trying to “catch up” on what the night did not deliver. That is the harder discipline, because a bad night often produces exactly the opposite impulse: extra coffee, a compressed schedule, an attempt to squeeze the day. Those are the days on which the activity budget is most fragile.

When to get medical help, not advice

Certain patterns of disrupted sleep in Long COVID are symptoms of discrete, diagnosable conditions that respond to specific treatments. Advice alone will not resolve them.

If you wake with a dry mouth, a morning headache, or the specific sense that you did not rest regardless of how long you were in bed, these are suggestive of sleep-disordered breathing — even in people who do not snore or carry extra weight, which in this population is not a reliable rule-out. Rates of obstructive sleep apnea appear to be elevated in the post-COVID population; Stavem and colleagues' 2021 analysis of persistent symptoms at 1.5–6 months reported increased sleep-related breathing problems at follow-up in patients who had recovered from the acute illness [9]. Obstructive sleep apnea is diagnosed with an overnight polysomnogram or a validated home-sleep-test device, and it is treatable. In a body that already has fragmented sleep architecture for other reasons, an untreated apnea on top is a hole that no amount of hygiene can fill.

If your legs at night produce the characteristic electric, crawly sensation that improves with movement and interferes with falling asleep, this is restless legs syndrome. It is more common in the Long COVID population than in the general population and is frequently associated with low iron stores. Patrick's 2007 review of restless legs pathophysiology highlights the iron–ferritin link, with target ferritin levels for symptomatic patients typically above 75 ng/mL — higher than the laboratory's "normal" cutoff, which reflects the general population rather than the subgroup with RLS [10]. A ferritin level is the first step, and it is a test your primary-care clinician can order at the next visit.

What this changes for you

Sleep in Long COVID is a physiology problem with several interacting mechanisms. Discipline does not reach the parts of the system that are actually failing.

The causation often runs the opposite direction from how it feels. Inflammatory signaling, nighttime sympathetic surges, and fragmented REM leave the nervous system unable to process what accumulated during the day; what gets labeled as anxiety is often the downstream effect of that incomplete processing. The loop is real — and the loop has several entry points, which is exactly what makes it tractable.

Do not try to fix all of it at once. Choose the one mechanism that best matches your picture and work on that one first. If your nights fit the dysautonomic pattern — evenings flat, midnight alert, early-morning wakings with a racing heart — start with head-of-bed elevation and a low-intensity extended-exhale practice, and watch what happens over two weeks. A second pattern worth testing: when nights deteriorate reliably after high-histamine dinners, run the low-histamine probe from Chapter 7 and watch the sleep column of the log. If you snore, wake with a headache, or wake with a dry mouth, order a sleep study before experimenting further. An untreated apnea is the wrong substrate on which to test anything else.

Optional self-observation prompt

For the next seven nights, keep a three-line sleep log — not in an app, in a notebook or phone note. The three lines are: the time you went to bed and the time you got up; the number of awakenings and roughly what time each one was; a subjective score out of ten for how you feel in the morning.

At the end of seven nights, look at the pattern. If awakenings cluster in the middle of the night with a racing heart, the first low-cost test is head-of-bed elevation for ten nights. Awakenings clustered after particular dinners point the other way: try a short low-histamine window as described in Chapter 7. If you wake with a dry mouth or morning headache, bring the log to your primary-care clinician and ask about a sleep study.

Change one variable. Observe for two weeks. Then decide. The usual trap in sleep work is stacking five interventions at once, feeling better, and having no way to know which one did the work.

Sources

1. Nedergaard M, Goldman SA. 2016. “Brain drain.” *Scientific American* 314(3):44–49. DOI: 10.1038/scientificamerican0316-44 Overview of the glymphatic system: cerebrospinal fluid moves through perivascular spaces during slow-wave sleep, clearing metabolic byproducts including amyloid-beta fragments accumulated during waking activity.
2. Walker MP, van der Helm E. 2009. “Overnight therapy? The role of sleep in emotional brain processing.” *Psychological Bulletin* 135(5):731–748; and Goldstein AN, Walker MP. 2014. “The role of sleep in emotional brain function.” *Annual Review of Clinical Psychology* 10:679–708. DOI: 10.1037/a0016570 REM sleep selectively reduces next-day amygdala reactivity to emotionally charged memory content while preserving the factual content of the memory.
3. Imeri L, Opp MR. 2009. “How (and why) the immune system makes us sleep.” *Nature Reviews Neuroscience* 10(3):199–210. DOI: 10.1038/nrn2576 Interleukin-1 beta and tumor necrosis factor

alpha promote non-REM sleep at low doses and fragment it at sustained elevated doses, acting through receptors in the basal forebrain and preoptic area.

4. Thakkar MM. 2011. “Histamine in the regulation of wakefulness.” *Sleep Medicine Reviews* 15(1):65–74. DOI: 10.1016/j.smrv.2010.06.004 Histaminergic neurons of the tuberomammillary nucleus form part of the ascending arousal system; firing is maximal during waking, suppressed during REM, and nearly silent during slow-wave sleep.
5. Zaccaro A, Piarulli A, Laurino M, Garbella E, Menicucci D, Neri B, Gemignani A. 2018. “How breath-control can change your life: a systematic review on psycho-physiological correlates of slow breathing.” *Frontiers in Human Neuroscience* 12:353. DOI: 10.3389/fnhum.2018.00353 Slow breathing with lengthened exhale increases parasympathetic tone and baroreflex sensitivity; resonance near 0.1 Hz maximizes heart-rate-variability amplitude.
6. Harding EC, Franks NP, Wisden W. 2019. “The temperature dependence of sleep.” *Frontiers in Neuroscience* 13:336. DOI: 10.3389/fnins.2019.00336 Core body temperature decline is a permissive signal for slow-wave sleep initiation; ambient bedroom temperature of 18–20°C supports this process.
7. Cooper VL, Hainsworth R. 2008. “Head-up sleeping improves orthostatic tolerance in patients with syncope.” *Clinical Autonomic Research* 18(6):318–324. DOI: 10.1007/s10286-008-0494-8 Twelve patients slept with the head of the bed raised by 10 degrees for three to four months; eleven showed improved orthostatic tolerance and plasma volume rose significantly. The authors note the measure may not be tolerated long term by patients or bedpartners. Support for plasma volume maintenance.

8. Sachse C, Brockmöller J, Bauer S, Roots I. 1999. “Functional significance of a C→A polymorphism in intron 1 of the cytochrome P450 CYP1A2 gene tested with caffeine.” *British Journal of Clinical Pharmacology* 47(4):445–449. DOI: 10.1046/j.1365-2125.1999.00898.x Polymorphic CYP1A2 variants account for wide individual variability in caffeine metabolism rate and half-life.
9. Stavem K, Ghanima W, Olsen MK, Gilboe HM, Einvik G. 2021. “Persistent symptoms 1.5–6 months after COVID-19 in non-hospitalised subjects: a population-based cohort study.” *Thorax* 76(4):405–407. DOI: 10.1136/thoraxjnl-2020-216377 Elevated rates of sleep-related breathing complaints reported at follow-up in post-COVID cohorts.
10. Patrick LR. 2007. “Restless legs syndrome: pathophysiology and the role of iron and folate.” *Alternative Medicine Review* 12(2):101–112. PMID: 17604457 Target ferritin above approximately 75 ng/mL is associated with symptomatic improvement; laboratory “normal” reference ranges reflect general-population distributions rather than the RLS subgroup.
11. Besedovsky L, Lange T, Born J. 2012. “Sleep and immune function.” *Pflügers Archiv — European Journal of Physiology* 463(1):121–137. DOI: 10.1007/s00424-011-1044-0 Landmark review establishing that slow-wave sleep supports consolidation of immunological memory, including augmented T-cell responses to prior antigen exposure.

Chapter 9. Data over feelings

At month eight, Catherine has more observations than she can hold in her head.

There is the morning pulse log she started after her cold-open support-group test — three rows a day for weeks, eventually three mornings a week once the protocol settled. There is the pacing diary she kept for the two weeks after Thursday's boom-and-bust, the one with activity blocks in one column, heart-rate peaks in the next, and a lagged-state score twenty-four and forty-eight hours later in the third. There are the ten nights of head-of-bed elevation logged in a notebook by the bed. There are the scraps: a phone note from a day she felt clearly better than the day before, a screenshot of her smartwatch readout on a bad afternoon, a Post-it on the fridge that says *ate leftover chicken → bad night*.

Her primary-care clinician has finally referred her to an internal-medicine physician who takes Long COVID patients, and the appointment is in eleven days. Catherine has spent an hour and a half this morning trying to gather what she has into something she can bring. She keeps running into the same obstacle. The observations are real; the notebook is real; the patterns are real. None of it is yet a document.

This is the chapter on turning it into one.

The problem is the format, not the doctor

People with Long COVID, on average, see clinicians about thirty times a year [1] — thirty separate attempts to explain what is happening, and thirty chances to try to compress eight nonlinear, shifting, delay-prone months into a ten-minute visit.

A lot of those visits end with the patient feeling unheard. The assumption people often carry out of them is that their clinician didn't care or wasn't listening. That is sometimes true. More often the issue sits upstream of the individual clinician. Modern outpatient medicine is built around a ten-minute snapshot of a condition that happens to be visible in the room. Long COVID and ME/CFS live in patterns and time series. Bringing a single snapshot to a clinical encounter designed for single snapshots is working as intended; it is also the wrong tool for this illness.

Data work here is about translation — not self-diagnosis, and not performance for a sceptical audience. You gather the sort of information the ten-minute snapshot cannot capture — the orthostatic response on Tuesday morning after Sunday's walk, the lagged crash on day two after a long phone call, the sleep disruption that clusters around particular dinners — and you hand it over in a form a busy clinician can absorb in sixty seconds.

Why memory is a poor database

The phenomenon has a name. *Recall bias* is the systematic distortion that happens when people report on past symptoms from memory rather than from real-time records. Stone and colleagues' 2000 *British Medical Journal* study — one of the canonical papers in the field — asked patients to keep a paper diary and compared what they actually wrote at the time with what they later reported [2]. Retrospective accounts differed from the real-time entries by about twenty to forty per cent. People remember peaks and endings well; they lose sequence and timing badly.

In ordinary illness this is an inconvenience. In illness with a delayed post-exertional component it is the thing that hides the mechanism. Post-exertional malaise, by definition, shows up a day or two after the trigger. If you rely only on memory, you will blame the crash on what was in front of you when the crash arrived — a stressful call, a skipped meal, the wrong weather — and miss the long phone call two afternoons earlier, the crowded waiting room the day before that, the stretch of standing you did on Sunday. Layer brain fog on top, and the job becomes harder still.

I made this point in Chapter 6 and it is worth restating here, because Chapter 9 is where it finally gets an instrument: the delay itself is the puzzle. If the delay could be pinned to the exact hour reliably, medicine would have solved this illness already. What a diary does is not give you exact hours. It gives you a forty-eight-hour window around each event, which is the time scale the illness actually runs on, and it gives you a log that does not forget.

What to track — the minimum set

A useful diary works like a flight recorder rather than a memoir. The goal is to catch patterns the memory cannot hold, in a form you can scan with tired eyes.

The minimum set that produces results within two weeks has four elements.

Morning pulse, supine and standing. The full NASA Lean Test protocol is in Chapter 4 and remains the gold-standard home measurement. For the daily record the compressed version is enough: five minutes lying quietly, then the pulse at two minutes of standing. Write both numbers down along with a subjective score out of ten. Three mornings a week is more informative than seven. Doing it on two consecutive days and then on one day later in the week gives you a short time series and a longer one in the same log.

Activity by form, not by label. The most common diary mistake is writing “walk” or “work” or “phone call” and moving on. The form of the load is as important as the duration. *Walked for twenty minutes* is a different load from *stood quietly on a train platform for twenty minutes*, which is a different load again from *sat at a laptop for twenty minutes*. Record each block of activity with its body position and its duration, and if any load is unusually cognitive or emotional, note that. This is a detail that tends to look excessive on paper for the first three or four days and then suddenly becomes the thing the diary is for.

Evening scores, split two ways. Before bed, give the day two numbers: one for physical load and one for cognitive load. This separation matters because the two categories do not always track each other, and they can produce different kinds of crash on different time scales. A day that looked light physically but included a forty-five-minute emotionally difficult conversation is a day your body will remember differently from a day that looked heavy physically but was conversationally empty.

Lagged state twenty-four and forty-eight hours later. This is the column that most new diary-keepers skip and most patterns depend on. Each evening you also note, next to yesterday's and the day-before-yesterday's entries, whether a crash has arrived. Heavier legs than usual? Thicker brain fog than the day-before-yesterday's score would predict? Heart-rate ceiling fired earlier than expected? A log without this time-lag column is a graph whose axes aren't labelled; the numbers exist, but the relationships don't.

The four columns together are enough to build the pattern map Chapter 3 introduced, across two weeks of actual data rather than across remembered days.

Pulse as an objective anchor

Of all the measurements you can take at home, pulse is the most informative and the least subjective. It does not require a lab, it does not require interpretation, and it tracks the body's current circulatory state closely enough that small changes in the number tend to correspond to real changes in what is happening.

The orthostatic tax — the difference between your supine pulse and your pulse after several minutes standing — is the single most useful derived number in this population. If you are at sixty-four lying down and one hundred at the five-minute mark, your tax is thirty-six beats. That figure often correlates with how the day will go more cleanly than any subjective score. It also responds to things you can act on: hydration the night before, quality of the previous night's sleep, phase of the menstrual cycle, temperature of the room. Three measurements taken across three different days are more informative than one "best" measurement.

A few details on how to take the measurement during daily life rather than just during a formal test. A chest-strap heart-rate monitor is more accurate than a wrist-worn one, particularly during movement. Wrist-worn watches are good for trend tracking; they lag real spikes and smooth short bursts. For real-time pacing — the eighty-seven-beat ceiling I described in Chapter 6, for instance — a basic chest strap with an electrocardiographic sensor reads more truthfully than a premium smartwatch. This matters less for weekly or monthly trend lines. It matters more when the instantaneous reading is what tells you to stop.

Hours of upright activity — one number for the doctor

There is one more number worth logging for doctor-facing purposes, and it is the one most likely to be understood on sight by a clinician who has never read this book. It is called Hours of Upright Activity, or HUA, and it comes out of the Bateman Horne Center's clinical care guide for this population.

HUA is the cumulative time each day with your feet on the floor — standing, walking, or sitting without your legs elevated. Healthy adults live at twelve to sixteen hours. People with moderate ME/CFS are often below five. People with severe disease are often below two. You do not need a tracker to record it; a phone note tallied across the day is enough.

Written next to your orthostatic tax and your standing pulse, HUA is the shortest statement of functional status you can hand a clinician. It is one number, in familiar units, already calibrated to the illness the rest of your diary is documenting.

HRV — a useful signal, a poor dictator

Heart-rate variability, or HRV, is a measure of the time differences between consecutive heartbeats. I glossed it briefly in Chapter 8 and deferred the detailed treatment to here. In a healthy body, a stable morning HRV trend tends to correspond to a nervous system that is recovering well and holding some flexibility; a falling trend tends to correspond to accumulating load or illness. That correlation is real, and for many people it is useful.

In bodies with Long COVID and dysautonomia, the correlation bends.

The "readiness" algorithms in consumer wearables were built and validated on healthy populations, and on athletes in particular. Two recent studies are worth naming. Da Silva and colleagues' 2023 paper in *Scientific Reports* found reduced resting and deep-breathing HRV in Long COVID patients compared with controls — meaning the underlying signal is different enough that the baseline assumptions of the algorithm do not necessarily apply [3]. Ruijgt and colleagues' 2026 study in *Sports Medicine* went further, looking at wearable HRV across sleep and daily activity in Long COVID and finding that the devices' standard "recovery" and "readiness" categories did not correspond reliably to whether patients were actually in a state that would tolerate load [4]. Put plainly: the device is seeing what it was trained to see, in a body that is not the body it was trained on.

It is worth being precise about what this means in practice. A tracker can distinguish rest from what might be called “emergency economy” by looking at heart rate directly — a chronically elevated resting pulse in a body at rest is, in fact, readable. What a tracker cannot reliably distinguish is true parasympathetic relaxation from a parasympathetic branch that is maximally activated fighting infection. In the first state, the numbers are green and the body is recovering. In the second, the numbers can also be green — and the body is mounting a defense. The wearable sees both as “ready.” One of them is, and one of them is not.

There is a second bend in the HRV signal that is worth knowing. Stay sleep-deprived for more than a day, even as a generally healthy person, and your HRV can start to rise, not fall. This looks counterintuitive on the dashboard. It is not a “good” sign. What it reflects is the parasympathetic branch trying to compensate during waking hours for what the body should have been doing overnight. The tracker shows a number that the algorithm associates with recovery, while the body is in fact in deficit.

The rule I would draw from this is simple. HRV is a vote in a committee, not the chair. Pair it with three other inputs: your orthostatic tax from the same morning, your subjective body report, and what happened twenty-four and forty-eight hours ago. If any three of those four disagree with the HRV number, the HRV number is the one to down-weight.

One pattern does travel well across Long COVID and ME/CFS, and it is worth using. Several consecutive mornings of falling HRV, in the absence of an obvious explanation like travel or a bad night, is a reliable signal that the coming days need less load — not more. The information is most useful as a direction and a trend; it is least useful as an instruction about a single day. Trusting the number on any given morning when you feel like you’ve been hit by a truck is a mistake in both directions; your body is reading data the watch cannot see.

Why your tracker's math does not fit this body

The algorithms running on your wrist were built to solve a specific set of problems for a specific type of physiology. When a fitness tracker calculates active calories, estimates a recovery score, or tells you how much rest you need, it relies on a standard model of human metabolism. That model assumes a predictable, unbroken relationship between physical work, heart rate, and oxygen consumption. In a healthy body, carrying a heavy laundry basket up a flight of stairs requires a known amount of oxygen. The lungs pull that oxygen in, the blood delivers it, and the muscles extract it to manufacture ATP. The sequence is stable enough that a watch can read your pulse and calculate the rest of the chain.

In an energy-limiting condition, that chain breaks at the final link. Chapter 1 laid out the mechanics of this failure: the oxygen arrives at the tissue and goes unused. The cells shift to a less efficient metabolic pathway to meet the demand. Your tracker has no sensor for cellular oxygen extraction. It reads the elevated heart rate, assumes your muscles are doing the equivalent of a brisk jog, and applies the math of a healthy body to a system operating under different rules. The device is measuring what it was designed to measure; the baseline equation has just changed underneath it.

This mathematical mismatch explains why the dashboard on your phone often contradicts the reality in your living room. You can wake up to a bright green readiness score on the morning your legs feel like they are filled with wet sand. The software evaluates your overnight data against a massive database of healthy recovery patterns, looking for signs that a normal nervous system has cleared the fatigue of an ordinary day. It has no mechanism to detect the delayed systemic worsening of post-exertional malaise. The athletes and healthy volunteers who trained the algorithm do not experience that delay.

As of spring 2026, there is no validated algorithm capable of predicting a crash. You will not find a hidden setting in your app that flags an exertion threshold today to prevent a symptom flare forty-eight hours from now. The research literature contains group correlations and promising heuristics; it does not contain a personalized predictive model that works for this specific illness. The feature you want most — a watch that vibrates while you are unloading the dishwasher to tell you to sit down before you ruin your Thursday — does not exist yet.

This structural gap is the reason we rely on the heart-rate ceiling discussed in Chapter 6. That ceiling is an educated heuristic rather than a precise biological boundary. The only way to find your true anaerobic threshold is a two-day cardiopulmonary exercise test. As noted in the safety warnings of Chapter 1, that formal test is inaccessible for most and carries a real risk of triggering a severe, lasting crash. We use the estimated ceiling as a rough guardrail to keep you clear of the worst of it, rather than treating it as a medical measurement. The watch provides the raw pulse data, and you provide the interpretation.

This leaves you in a position that often feels frustrating. You own a sophisticated piece of hardware that tracks your heart rate and feeds that number into an irrelevant formula. The practical solution is to ignore the formula and keep the raw number. The pulse is an objective measurement; the recovery score is a guess built for someone else.

Building your own empirical model is the only way forward. You track the raw inputs — your standing pulse, your hours of upright activity, your daily physical and cognitive load — and you map them against the symptoms that arrive two days later. You are doing the math the software cannot do. When you write down that twenty minutes of standing in a grocery line on Tuesday produced heavy brain fog and an elevated resting heart rate on Thursday, you are establishing a personal threshold.

You are finding the edges of your current envelope by observing where it tears.

Your log will not give you exact predictions. It will give you a working perimeter, built on the actual responses of your own body rather than an idealized average. A diary serves as the necessary bridge between a device that sees only half the picture and a body that experiences the whole delayed reality, rather than acting as a primitive fallback for people who do not understand their smartwatches. This translation from raw experience into documented pattern is the exact job the next tool on the table was designed to handle.

The DePaul Symptom Questionnaire — PEM on paper

Most standard medical questionnaires do a poor job of capturing post-exertional malaise. Asking someone with PEM "how tired are you?" is a little like asking someone in a house fire "how warm are you?" The question is technically answerable and almost entirely useless. PEM is a delayed worsening of the entire system after a load that used to be tolerable — a different animal from tiredness, measured on a different axis.

The instrument that takes PEM seriously is the DePaul Symptom Questionnaire, developed by Leonard Jason and colleagues at DePaul University and validated across thousands of patients [5]. It was designed to capture the specific architecture of ME/CFS: post-exertional malaise, sleep quality, fatigue quality distinct from sleepiness, pain, cognitive dysfunction, and orthostatic intolerance — each scored on both frequency and severity, across a six-month window. It exists in a long form used in research and a short form usable in a clinic visit or at home. Copies are available through the DePaul research centre's website for patient use without cost.

The reason to run it before an appointment, rather than during one, is that filling it out carefully takes time and pattern recognition. You cannot sit down in a waiting room and recall six months of cognitive dysfunction frequency accurately; you can sit at a kitchen table at month eight with your diary open and fill it out with a fair degree of reliability. The completed form turns into a compact document that meets clinicians on their own ground. “I have symptoms consistent with PEM, documented at moderate frequency and moderate-to-severe intensity on the DePaul Symptom Questionnaire” is a sentence a clinician can put in a chart and act on. “I feel tired in a way that isn’t normal” is a sentence they have heard a thousand times in twenty other illnesses.

The form is particularly important before any of three conversations: a physical-therapy referral, a return-to-work plan, or any proposal for a graded increase in exercise. PEM is the contraindication the clinician needs to see documented before any of those programmes gets written, and the most efficient way to document it is a validated questionnaire filled out honestly.

The one-page doctor summary

A ten-minute appointment has a bandwidth problem. Your illness is a documentary in high resolution; the appointment is a fax machine. Your job is to compress the signal well enough that it arrives intact.

One page. Not two. The structure I would use, and the structure I have seen work repeatedly in this population, is as follows.

The top line is the main problem in a single sentence, with the suspicion and the ask explicit. *Suspected post-infectious dysautonomia with post-exertional malaise following COVID in March 2021. Requesting an orthostatic evaluation and a discussion of next steps.* This is the sentence the clinician reads first and uses to orient the entire appointment.

The second line, one short row each, is red flags today — chest pain, syncope, falling oxygen saturation, leg swelling — present or absent. This reassures the clinician about acute safety before the conversation moves to the chronic picture.

The third block is the timeline, three lines: what you could do before the illness; the date and trigger; how long you have been in the current state.

The fourth block is three patterns, named in plain language: for example, intolerance of standing still, delayed crashes after activity, sleep disruption. Three, not five; this is the part the clinician will remember.

The fifth block is objective data: supine and standing pulse, the orthostatic tax, and one or two observations from the diary that demonstrate the delayed pattern. Keep this compact. A single sentence of the form *Supine 68, standing 104, tax 36; crash typically arrives at forty-eight hours after cognitively demanding days* is stronger than a paragraph.

The sixth block is function before and function now. *Previously walked eight thousand steps a day, worked full-time in a classroom; currently cannot stand in the shower for more than three minutes.* This gives the clinician a sense of delta that numbers alone cannot carry.

The seventh block is the specific ask. One, two, or at most three items. *An orthostatic evaluation; a referral to autonomic medicine if the tachycardia pattern replicates in clinic; a discussion of whether low-dose pharmacologic support is appropriate.* The specific ask prevents the appointment from ending with “let’s see how you do” by accident.

Send the page through the clinic’s patient portal twenty-four to forty-eight hours before the visit if possible, and bring a printed copy in any case. A clinician who walks into the room having already seen a structured one-pager spends the appointment on your problem instead of on decoding it, and the rest of the visit becomes available for judgement and planning. The full chapter on this conversation is Chapter 10 —

the purpose of the one-pager here is to make sure the data work of this chapter lands somewhere. A blank template of the page, with a filled-in example, is in Appendix 2; this section is the reasoning behind it.

When the data disagree with each other

Sooner or later — usually sooner — the numbers will disagree. Pulse says one thing, HRV says another, the diary says a third, and your body says a fourth. That disagreement is the correct reading of an illness with several moving parts, rather than a failure of the instruments.

The practical rule, inside the noise, is that the body wins over the gadget. Wearables read indirect signals. Your body reads direct ones. When the wrist shows green and you feel like the floor is moving, the correct response is to believe the floor.

The second rule is that trends matter and single points do not. One bad morning is weather. Five consecutive mornings of deterioration after a week of higher load is a signal. One good HRV reading is an anecdote. Three weeks of slowly rising HRV at consistent wake times is a trend. The strongest mental move I can recommend here is refusing to act on single data points — letting the log accumulate enough entries that a line has to be drawn through more than one of them before the line means anything.

The third rule is that monitoring works better in seasons than continuously. Two weeks of active tracking when a new intervention goes in; a month of rest from tracking; two more weeks when something else changes. Continuous monitoring in this illness tends to become its own source of anxiety. What the research can tell us here is narrower than the advice: Lank and colleagues followed sixty-three people with neurologic Long COVID through a three-month window on a symptom-tracking application, and the finding that matters for you is how uneven recovery looked from inside the data — the people who improved were the ones whose scores fluctuated most, while flat lines belonged to those who did not [6]. Participants themselves found the app easy to use. The case for seasons rather than continuous tracking rests on what patients report about attention as a cost, and on the arithmetic of this illness, rather than on that study. The log exists to serve you. Step away from it when looking costs more than it returns.

Data as evidence, not self-diagnosis

One final framing, because it is the one that most often gets misread on both sides of the desk.

The data you gather at home is the information the clinical encounter structurally cannot capture — not an attempt to diagnose yourself on the internet, and not a performance of suffering for a sceptical audience.

A standard visit measures you at rest, once, under fluorescent lights, while you are mobilising every reserve you have to arrive, sit up, and appear like someone whose problems are legible. For an illness with delayed symptoms and positional triggers, this is close to the worst possible measurement window. Your body on a Monday morning in a waiting room is your body at its most guarded. The relevant data lives somewhere else entirely — on a Tuesday morning after Sunday's errands; in the second half of a phone call, not the first; on day two of a trip, not the day you arrived.

Your diary is how that data reaches the room — field data, on an illness that manifests in the field.

By month eight, and by the structure of the illness rather than by choice, you are the field researcher of your own case.

Where Catherine is at the end of this

At the end of the week Catherine spends thirty-five minutes with the DePaul Symptom Questionnaire at her kitchen table with her diary open.

The moderate-to-severe PEM score is documented; the sleep-disruption pattern is documented; the orthostatic-intolerance score is documented; the cognitive-dysfunction score comes out heavier than she expected, and she realises that her month-six habit of declining phone calls was a pacing strategy she had not yet named as cognitive pacing.

She writes the one-page summary the following morning. The top line reads: *Suspected post-infectious dysautonomia with post-exertional malaise following COVID in March 2021. Requesting an orthostatic evaluation, a referral to autonomic medicine if indicated, and a conversation about low-dose pharmacologic support.* The three patterns she lists are orthostatic intolerance, delayed crashes at forty-eight to seventy-two hours after both physical and cognitive load, and early-morning wakings responsive to head-of-bed elevation. The objective-data line is *supine sixty-eight, standing one-oh-four at five minutes, tax thirty-six; repeatable across three mornings; DePaul PEM score attached.* The functional-comparison line is *previously taught a full classroom load with two sons and a Labrador; currently working at fifty per cent, using a pacing ceiling of eighty-seven beats, unable to stand in the shower longer than three minutes*.

She sends the page through the portal on Sunday evening. She has not had the appointment yet. She has, for the first time in eight months, prepared for a clinical conversation she can actually have — one that starts with a document instead of starting with her trying to summarise eight months of experience from memory while a clinician watches a clock.

The change is not that her condition has improved. The change is that the data she has been accumulating all along has finally been turned into a document the system can read.

Optional self-observation prompt

Tomorrow morning, take three numbers: your supine pulse after five minutes lying quietly, your standing pulse at two minutes, and your standing pulse at five minutes. Write them down with the time and the date. Do the same thing the next two mornings.

On day four, look at what you have. If your tax — the difference between supine and standing — is above twenty-five to thirty beats across all three mornings, you have the beginning of a pattern and a piece of objective data that belongs in any subsequent clinical conversation. If it is below that threshold some mornings and above it on others, you have a different kind of pattern — one that will reward two weeks of more careful tracking on both sides of the meal, fluid, and sleep variables.

Either way, on day four you have something you did not have on day one: a number that is yours, reproducible, and translatable into clinical language. The rest of this chapter's tools — the diary, the HRV pairing, the DePaul form, the one-pager — build out from there.

Sources

1. Lin L-Y, Henderson AD, Carlile O, et al. 2024. "Healthcare utilisation in people with long COVID: an OpenSAFELY cohort study." *BMC Medicine* 22(1):255. DOI: 10.1186/s12916-024-03477-x Matched cohort in NHS England data: 52,988 people with a long COVID diagnosis against 264,867 comparators. Healthcare utilisation 49% higher (RR 1.49, 95% CI 1.48–1.51), with a modelled mean of 29.2 healthcare contacts per year against 16.0 in the comparator group. "Contacts" here include prescriptions as well as consultations and admissions.
2. Stone AA, Shiffman S, Schwartz JE, Broderick JE, Hufford MR. 2002. "Patient non-compliance with paper diaries." *British Med-*

ical Journal 324(7347):1193–1194; and Stone AA, Shiffman S. 2002. "Capturing momentary, self-report data: a proposal for reporting guidelines." *Annals of Behavioral Medicine* 24(3):236–243. DOI: 10.1136/bmj.324.7347.1193 Retrospective symptom reports diverge from concurrent real-time entries by approximately 20–40%; recall systematically distorts symptom timing and intensity, with peaks and endings remembered more reliably than sequence and duration.

3. da Silva ALG, Vieira LP, Dias LS, Prestes GS, Back GD, Goulart CL, Arena R, Borghi-Silva A, Trimer R. 2023. "Impact of long COVID on the heart rate variability at rest and during deep breathing manoeuvre." *Scientific Reports* 13:22695. DOI: 10.1038/s41598-023-50276-0 Reduced resting HRV and reduced HRV during deep-breathing manoeuvre in Long COVID patients compared with healthy controls, consistent with altered parasympathetic regulation.
4. Ruijgt TM, Slaghekke A, Ellens A, Janssen KW, Wüst RCI. 2026. "Wearable Heart Rate Variability Monitoring, Autonomic Dysfunction and Post-exertional Malaise in Long COVID: An Observational Study." *Sports Medicine*, online first. DOI: 10.1007/s40279-026-02487-4 Consumer-wearable HRV was lower during sleep and everyday activity in Long COVID patients; algorithmic "recovery" and "readiness" categories, validated in healthy populations, did not map reliably to whether patients could tolerate load.
5. Jason LA, So S, Brown AA, Sunnquist M, Evans M. 2015. "Test-Retest Reliability of the DePaul Symptom Questionnaire." *Fatigue: Biomedicine, Health & Behavior* 3(1):16–32; and Jason LA, Sunnquist M, Brown A, Evans M, Newton JL. 2014. "Are Myalgic Encephalomyelitis and Chronic Fatigue Syndrome Different Illnesses? A Preliminary Analysis." *Journal of Health Psychology* 21(1):3–15.

DOI: 10.1080/21641846.2014.978110 DePaul Symptom Questionnaire measures post-exertional malaise, fatigue quality, sleep, pain, cognitive dysfunction, and orthostatic intolerance across a six-month window, scored on both frequency and severity; long and short forms validated in multiple ME/CFS cohorts; distributed without cost through the DePaul Center for Community Research.

6. Lank GK, Budhiraja S, Gaelen JI, Mukherjee S, Singer T, Venkatesh A, et al. 2026. “Characterizing Neuro-PASC outcome with the mobile Neuro-COVID recovery care companion application.” *BMC Neurology*. 2026;26(1):24. DOI: 10.1186/s12883-025-04577-8 PMC12794446. Sixty-three patients tracked through a mobile application across a three-month window; 36 (57.1%) were classified as non-improvers. Improvers showed greater variance in subjective recovery than non-improvers (7.01 vs 3.79; p=0.0004). Both groups rated the application easy to use and satisfactory.

Chapter 10. Talking to your doctor

Catherine walked into her appointment with a single sheet of paper in her hand.

She had printed the one-pager we built in the last chapter the night before — timeline, three symptom patterns, fourteen days of lying and standing heart rate, the line about losing half her afternoons to a grocery trip. She had rehearsed her opening sentence in the car. Her resting heart rate was up, which she noticed and filed away rather than argued with.

When the doctor came in and asked what was going on, Catherine slid the page across the desk. The doctor looked at it for maybe twelve seconds, then asked a question nobody had asked her in eight months:

"Where did you get this data?"

That was the whole pivot. It was the first appointment since her illness began that did not end with a psychology referral.

This chapter is about how to make that kind of opening happen on purpose. The page itself is not magic, and a clinician who reacts that way is not unusually kind. The page works because it does something a spoken account cannot do inside a ten-minute visit: it gives the clinician shape before it gives them feelings.

What the clinician is actually trying to do in ten minutes

Your appointment is a compression problem. You are carrying something that has been unfolding for months or years across half a dozen body systems, and the slot on the other side of the desk is built to absorb a single chief complaint in ten to fifteen minutes. If you arrive with the whole story and expect it to fit, the room will shrink it for you in ways you did not choose.

Inside that slot, a clinician is running a small, fast list:

- Is anything dangerous happening right now?
- What pattern does this fit?
- What common, treatable lookalikes do I need to rule out?
- What can I actually measure today?
- What is the next useful step?

If your words land as a flood of symptoms across six systems, the list above has nothing to grab onto. The clinician hears noise where you meant signal. The format itself is the problem here. Primary care was built around compact complaints like "my knee hurts when I run," and it does not have a standard playbook for a system running a slow fire across heart, gut, vasculature, and cognition at once.

Your job inside those ten minutes is to help the clinician see the shape. Nobody is going to score you on how much you are suffering, and the person across the desk has already seen someone in pain this morning. Shape means: a trigger, a pattern, a before-and-now, and one clear ask. That combination is what lets a clinician stop searching and start working.

The twenty-second intro

I ask everyone I talk with about this to write a single paragraph before their next visit. One paragraph, read aloud in about twenty seconds, that contains four elements and nothing else.

Catherine's looked like this:

"After a COVID-like illness in March 2021, standing up started causing palpitations, dizziness, and nausea. Now any sustained exertion triggers a delayed crash forty-eight to seventy-two hours later. Before this I worked full-time and walked eight thousand steps a day; now I sit to shower and cannot stand in a grocery line longer than seven minutes. I am here today to ask for orthostatic vitals, basic labs, and a working plan for what looks like post-viral dysautonomia with post-exertional malaise."

Four moves in four sentences.

Trigger. A specific onset tied to a specific event, with a date if you have one. "It started gradually" gives a clinician nothing to hold. "It started within six weeks of a flu-like illness in March 2021" puts the picture inside a known medical frame — post-infectious — and that frame carries its own reading list.

Pattern. One or two dominant mechanics, stated in mechanics language. Upright intolerance. Delayed post-exertion crash. Post-meal reactivity. Cognitive drop under load. These words map onto conditions the clinician can look up during the visit. "Fatigue" does not map onto anything.

Function, before and now. Two anchors, side by side. The "before" is a life the clinician can imagine. The "now" is the specific thing you used to do and can no longer do. Numbers are useful wherever they fit: steps, minutes upright, flights of stairs, work hours. A functional delta is how disability gets documented, and the documentation itself matters.

One specific ask. Not a diagnosis. Not a cure. The single next useful action for today. Orthostatic vitals with blood pressure at lying, one minute standing, three minutes, ten minutes. A named lab panel. A referral to a specific type of specialist. "I want to feel better" is not an ask. "I would like a ten-minute NASA lean test performed today — the one we walked through in Chapter 4" is.

If brain fog is active on the day of your appointment, write the paragraph out and hand it across the desk. Adrenaline and illness both shrink working memory. A page does not flinch.

SOCRATES, adapted for a body that refuses to stay in one organ system

SOCRATES is a mnemonic clinicians already know. It was designed to interview pain, but it happens to fit any localizable symptom, and post-infectious illness is survivable inside its columns if you adapt them slightly.

The standard frame:

Letter	What the clinician is listening for	What you write
S — Site	Where is it happening?	Body system: chest, legs, head, gut, “cognition,” or “upright.”
O — Onset	When did it start?	The date, the trigger (e.g., infection), gradual or sudden.
C — Character	What does it feel like?	Pounding, air hunger, pressure, leaden limbs, tingling.
R — Radiation	Does it move?	Chest to neck, headache behind the eyes, dimming vision.
A — Associated	What comes with it?	Nausea, heat intolerance, tremor, sore throat during crashes.
T — Timing	When and how long?	Constant or episodic? Post-meal? Twenty-four hours after effort?
E — Exacerbating / Easing	What makes it worse and what helps?	Standing, heat, showers, stairs / lying flat, salt, rest.
S — Severity	How does it limit you?	Function: minutes upright, distance walked, work hours lost.

Three upgrades for post-infectious illness.

Make timing work hard. This is where PEM lives, and PEM is the pattern most clinicians have never been trained to recognize. “Exertion makes me tired" is too thin. "If I walk more than three thousand steps, I crash forty-eight to seventy-two hours later for two days" names a delay, a threshold, and a duration. The delay is diagnostic. A clinician who hears a twenty-four-hour lag between effort and worsening has reason to think about a different list of conditions than a clinician who hears "I get tired."

Use function instead of adjectives. Adjectives float. "I feel awful" is accurate, and a skilled clinician will not dismiss it, but adjectives cannot be measured. "I can stand at the counter for six minutes before I need to sit" is legible. Minutes, meters, steps, hours, flights — the numbers you already tracked in Chapter 9 become your grammar here.

Present the cluster as a cluster. This is the upgrade that changes the visit most. If you list IBS-type symptoms, migraines, post-meal palpitations, heat intolerance, and a gritty alertness you cannot turn off, a clinician trained to find one thing will pick the one that fits their specialty and send you home. If you present those symptoms as a single pattern — a whole-body amplification that looks like the nervous system stuck in a state of high alert — you are helping the clinician see what the evidence base now says. As laid out in Chapter 3, the 2026 *JAMA Network Open* analysis found that 86.6% of POTS patients also met criteria for central sensitization syndrome — the same study is the evidence base behind the cluster argument [1]. Those symptoms describe one system-wide signal amplifier with five different surfaces showing, and presenting them that way saves everyone time.

Four scripts for when the room gets crooked

Four clinical situations account for most of the moments where post-infectious patients lose the visit. Having a short line ready for each one keeps the appointment inside the frame you walked in with.

When it arrives as "this is probably just anxiety."

In the largest survey of this population ever run — 4,835 people with a physician diagnosis of POTS — 77% had met a doctor who suggested their symptoms were psychiatric or psychological before anyone arrived at the right diagnosis, and three-quarters had been misdiagnosed outright [2]. The figure worth setting beside those two is the third one: 28% reported actually having a psychiatric or psychological problem at the time. Assessed formally, people with POTS carry no higher lifetime rate of psychiatric disorders than the general population [2]. Knowing those numbers will not change the doctor's mind, but it will change yours. The goal is to move the conversation to measurement rather than to argue about the word "anxiety."

"I understand anxiety could be part of the picture, and I am open to addressing it. The reason I am here today is that I have physical symptoms triggered specifically by standing up and relieved specifically by lying down. Can we measure orthostatic vitals now so we have a physical baseline?"

Notice what that does. It acknowledges the clinician's hypothesis instead of fighting it, reframes your request as an additional data point rather than a rejection, and asks for a procedure that takes about ten minutes and that the clinic can do today.

When it arrives as "your labs are normal."

Normal labs are characteristic of post-infectious illness. Most of what breaks in these conditions does not show up on a complete blood count or a standard metabolic panel. The script below accepts the normal labs and moves forward:

"That is useful to know. Given that the basic labs are clean and the symptoms are still severe and pattern-specific, what conditions would you consider next? I am particularly interested in dysautonomia workup and post-viral syndromes."

Naming the next candidate categories by their proper names — dysautonomia, post-viral — gives the clinician permission to look them up during the visit if they need to.

When it arrives as "you just need to move more."

This is the most dangerous script because it sounds reasonable and is wrong in a specific, testable way. In PEM, graded exercise that would help almost any other deconditioned person can trigger a crash that sets the patient back weeks or months. The script does not argue the general value of movement:

"I would like to move more, and I am willing to work up to it. The piece I want to plan around is that effort produces a delayed worsening forty-eight to seventy-two hours later. How does your plan account for that lag, and what would we do if it shows up?"

That question is mechanical rather than confrontational. A clinician who has a good answer will give you one. A clinician who does not will usually pause, and the pause itself is information you can use.

When it arrives as a refusal to order a test you need.

There is one question that works better than any argument here. I learned it from patients who went through ten-visit odysseys before they got a tilt table or a proper NASA lean test, and it is the closest thing I have found to a master key:

"Can you tell me what specific finding or symptom would make that test appropriate, in your view?"

The question is disarming. It does not contradict the refusal. It asks the clinician to name their threshold out loud. Sometimes the threshold is reasonable, and you can meet it by coming back with more data. Sometimes the threshold is vague, and naming it out loud moves the clinician toward action. Either way, the question gives you something concrete to carry into the next appointment, either here or with another provider.

What to ask for — and why

There is a short list of tests and measurements that earn their place in a post-infectious workup. You do not need every test. You need the ones that rule in the common mechanisms and rule out the common lookalikes.

For upright symptoms, ask for orthostatic vitals done properly — heart rate and blood pressure measured lying down, at one minute standing, at three minutes, and at ten minutes. A ten-minute NASA lean test is simple, free, and legible on a chart note. Ask for an ECG. Ask for ferritin (not just hemoglobin), vitamin B12, and thyroid function including TSH and free T4. Low ferritin and borderline thyroid abnormalities can make dysautonomia look much worse than its underlying severity, and catching them early saves months of guesswork.

For delayed post-exertional worsening, the request is documentation more than imaging. Ask for PEM to be named in the chart by its proper words — “delayed post-exertional symptom exacerbation” — because that note is what lets future clinicians find the pattern without having to rediscover it. If physical therapy is on the table, ask the mechanical question: “How does this plan account for twenty-four-hour post-exertion worsening?”

For cognitive symptoms, bring specifics. “Brain fog” is a container. What your clinician can work with is more precise: I lose my place in a paragraph after the second page; I cannot find common nouns under time pressure; I lose the thread of a conversation when a second speaker joins. Those three patterns point to different kinds of cognitive support, and a vague label does not.

For suspected mast cell involvement, describe the timing of the symptoms you are flagging — ninety minutes after meals, after hot showers, after specific exposures — rather than asking for a diagnosis by name on your first visit. The timing is what makes the case, and the diagnosis is what comes after.

Green flags and red flags

A good clinician for post-infectious illness does not have to be an expert on your condition. What they need to be is a collaborator who takes your data seriously and keeps the plan moving.

Green flags are quiet. The clinician says some version of "I am not sure what this is yet, but your pattern is consistent." They are willing to look at the home data you brought and to write at least part of it into the chart. Telehealth appears in the follow-up plan so you do not burn orthostatic budget on a waiting room. A specific next step gets named before you leave.

Red flags are also quiet, and they are easier to miss because they come wrapped in confidence. The clinician refuses to measure orthostatic vitals during a visit for orthostatic symptoms. A mental health diagnosis arrives before any tests have been done. The exercise plan shows up without a single question about delayed worsening. Your data goes unacknowledged.

One more signal worth naming: distraction. A visit in which the clinician spends most of the ten minutes looking at a screen that is not your chart is a visit that cost you energy you will not get back. Your time is expensive to you in a literal physiological sense. Spend it where it returns something.

If the match is wrong, move. The internal-medicine system in 2026 is imperfect but not closed — most regions have at least one clinician who has built a post-viral caseload, and the patient communities will tell you who they are. Looking for the right collaborator is a form of the same calibration you already do with your symptoms.

Your position: navigator

I want to be careful here, because it is easy in a chapter like this one to drift into a tone that treats clinicians as adversaries. That tone does not match my own experience of working with specialists over the last two years.

Most clinicians entered medicine to help people. The ones who miss post-infectious illness are usually missing it because their training did not include it, because the slot they are working inside is ten minutes long, and because the format of most visits is hostile to the specific shape these illnesses take. When you arrive with a page that fits the format, many of those same clinicians will do excellent work.

Your position inside the treatment relationship is navigator. The condition does not yet have a single-test diagnosis or a single-pill treatment, which means the person who knows the whole terrain is you. The clinician's job is to add tools, rule things out, and prescribe. Your job is to know what works for your system under what conditions, and to bring that knowledge into the room clearly.

That division of labor reflects what the evidence base currently allows, rather than any personality fit. A 2026 narrative review in *Frontiers in Medicine* assessed fourteen of the most-discussed candidate treatments for long COVID and found that only six of them had any long-COVID-specific randomized-trial evidence at all; the rest rested on mechanism plausibility or case reports [4]. Your navigator position is structural rather than compensatory — the condition genuinely does not yet have a consolidated evidence base, and the person who keeps track of what has worked for your specific body, under what conditions, is filling a gap the literature has not yet filled.

Two of those research arms are worth naming for orientation. The first is the antiviral category — Paxlovid (nirmatrelvir-ritonavir), simnotrelvir, ensitrelvir — being trialled both for reinfection prevention in long-COVID patients and, in a smaller line of work, for treatment of persistent viral fragments in patients who never fully cleared the original infection. Whether any of those compounds lands in routine post-viral care is not yet decided; what matters for navigation is that the trials are running. The second arm is metformin, an old diabetes drug that lowered the rate of subsequent long COVID by about 41% over ten months of follow-up in one large outpatient trial called COVID-OUT — 6.3% of the metformin group against 10.6% on placebo — with a stronger effect among the patients who started it earliest after their symptoms began [5]. That window has closed for anyone already months or years into the illness; the finding is included for completeness rather than action.

A team called Hitch and Wrench, writing about the management of long COVID, put the history itself at the center of the workup:

"While no diagnostic test exists for Long COVID, symptoms are well described and can be elucidated with a careful history. This allows for an early diagnosis, facilitates early management, significantly reduces concerns about other causes of the person's symptoms, and improves patient outcomes." [3]

A careful history is what you are bringing. The pages you wrote in Chapter 9 are the raw material; the twenty-second intro and the SOCRATES grid are how you compress that material to fit a ten-minute window.

Optional self-observation prompt

Write your twenty-second intro tonight. Not in your head. On paper, or in a notes app you can open at the clinic.

Four sentences, nothing more:

1. The trigger and the date.
2. The one or two dominant mechanics.
3. Function, before and now, with at least one number.

4. The single ask you want from the next visit.

Read it out loud. Time yourself. If it runs past twenty-five seconds, cut. If the ask is plural, pick the one that matters most and put the others on a separate list for later.

Bring the page. Slide it across the desk the way Catherine did. If the clinician reads it, you have saved yourself the most expensive minutes of the visit. If the clinician does not read it, you still have it in the room as a reference, and you have the one-pager from Chapter 9 underneath it as backup.

Catherine left her appointment with an order for orthostatic vitals, a referral to a cardiologist who sees dysautonomia, and three words written in her chart: "post-viral syndrome, provisional." Provisional is a label the system can act on; that is the entire function of the word. A label the system can act on is the next useful step, and in a ten-minute visit, the next useful step is the whole job.

Sources

1. Mathew GT, Novak P. "Prevalence of Central Sensitization in Postural Tachycardia Syndrome." *JAMA Network Open*. 2026;9(1):e2553694. DOI: 10.1001/jamanetworkopen.2025.53694 264 of 305 patients with confirmed POTS (86.6%) met criteria for central sensitization syndrome.
2. Shaw BH, Stiles LE, Bourne K, Green EA, Shibao CA, Okamoto LE, Garland EM, Gamboa A, Diedrich A, Raj V, Sheldon RS, Biaggioni I, Robertson D, Raj SR. "The face of postural tachycardia syndrome — insights from a large cross-sectional online community-based survey." *Journal of Internal Medicine*. 2019;286(4):438–448. DOI: 10.1111/joim.12895 Survey of 4,835 people with a physician diagnosis of POTS. 77% (n=3,471) encountered a physician who

suggested their symptoms were due to a psychiatric or psychological problem before their POTS diagnosis; 75% (n=3,421) were misdiagnosed prior to diagnosis; 28% (n=1,247) reported actually having a psychiatric or psychological problem at that time. Median diagnostic delay 24 months (IQR 6–72), mean 4.9 ± 7.1 years. See also Ch 02 ref [4].

3. Hitch D, Wrench J, eds. 2026. *The Rehabilitation and Management of Long COVID: A Handbook for Clinical Practice.* London: Routledge. DOI: 10.4324/9781003528104
4. Baptista SN, Atkins T, Chakraborty S, Bakhit M, Glasziou P, Byambasuren O. "Candidate treatments for long COVID: a narrative review of expert and patient-driven priorities." *Frontiers in Medicine*. 2026;13:1734600. DOI: 10.3389/fmed.2026.1734600 Narrative review of fourteen candidate long-COVID treatments — six with any long-COVID-specific RCT evidence, eight relying on mechanism or observational data.
5. Bramante CT, et al. "Outpatient treatment of COVID-19 and incidence of post-COVID-19 condition over 10 months (COVID-OUT): a multicentre, randomised, quadruple-blind, parallel-group, phase 3 trial." *The Lancet Infectious Diseases*. 2023;23(10):1119–1129. DOI: 10.1016/S1473-3099(23)00299-2 Relative risk reduction of 41.3% at day 300; cumulative incidence 6.3% on metformin against 10.6% on placebo. Enrolment required treatment within three days of symptom onset and seven days of a positive test; a prespecified subgroup starting under four days after symptom onset showed a larger effect, which the trial reports as a direction rather than a separate figure.

Chapter 11. Psychological support ≠ "it's all in your head"

Catherine left the appointment we opened Chapter 10 with carrying three new things: an order for orthostatic vitals, a referral to cardiology, and the phrase *post-viral syndrome, provisional* written into her chart. It was also the first appointment in eight months where a psychology referral was not what the clinician reached for first. That second fact is what this chapter is about.

Psychological support has a place in this illness, and sometimes it has an essential place. The problem is when it is offered *instead of* a workup. Some referrals to therapy will come. Some of them should. The job of this chapter is to help you tell the difference between the kind of psychological support a person carrying a chronic illness actually needs, and the kind that is a soft-lit version of being sent home without help.

Let me start by drawing one line very clearly.

Gaslighting in a cardigan

"Your symptoms are psychosomatic" is medical gaslighting. "You might benefit from therapy" is a normal recommendation for a person who has lost the body they used to live in, the rhythm of their days, their confidence, and a working share of the identity they had a year ago.

The first sentence dismisses you. The second sentence supports you. Clinicians often dress up the first as the second, and that is why the word *therapy* puts so many people with Long COVID, ME/CFS, POTS, and fibromyalgia on guard. They have seen it offered too many times as a replacement for medicine rather than a partner to it.

So when a clinician suggests therapy, ask yourself one question: is this in addition to a medical workup, or instead of it?

If it is instead of a workup, it is a dismissal wearing softer clothing.

Three injuries

The illness is the first injury.

Being disbelieved for months about the illness is the second.

Starting to disbelieve yourself is the third. It is also the most dangerous.

External dismissal can be argued with. You can find a different clinician, bring data (as Catherine did in the last chapter), and keep looking until someone reads the page. But when you stop trusting your own body, the internal search engine shuts down. You stop tracking patterns, cancel the next appointment, and start explaining every symptom as a character flaw. Then you push through things that your body has already taught you not to push through, which leads, reliably, to crashes you could have avoided.

Self-gaslighting sounds very reasonable from the inside. That is how it gets past your guard.

Everyone gets tired.

I’m just out of shape.

Maybe I’m exaggerating.

I looked fine at lunch — it can’t really be that bad.

A fluctuating, invisible illness with delayed symptoms is perfect fuel for those sentences. In PEM especially, the link between effort on Monday and collapse on Wednesday is broken in time, which means the simplest cause-and-effect you usually use to understand your own body doesn’t apply. A person without PEM who walks too far gets sore tomorrow and says *I walked too far yesterday*. A person with PEM who walks too far crashes on day three and cannot always connect the dots. The illness itself manufactures doubt.

Support is not an explanation of cause

Every serious chronic illness carries a psychological cost. Cancer does. Stroke does. Multiple sclerosis does. Nobody listens to a stroke patient describe exhaustion and concludes the blood clot was imaginary.

It is only when medicine lacks a clean, standard test that emotional support starts getting repackaged as an explanation for the cause of the illness. That move depends on a category error: your brain processes every symptom you feel. It also processes a kidney stone. *The brain is involved in the experience of a symptom* and *the cause of the symptom is psychological* are two completely different statements, and a lot of clinical damage comes from collapsing them into one.

This distinction matters a lot in post-infectious illness, where routine labs often come back unremarkable — the CBC is fine, the thyroid panel is fine, the chest x-ray is fine — while standing up takes twenty minutes of recovery, a shower costs half the day, and a trip to the grocery store leaves an invoice on your desk the day after.

A clinic visit is also a poor camera angle on this kind of illness. To get yourself to the appointment, you probably showered, dressed, took your meds, got into a car, sat upright under fluorescent light, and ran on whatever reserve you had left. You spent the best hour of your day just arriving. A clinician with seven minutes to form an impression can easily mistake that performance for your baseline function.

So when a doctor says "you might benefit from therapy," the useful response is a question:

Help with what, exactly? And in addition to what, or instead of what?

If the answer is "help with the grief of what you have lost, the strain on your relationships, the fear of your next crash, and the weight of being disbelieved for months" — that is ordinary, sensible support, and it belongs in the plan.

If the answer is "because your labs are normal and I have nothing else to offer" — that is gaslighting with softer lighting, and you are allowed to decline it.

Being believed changes your biology

Being accurately understood is a form of biological input.

There is a supervisory region in the front of the brain, the medial prefrontal cortex, which helps regulate the body's stress response. Bessel van der Kolk, whose long clinical work with trauma survivors has shaped how many clinicians think about this, describes it as the brain's watchtower — the structure that helps the rest of the system put incoming signals into context rather than treating every signal as a five-alarm fire [1]. When someone gets your state right — when a clinician reads the page, when a friend says *yes, that sounds like how this illness actually works* — the watchtower has less to argue with. Internal static drops. Working memory loosens. You have more access to words, to choices, to the next step.

Neuroscience has recently added a second way to describe this same mechanism, and the two pictures fit together usefully. In that second view, fatigue itself is not just the experience of low energy — it is the brain's running report on its own ability to keep the body in balance. When every effort produces a larger swing than it used to — every flight of stairs, every tense meeting, every normal-sized meal — the brain registers that its forecasts about the body have stopped being reliable. The signal it sends out of that registration, to keep you from pushing through into damage, is the sensation of fatigue. The overwhelmingness is a message from a system that has lost confidence in its own ability to predict what will happen next. Which is why being heard by a clinician can lift the fatigue by a measurable notch within minutes. Nothing has changed about the muscles or the immune system. Something has changed about what the brain thinks it is being asked to manage [5].

This does not cure a dysautonomic tachycardia or repair a mitochondrial enzyme defect. It does something narrower and more practical: it stops the nervous system from spending energy arguing with reality, and when your nervous system is already running near its ceiling, that matters.

Think about a rushed appointment in a specialty clinic. Fluorescent lights. A waiting room that ran long. Twelve seconds of speech before the clinician cuts in. A skeptical face across the desk. You leave feeling worse than when you arrived — and nothing psychological had to be imagined for that to be real. Your body read the room. Your sympathetic nervous system interpreted the signal. Stress worsens asthma, and no one claims asthma is caused by stress. The same is true here. Lowering the alarm reduces suffering in any damaged system, and that is why psychological support helps even when the illness is entirely physical.

This is also, quietly, part of why bringing someone with you to medical appointments tends to make the appointments go better. You can advocate for yourself, and you do. Stress also narrows working memory, and a second person acts as both witness and buffer, so you get to have the conversation with more bandwidth than you would have had alone.

Grief is not depression

A low mood in a chronic illness is, a lot of the time, grief.

You are grieving your health, and more than your health. You are grieving your spontaneity, your career, your sex life, your social life, your identity, and the version of you who could stand up from a chair without thinking twice. That is an accurate reaction to a real loss. Calling it depression and reaching for an antidepressant can be reasonable, but it is often imprecise, and imprecise framing leads to imprecise support.

The distinction has practical meaning. In grief, *wanting* is usually still there. You want to live again, you want your old capacity back, you want the version of Saturday afternoon you remember. You cannot have it right now, which hurts, but the wanting is intact. In clinical depression, the wanting itself goes dark. The desire to do anything at all recedes from the inside, with no body standing in the way.

Both can be true at once — a chronic illness often produces both, in layers — but the right support depends on which layer is active. Grief responds to being witnessed, to being allowed to name what is gone, and to slow practical adjustment. Clinical depression often needs a more direct intervention.

Anger is worth naming here too, because it has nowhere to land. There is no one you can reasonably be angry at: not the virus, which does not care; not always a specific clinician, because many of them are doing their best inside a system that is genuinely difficult; not the insurance company, though some of them invite the feeling. And so the anger — which is accurate and appropriate — comes home and finds the nearest target, which is usually you. That is why so many patients with this kind of illness are harder on themselves than any outside critic. The anger needed an address and it found the one closest to hand.

A large part of what good psychological work does, early on, is give that anger a different address — somewhere to land that is not you.

A program that will not switch off

In Chapter 7 I described sickness behavior: the coordinated behavioural program that runs when your body is fighting an acute infection. You slow down, you sleep more, you lose appetite, you withdraw from the group. Pro-inflammatory cytokines act on the brain and produce a state that feels exactly like being told *lie down, do not come to the well, do not eat with the others today*. The evolutionary point of this program, as I explained in the food chapter, is kin protection — the sick animal that isolates itself does not infect the others, and that benefit was large enough across evolutionary time that the cost to the individual was tolerable [2].

In an acute infection, the program runs for a few days and then switches off. In a post-infectious illness like Long COVID or ME/CFS, that switch can jam in the on position. Immune signaling keeps telling the brain *you are sick, stay down*, long after the acute infection has cleared. The result from the outside looks a lot like depression. From the inside it feels like depression. The mechanism, though, is immunological, not psychological.

This is one of the reasons medicine regularly tangles itself up here. What a person with Long COVID feels, sitting at home on a bad afternoon, can be a combination of inflammation, disrupted sleep, pain, reduced blood flow to the brain, and social isolation, all at once. Each of those alone can produce a state that maps onto depression. Together they produce something that looks textbook and responds poorly to a textbook reading.

There is direct clinical evidence that immune signals can produce depression on their own. Patients receiving interferon-alpha as therapy for viral hepatitis develop clinical depression at striking rates — a large share of them, depending on the study — and those rates reflect immune molecules acting directly on the brain, changing mood by the same routes that a depressive episode runs along [3]. In Long COVID, chronic low-grade inflammation produces a quieter, mechanistically related picture. Sometimes the depression-looking state is itself part of the illness.

There is also, now, a way of seeing this from outside the patient. A specific kind of brain scan — a PET scan tuned to pick up the nervous system's own immune cells when they are activated — has begun to make the quiet part visible. In people who had COVID and did not come back to baseline, some show raised low-grade inflammation on these scans two years after the infection, concentrated in the regions that handle attention, memory, and mood [6]. The part that matters most is who shows it. In a cohort of forty-seven people scanned at the two-year mark, the raised signal appeared in part of the group rather than across it, varied widely from one person to the next, and did not track how severe anyone's symptoms were [6]. In a second cohort, ten of forty-five showed it [7]. What that describes is an inflammatory subtype: a real mechanism running in some bodies, which a clean scan does not rule out and a loud scan does not measure. The technique itself is also young. TSPO, the protein these scans bind to, sits on more than one kind of cell — microglia carry it, so do astrocytes and the lining of small vessels — so a bright scan reports immune activity in the tissue without telling you which cell is doing the work [8]. Held together, that is enough to say, with sources behind the sentence, that the brain-level part of this illness is not a metaphor. The inflammation your nervous system is reading signals from is being picked up, in some of these patients, by a machine.

There is a second line of imaging that fits next to this one. A different kind of PET scan — one that measures how much fuel the brain is actually burning, by tracking glucose uptake — has picked up a matching pattern in these patients: regions of the brain running at a reduced rate, and doing so at a level that does not fully line up with how much blood is getting there. The first systematic description of the pattern came from a 2021 cohort scanned weeks to months after their initial infection [9]; subsequent work, including a 2025 review in *Immunometabolism*, has found the same hypometabolic signature in independent cohorts and at follow-up windows out beyond a year [10]. In plain language: the fuel is arriving at the cell, and the cell is not using it well. Chapter 1 walked through this pattern on the muscle side — oxygen available, extraction impaired. The brain appears to show its own version of the same problem.

There is a second way of thinking about all of this that is worth holding alongside the kin-protection frame from the food chapter. The kin-protection account is a story about why this program exists at all, written on an evolutionary timescale. There is a parallel story, written on the timescale of an afternoon, about how the same program operates inside a single body. In that story, the immune system and the brain are running their own energy budgets, and when both are stretched thin, the available energy gets routed toward whichever process the body reads as most necessary for staying functional. Fatigue, in this view, is what a body looks like when it is making real-time trade-offs between competing protective priorities. The point of naming this is not to assign blame — to your body, to your habits, to anything — but to recognise a system doing its best inside constraints it did not choose.

This is where a useful self-question comes in. When your mood is heavy and your day is flat, ask:

Do I want to do things but my body is blocking me? Or has the wanting itself gone out?

The first answer points toward sickness behavior and physical limits and grief. The second answer points toward depression that needs its own seat at the table. Both can be true at once. But the shape of the support you need is different depending on which one is driving.

Brain retraining: a tool, not a religion

Neuroplasticity programs sit in an awkward place in the Long COVID conversation. Some people overpraise them. Some people dismiss them too quickly. Serious randomized controlled trials remain thin on the ground.

The two most talked-about are DNRS (the Dynamic Neural Retraining System) and the Gupta Programme. Both rest on a similar idea: the limbic system — the threat-detection machinery of the brain — can, after an infection or an injury, get stuck in a posture of over-reactivity, and repetitive exercises (visualisations, attention redirection, very gradual expansion of activity) can, over time, retrain that pattern.

Here is the honest dual truth. Some people do appear to recover using these programs. The high-quality evidence base is weak. Both of those statements can be true at the same time, and the responsible thing is to hold them together rather than picking whichever one matches your mood.

At the time of writing, neither program has been through a rigorous randomized controlled trial. The cost can be substantial — several hundred dollars for an online course, sometimes more. If you are considering this approach, treat it as an experiment: set the success criteria in advance, set the timeframe in advance, and decide how long you will run the test before you evaluate. That is not the same thing as committing your identity to it.

If your illness clearly includes a hyper-reactive threat loop — the body lighting up at the smallest sensation, sensory overload triggering crashes, a persistent sense of being in danger that has no matching external threat — then a program that targets that loop may help. The nervous system is a physical structure, and physical structures can be trained.

Where this approach becomes dangerous is the point where the phrase *this might help* is slid over into *this is all in your head*. That slide is lazy, and it is cruel, and it confuses a tool with a diagnosis.

Brain retraining is not a replacement for pacing, sodium, medication, sleep work, or medical workup. It will not dissolve a microclot or clear a viral reservoir by force of attention. If your thermostat is stuck, sometimes working with the nervous system is one way to move the dial. If you try it, do not let anyone talk you into pushing through a crash on the grounds that the crash is a sign of progress. And if a month or two goes by without a real change — without, for example, a measurable improvement in how long you can stand before your pulse jumps — it is not your tool.

Choosing a therapist who understands the biology

A therapist is a tool, not a worldview. You are allowed to be picky.

A good therapist does not need to be a Long COVID specialist from the first session. They do need to understand a small set of non-negotiables:

The illness is real and has a physiological basis.

Pacing is the management of a physiological ceiling. A therapist who treats it as avoidance will push you toward a crash.

Symptom relief is not the same thing as a psychological origin for the symptom. The fact that a slow walk in the garden helps does not mean the illness was psychological to begin with.

Push-through advice is dangerous for a person with PEM. A therapist who treats pacing like a phobia, and pushes for *exposure to activity* as a way of building resilience, can push you into a crash by calling it courage.

A good therapist sees the line between avoidance (fear narrowing the world) and pacing (respecting a physiological limit). That line matters. If they do not see it, they will pressure you in the wrong direction at exactly the wrong moment.

You can set the tone in a first session with something like this:

I have a systemic post-viral illness. I want help with grief and with the impact of being disbelieved for months. I do not want my physical symptoms reframed as anxiety.

If that makes the therapist defensive, you have just saved yourself months and several thousand dollars.

Cognitive behavioural therapy for adapting to a chronic illness, for working with grief, for calming a genuine anxiety loop, is reasonable and often helpful. The problem comes when CBT is applied with the underlying belief that *wrong thoughts caused this illness* and that *changing thoughts will cure it*. In that form, it is GET with a therapist’s office instead of a treadmill.

Peer support: borrowed reality

Sometimes the first person who understands what is happening to you is another patient.

Patient communities have played a role in Long COVID that is hard to overstate. Long COVID was named by patients, in a Twitter thread, before medicine had a box to put it in. The repeating patterns of this illness — the grocery-store stack of noise and standing and light, the Tuesday crash after Sunday’s brunch, the three-day hangover after one cognitively hard meeting — were mapped by patients talking to each other, often long before a clinician could recognise them as a recognisable syndrome. Callard and Perego, writing in 2021, argued that patients did not just describe this illness; they helped make it visible as an illness in the first place [4].

What peer support does, psychologically, is return your own reality to you. When ten other people describe the exact shape of your Tuesday — *yes, the grocery store is harder than a short walk, because it stacks standing and noise and light and decision-making* — your nervous system stops asking whether you made the pattern up. That is borrowed reality. It is a real psychological function. Another person’s experience that lines up with yours is data. It affects how accurately you track your symptoms, how confidently you describe them to a clinician, and how quickly you spot a trap like a good day that is about to become a bad week.

With one caveat worth keeping.

Not every patient community is equally useful. A community that helps people trade patterns, share tools, and find clinicians is a resource. A community that cultivates despair, or that sells unverified protocols at a markup, is not. Choose the environment as carefully as you would choose a doctor. An hour in the wrong community can undo a week of careful work on your own head.

Give your people a dashboard, not a lecture

Invisible illness breaks relationships through a gap in the model. The person who loves you sees you laughing at dinner and cannot reconcile that with you being unable to stand up tomorrow. What is missing is a working picture of what this illness does to a body.

Give them a working picture — not a lecture on cytokines and cerebral blood flow.

The next chapter walks through three metaphors in detail: the battery, the tokens, and the usable hours. Each of them makes the invisible part of the illness legible to someone who is not inside it. What matters, whichever you use, is that the metaphor includes recovery time. Your capacity is measured by the cost you pay the day after, not by how you look while you are using it.

And please, stop using your own smile as evidence against yourself. A smile at dinner is proof that a human being can, for forty minutes, pull together enough reserve to show up at a table. It is what living with the illness looks like.

One last distinction

Psychological support is for the human being carrying the illness, and it has nothing to say about what caused the illness.

You can have inflammation and pain and grief and anxiety all at once. That is what a complicated chronic illness looks like from the inside, and the right kind of psychological support will never ask you to stop believing your body in order to get it.

It will just help you stop abandoning your body on the way.

*If you or someone you love is in psychological crisis, call or text **988**** (the Suicide & Crisis Lifeline). It runs twenty-four hours a day, it is free, it is confidential, and it is available by phone, text, and online chat at 988lifeline.org.*

Optional self-observation prompt

Pick one metaphor from the next chapter — battery, tokens, or usable hours — and explain it to one person in your life. Not the whole story of the illness. Just the metaphor.

Then watch what happens to the conversation when the other person has a working model instead of a set of guesses.

Sources

1. van der Kolk B., 2014. *The Body Keeps the Score: Brain, Mind, and Body in the Healing of Trauma.* New York: Viking. The medial prefrontal cortex is described here as the brain’s “watchtower” — a regulatory region that helps the stress response come back into proportion when a person feels accurately understood.
2. Dantzer R., 2001. “Cytokine-induced sickness behavior: mechanisms and implications.” *Annals of the New York Academy of Sciences* 933: 222–234. DOI: 10.1111/j.1749-6632.2001.tb05827.x Sickness behavior framed as a coordinated behavioural program driven by pro-inflammatory cytokines, with an evolutionary function of reducing transmission within the group.

3. Capuron L., Miller A.H., 2011. "Immune system to brain signaling: neuropsychopharmacological implications." *Pharmacology & Therapeutics* 130: 226–238. DOI: 10.1016/j.pharmthera.2011.01.014 Direct effects of cytokines and interferon therapy on brain function, including depressive states as a neurobiological consequence of immune activation.
4. Callard F., Perego E., 2021. "How and why patients made Long Covid." *Social Science & Medicine* 268: 113426. DOI: 10.1016/j.socscimed.2020.113426 Patient communities as a site of mutual validation and collective mapping of a novel illness.
5. Stephan KE, Manjaly ZM, Mathys CD, Weber LAE, Paliwal S, Gard T, Tittgemeyer M, Fleming SM, Haker H, Seth AK, Petzschner FH. "Allostatic Self-efficacy: A Metacognitive Theory of Dyshomeostasis-Induced Fatigue and Depression." *Frontiers in Human Neuroscience*. 2016;10:550. DOI: 10.3389/fnhum.2016.00550 Predictive-coding model in which fatigue functions as the brain's signal of low confidence in its forecasts about bodily homeostasis.
6. Visser D, Golla SSV, Palard-Novello X, Verfaillie SCJ, Verveen A, Koch DW, Rikken RM, van de Giessen E, Boellaard R, Knoop H, Tolboom N, et al. "Varying Levels of Inflammatory Activity in Brain and Body of Patients with Persistent Fatigue and Difficulty Concentrating After COVID-19: A TSPO PET Study." *Journal of Nuclear Medicine*. 2025;66(11):1787–1794. DOI: 10.2967/jnumed.124.269297 Forty-seven people scanned 25–27 months after infection with dynamic [^{18}F]DPA-714 PET and arterial sampling, all high-affinity TSPO binders by genotype. Raised cerebral binding appeared in a subset of those reporting severe fatigue and difficulty concentrating, with wide between-person variability and no correlation between binding level and symptom severity. The authors describe the finding as an inflammatory subtype.

7. Visser D, Pieperhoff L, Lorenzini L, Golla SSV, Verfaillie SCJ, Verveen A, et al. “Decreased functional connectivity in post-COVID syndrome patients with high neuroinflammatory activity.” *NeuroImage*. 2026;333:121951. DOI: 10.1016/j.neuroimage.2026.121951 Ten of forty-five participants (22%) showed neuroinflammation on [^{18}F]DPA-714 PET. **And:** VanElzakker MB, Bues HF, Brusaferri L, Kim M, Saadi D, Ratai EM, Dougherty DD, Loggia ML. “Neuroinflammation in post-acute sequelae of COVID-19 (PASC) as assessed by [^{11}C]PBR28 PET correlates with vascular disease measures.” *Brain, Behavior, and Immunity*. 2024;119:713–723. DOI: 10.1016/j.bbi.2024.04.015 Independent confirmation with a different tracer — twelve PASC patients against forty-three controls — with raised binding correlating with circulating markers of vascular dysfunction.
8. Nieuwland JM, Nutma E, Philippens IHCHM, Böszörményi KP, Remarque EJ, Bakker J, Meijer L, Woerdman N, Fagrouch ZC, Verstrepen BE, Langermans JAM, Verschoor EJ, Windhorst AD, Bontrop RE, de Vries HE, Stammes MA, Middeldorp J. “Longitudinal positron emission tomography and postmortem analysis reveals widespread neuroinflammation in SARS-CoV-2 infected rhesus macaques.” *Journal of Neuroinflammation*. 2023;20(1):179. DOI: 10.1186/s12974-023-02857-z Animal model, four rhesus macaques. Postmortem staining found TSPO expressed by microglia ($IBA1^+$), astrocytes ($GFAP^+$), and the walls of small vessels (collagen IV^+) — the basis for reading a TSPO signal as immune activity in tissue rather than as microglial activation specifically.
9. Guedj E, Campion JY, Dudouet P, Kaphan E, Bregeon F, Tissot-Dupont H, Guis S, Barthelemy F, Habert P, Ceccaldi M, Million M, Raoult D, Cammilleri S, Eldin C. “^{18}F-FDG brain PET hypometabolism in patients with long COVID.” *European Journal of Nuclear*

Medicine and Molecular Imaging. 2021;48(9):2823–2833. DOI: 10.1007/s00259-021-05215-4 First systematic FDG-PET cohort describing a characteristic hypometabolic pattern in long COVID, concentrated in olfactory and limbic regions.

10. Zhu Y, Quan P, Yamazaki T, Norweg A, Natelson B, Xu X. "Metabolic neuroimaging of myalgic encephalomyelitis/chronic fatigue syndrome and Long-COVID." *Immunometabolism*. 2025;7(4):e00068. DOI: 10.1097/IN9.0000000000000068 Mini-review of PET and MRI evidence on cerebral metabolism in post-acute infection syndromes — confirms the FDG-PET signal across multiple independent cohorts and at follow-up windows beyond one year, with discussion of the metabolic substrate (oxygen-extraction and mitochondrial pathways) underlying the imaging finding.

Chapter 12. Explaining to those around you

You sit at the kitchen table on a Friday evening, eat a normal meal, and carry a normal conversation. The person sitting across from you relaxes, relieved that you are finally looking like yourself again. Then Saturday morning arrives, the bill for Friday comes due, and you cannot get out of bed. The person who watched you laugh twelve hours earlier stands in the doorway holding a cup of coffee, trying to square the ghost in the bed with the person at the table.

That confusion comes from a lack of context rather than a lack of care. They are working from a rulebook with two settings: sick in bed, or fine. The bond between you is intact, and the lens they are looking through simply has no category for a condition where capacity swings from one morning to the next. They see a good evening and assume the storm has passed.

That missing framework creates a vacuum in the middle of the house. You feel dismissed when they suggest pushing through the fatigue, and they feel helpless when their attempts to encourage you hit a wall. Without a shared language for what is actually happening, bitterness begins to build in both directions.

Invisible illness damages relationships through a gap in models. The people around you cannot see the illness, and they do not have a working picture for it, so they fall back on the pictures they do have: sick or healthy, trying or not trying, real or exaggerated. They watch you laugh at a party and cannot square it with you being flat the next day. “Bad day” lands in their ear with no way to judge how bad, or why. Help is what they want to offer, and they have no idea how. In that vacuum, resentment starts to form on both sides.

The repair is small and specific. Put a working picture in their hands — something they can use in a real moment, when you say no to a dinner, or come home early, or crash after an ordinary Tuesday. That picture is the subject of this chapter.

Why "just explain" doesn't work

Most attempts to explain a chronic, invisible illness end one of two ways.

The first is over-explanation. Immune cascades, mitochondria, dysautonomia, PEM. The other person nods the whole time, and then a week later asks "can you just come for an hour?" again. The information didn't turn into understanding.

The second is under-explanation. *I feel bad. I'm tired. I don't have the energy.* Every word of that is true, and none of it gives the other person anything to work with the next time they are planning a weekend.

The problem isn't the amount of information. Information without a frame does not become understanding. What your people need is a map — something that fits in the pocket and can be pulled out in a decision moment, when a question about dinner or a weekend or a trip shows up.

Three working pictures

A good metaphor does the work a lecture can't. It doesn't try to explain the mechanism; it gives the other person a model they can actually use when the next decision shows up. Pick the one that fits your household, your job, or the person you are trying to reach.

Readers who have been in patient communities for a while will recognise a shared intellectual ancestor here. In 2003, Christine Miserandino wrote an essay at a diner table using handfuls of spoons to explain lupus to a friend, and that short piece — Spoon Theory — is still one of the most widely shared descriptions of what a limited energy budget looks like from the inside [1]. Patient-built tools like this one have done a great deal of the early, practical work of making invisible illness legible, long before formal medicine caught up with the vocabulary [2]. The frames below use different objects because different objects land with different listeners. The arithmetic is Miserandino's.

The battery. "I don't wake up at one hundred percent. Some days my day starts at forty, and that's a fact I cannot argue with. Heat, stress, standing, and noise drain me faster than it looks from the outside. And unlike a phone — I do not always recharge overnight. Sometimes a bad day means the next morning starts at twenty."

This image works because everyone already knows what a low battery means. A phone at six percent cannot do what a full one can, no matter what you want from it, and the word "laziness" never enters the conversation.

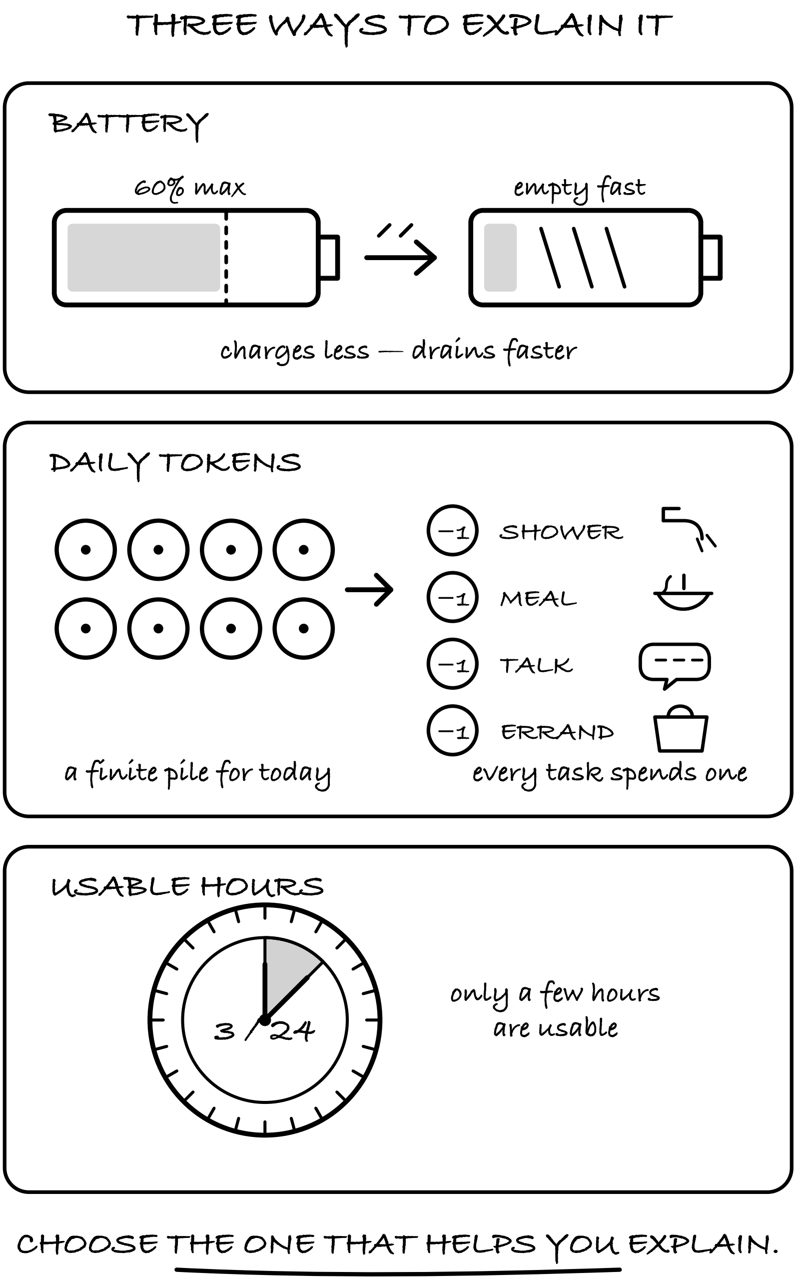
The tokens. "I get a limited number of tokens each day — call it ten. A shower costs two. A phone call costs one. A stressful conversation costs three. A trip somewhere costs four. If I spend twelve, I'm borrowing from tomorrow, and tomorrow I won't have ten; I'll have seven, or five. When I say no, I'm not saying no to you. I'm protecting tomorrow."

The last line — *I'm protecting tomorrow* — often changes how the refusal lands. It reframes saying no from withdrawal into resource management.

Dan was trying. You could see it. He had started quietly googling Catherine's symptoms at night, he was taking over more of the house, and he had stopped planning trips without warning. And sometimes — not often, but often enough to hurt — he would say something like *you seemed fine yesterday*, or *maybe try just going outside for a bit of air*. None of it was cruelty. He had no working model of her illness, and without one the only thing left available to him was encouragement.

The tokens gave him one.

The usable hours. “I have about three usable hours in a day — hours where I can think clearly, move reasonably, and hold a conversation. The rest of the day is either rest that keeps those hours possible, or recovery from whatever used them up. It doesn’t mean I suffer for the other twenty-one hours. It means I have a very small pool of active time, and I have to choose what to spend it on.”



Three ways to say it. Use whichever one lands with the person in front of you.

This one travels especially well at work, because it translates the illness into the same language managers already use to plan a week.

All three metaphors carry one idea the lecture version keeps losing: the cost is paid later. Capacity is measured by the bill that arrives the next day. How you look while you are spending is the loudest signal and also the most misleading one.

The delayed invoice is the hard part

Of everything your people will need to understand, the delayed bill is the most counterintuitive, because it breaks the visible link between cause and effect.

You laugh through dinner on Friday. On Saturday you can't get out of bed. The person watching sees two separate events — a normal Friday and a bad Saturday — and has to make sense of the gap. From the outside the gap looks like unpredictability, or sometimes even performance.

From the inside, Friday and Saturday are one event with a delayed payment.

The short script: "When you see me doing well, that is not the same as being well in the healthy sense of the word. It means I am spending a resource I will be billed for later. Like a credit card — you can use it now, but the statement comes tomorrow. If I can't get up on Saturday, Friday wasn't a mistake. Saturday is just the statement arriving."

One extra sentence helps: "When you see me struggling, don't look for what happened yesterday. Look at what happened the day before."

Good days, and what they are not

Good days become their own trap inside a relationship. Someone close to you sees a good day and starts hoping this is it — the illness turning the corner. When the next bad day comes, it reads as a setback, or inconsistency, or worse, proof that the bad days were exaggerated.

It helps to name the mechanism in advance, before a good day shows up. "When I have a good day, that's good news, but not the news you're thinking. It doesn't mean I've recovered. It means I have a little more resource today. That's exactly why, on a good day, I'll try to spend less of it than it feels like I could — so today doesn't turn into a bad week."

Have this conversation on a good day. On a bad day you don't have the bandwidth to hold it; on a good day you do. It is one of the few situations in which your own good days have a useful job to do for you.

Work is a different conversation

Talking with a manager, HR, or a colleague follows a different set of rules than talking with a partner. There is less room for emotion and more room for functional description, which is actually good news — functional language is the language your workplace already runs on.

Three principles help.

Speak in functional limits. "I need to limit time on my feet" travels further in an office than "I have POTS." "I need recovery time scheduled after intensive meetings" is more useful than "I have PEM." Your workplace runs on what you can and cannot do on a Tuesday, and that is the level you need to meet them at.

Be concrete in your asks. Skip the phrase "I need accommodations" and say something specific instead: "I need permission to sit during meetings," or "I need meetings spaced out rather than stacked back-to-back," or "I need the option of working from home after days with high physical load."

Put the agreement in writing. Write it down because brain fog makes spoken agreements unreliable on your end. The reason has nothing to do with trust; it has to do with cognitive bandwidth on hard days. A short email after the conversation — *Confirming what we discussed today* — protects both sides.

In the United States, the Americans with Disabilities Act requires employers with fifteen or more employees to provide reasonable accommodations — changes in working conditions or schedule — for a documented medical condition [3]. In 2021, the Department of Justice clarified that Long COVID can qualify as a disability under the ADA where it substantially limits a major life activity [4]. That gives you standing to ask, and an employer covered by the law has an obligation to consider the request in good faith. Case law in this area is still forming, so check current guidance at eeoc.gov when you file. If you are outside the US, an equivalent workplace-adjustment right probably exists under your local law; the framing above transfers even when the statute doesn't.

When someone close will not meet you there

Not everyone in your life will do the work of learning a new model. Some people will keep saying *you're just stressed*, or *maybe you should move more*, no matter how many times you explain.

That is painful. And it is real.

A few things are worth accepting.

You cannot make a person understand something they are unwilling to understand. Your responsibility is to give them the tools. What they do with the tools is theirs.

Chronic illness tends to reveal the shape of a relationship; it rarely invents a problem that was not already present. When someone close to you responds to your illness with dismissiveness or impatience, you are learning something real about the relationship, and the information matters even when it is hard.

One more thing worth saying plainly. In Chapter 11 I described a second injury that sits on top of the illness itself — being disbelieved. A person close to you who refuses to update their model is adding to that second injury, quietly and sometimes without meaning to. Recognising that as what it is helps you stop carrying the weight of trying to convince them one more time.

Your job here is to build enough understanding around you that support is available where it matters. Converting every person in your circle was never on the list. Sometimes the practical move is shifting attention toward people who are willing to learn, and lowering your expectations of the ones who are not. That is triage, and it is allowed.

A short letter for the harder cases

Sometimes a spoken conversation is the wrong medium. Brain fog gets in the way. An emotional moment narrows your working memory. The person in front of you reacts before you have finished a sentence. A written letter — composed in a calm window, read in their own time — often lands in a way the same words in person do not.

Here is a template. Adapt it to the person you are writing to, and to how much or how little they already know.

I want to help you understand what's happening with me — not because I need your sympathy, but because I want us to be able to plan our life together in a way that works.

I have a limited energy budget, and it's smaller than it looks from the outside. When I say no to something, or ask for rest, I am not pulling away from you. I am managing the resource that lets me participate in our life at all.

Feeling well in a moment does not mean I have recovered. Good days and bad days are not random. They are connected to what happened forty-eight to seventy-two hours before.

I don't need you to understand everything. I need you to trust me when I tell you what I can and cannot do right now.

That kind of letter works because it gives information in a form the other person can sit with. A letter written in a calm hour often reaches people more clearly than ten conversations in a bad one.

Receiving help doesn't create a debt

The last conversation of this chapter — possibly the hardest one — is with yourself.

Chronic illness changes the arithmetic of a relationship. You give less than you used to. You lean on other people more than you are used to. For people who have always been the dependable one, the person others leaned on, this is a strange and uncomfortable shift.

Something worth holding onto: accepting help is part of how a relationship functions when one person is carrying an illness, and it carries no hidden tab back to you. Your job inside the relationship right now is honesty about what is happening, clarity in what you are asking for, and a willingness to receive support without apologising for needing it.

Your value in a relationship was never measured by how many things you could do while standing up. It isn't now either.

One more direction, and it goes the other way. The person you are explaining this to — the partner, the parent, the close friend absorbing the load — is also inside this illness now, even though they are not the one carrying it. Long stretches next to a chronic condition carry their own cost: isolation from the social rhythms that used to be shared, financial strain, an eroded ability to rest. There are two people in this household now, and only one of them is the patient. That does not shift work back onto your side of the ledger; your partner may also need a break that belongs to them, or peer support among others who are watching someone they love navigate the same terrain [5].

Intimacy runs on the same budget

This topic rarely gets written about, and it matters to most people living with these conditions inside a partnership.

Sex is a physical load, an orthostatic load, and an emotional load all at once. Under POTS, some positions — the vertical ones especially — are considerably costlier than others. Under PEM the delayed invoice applies here too, and an intimate evening can quietly cost the next morning.

The practical frame is the same as any other activity that draws from the budget. Time of day matters; many people with these conditions feel better in the evening than in the morning. Horizontal positions reduce the orthostatic cost. Hydration before and after is as ordinary a preparation as it would be before any other physically demanding activity. And — please — talk about it with your partner. Silence creates more distance in a relationship than an honest conversation about the fact that you have limits, and that desire is still there.

Optional self-observation prompt

Take the letter above. Adapt it for one person in your life — a partner, a parent, a close friend, or a manager. Just the letter, one page, nothing else attached.

Send it.

Then watch what happens to the next conversation, once the person on the other end has a working model instead of a set of guesses.

Sources

1. Miserandino C., 2003. “The Spoon Theory.” butyoudontlooksick.com. The original essay that introduced the idea of a fixed, countable daily energy allowance to chronic-illness

communities. The metaphors in this chapter use different objects; the arithmetic is Miserandino's.

2. Callard F., Perego E., 2021. "How and why patients made Long Covid." *Social Science & Medicine* 268: 113426. DOI: 10.1016/j.socscimed.2020.113426 Patient communities as the origin of the name "Long COVID" and of the initial collective mapping of the illness; cited here in support of patient-led tool-building as the place where much of the early language for invisible chronic illness was worked out.
3. U.S. Equal Employment Opportunity Commission. *Americans with Disabilities Act*, 1990, as amended. Employers with fifteen or more employees are required to provide reasonable accommodations — changes in working conditions or schedule — for a documented medical condition. https://www.eeoc.gov/statutes/titles-i-and-v-americans-disabilities-act-1990-ada
4. U.S. Department of Justice, Civil Rights Division, and U.S. Department of Health and Human Services, Office for Civil Rights, 2021. "Guidance on 'Long COVID' as a Disability Under the ADA, Section 504, and Section 1557." Long COVID can qualify as a disability under the ADA where it substantially limits a major life activity. Current employment guidance is maintained by the EEOC at eeoc.gov. https://www.hhs.gov/civil-rights/for-providers/civil-rights-covid19/guidance-long-covid-disability/index.html
5. Bateman Horne Center. *ME/CFS & Long COVID Clinical Care Guide*, §17 "Caregiving." Documents the three main load components for household caregivers of patients with energy-limiting chronic illness — social isolation, financial strain, and eroded capacity for recovery — and points to caregiver peer-support structures as an appropriate intervention. https://batemanhornecenter.org/clinical-care-guide/

Conclusion. Recovery as a project

The vocabulary of healing is full of passive verbs. People tell you to rest, give it time, and let the illness run its course. That language assumes your body already knows how to fix the problem and only needs you to stay out of the way. When a nervous system is stuck in a defensive loop, waiting it out does not move the dial. Rest matters here as much as anything in this book, and it works as a deliberate instrument rather than as an absence of activity. The alarms keep ringing, and the cost of a short errand still arrives two days later.

Applying the word "recovery" to an energy-limiting condition demands an active stance. The pacing, the hydration habits, the diary — everything this book set up is something you do. You are constructing a stable floor under your current reality so you can walk to the mailbox or stand at the stove without losing the rest of the week. This project is about taking control of the daily inputs you give your body rather than waiting for your old baseline to reappear.

What recovery actually looks like

The word *recovery* is borrowed from acute medicine. You get the flu, you recover. You break an ankle, you recover. In both cases there is a before, a during, and an after, and the after looks essentially like the before. The curve is a U.

The conditions this book is about do not follow that curve. The Long COVID trajectory literature that has accumulated since 2022 — European cohort studies, the RECOVER program data, and the growing body of work out of the Bateman Horne Center and other specialist clinics — converges on a shape with a wide distribution: some people reach a stable plateau inside the first year, some are still climbing at three years, and a meaningful minority remain in the fluctuating phase for longer. The SARS-1 long-horizon follow-up literature from Hong Kong and Toronto, tracking patients for three to four years after the 2003 outbreak, describes a structurally similar pattern: most patients improved meaningfully over one to three years, a subset stabilized at a reduced baseline, and the best predictor of trajectory was whether they were protected from repeated crashes during the climb.

Neither dataset describes a U. They describe a slope with switchbacks.

I want to be plain about what this means for you. If you have been sick for six months, the odds are that your functional ceiling a year from now will be meaningfully higher than it is today, provided you stop crashing. If you have been sick for two years, the climb is slower and the switchbacks are more pronounced, but it is still a climb. If you have been sick longer — and some readers of this book have — the work shifts from regaining a former ceiling toward widening the floor. The floor is where life happens. The floor is what “a good Tuesday evening” is made of.

“Recovery” in this book means building a life that is good inside the current constraints while the constraints themselves slowly change — rather than a return trip to the person you were before.

That is a project, not an event.

The Stockdale Paradox

In 1965 James Stockdale was shot down over Vietnam and spent seven years as a prisoner of war. When Jim Collins interviewed him for *Good to Great* in 2001, Collins asked him how he had survived when others around him had not. Stockdale’s answer became one of the most quoted lines in modern management literature, and it has traveled well outside that field because it describes something real about long-horizon endurance.

Stockdale said the prisoners who died first were the optimists. They were the ones who told themselves they would be out by Christmas. Then Christmas came and they were still there, and they told themselves they would be out by Easter. Easter came. Then the Fourth of July. They died, Stockdale said, of a broken heart.

What saved him was a different mental posture. He held two things at once: unwavering faith that he would eventually prevail, and the discipline to confront the most brutal facts of his current reality. He held no forecast for when he would get out, refused to pretend the situation was better than it was, and kept intact the underlying conviction that he would, in the end, walk out of there.

Collins called this the Stockdale Paradox. Two things, held together, neither one giving way.

The reason I am quoting a POW story at the end of a medical book is that the shape of the mental task is the same. You are being asked to hold unwavering faith — that your thermostat can be renegotiated, that your floor can widen, that you can build a life that is good — while looking clearly at today, which may still be very hard. The readers who run into the deepest trouble are the ones who cannot hold both. They either collapse into the brutal facts and give up, or they escape into an optimism that the body keeps disappointing, and each disappointment costs them something they can't afford to lose.

Hold both. Today is what it is. The climb is real.

One lever for tomorrow

I have spent twelve chapters handing you tools. Pacing. Orthostatic protocols. Food as a modulator. Sleep architecture. HRV as a signal. Scripts for doctors. Scripts for family. A way to think about medication. A way to think about the nervous system.

If you try to install all of them at once you will fail, and the failure will feel like the illness winning. That kind of failure is a volume problem — the illness is not the agent here.

Pick one lever for tomorrow. Not five. Not a program. One.

The lever should be small enough that you can do it on your worst day of the next two weeks, not your best. If pacing is the lever, pacing means one thing — a planned rest before a predictable trigger, say. If sleep is the lever, it means one thing — phone out of the bedroom, or a hard stop on screens at a specific hour. If food is the lever, it means one thing — protein at breakfast, or removing one specific trigger. If the lever is "breathing before standing up," that is enough.

Do it for two weeks. Measure whatever you are already measuring. If the lever is working, your HRV trend, your step count, your payback patterns, or your subjective energy on a 1–10 scale will tell you something. If it is not working, you have lost two weeks and learned one fact. If it is working, you add a second lever and hold the first.

Two weeks. One lever. Repeat.

This is how the book becomes a life.

You are a field researcher

Somewhere around month four or month five — it is different for everyone — something quiet happens. You become the person in the room who knows the most about your own physiology: more than the specialist who saw you once for eleven minutes, more than the husband who loves you and is doing his best, more than the mother who remembers you as you were at fifteen.

You have data on yourself that nobody else has: a baseline heart rate trend, a standing-test record, a payback map, a trigger list, a sleep-architecture file, a food response log, and above all a year of lived attention to a single nervous system operating under load. No physician on Earth has more data on your particular case, because no physician on Earth has been in the room for all of it.

This is the working vantage point from which the next phase of your illness is managed.

Hold that position seriously. You are a field researcher running a single-subject longitudinal study with imperfect instruments, a small budget, and the most important question in the world — how does this body of mine regain ground? The results of that study do not need to be publishable. They are for you and the people you love, and eventually, if you want, for the next person who walks the same road and needs a map.

The broader patient community is already doing this at scale. The RECOVER initiative's patient-partnered research design — in which patient representatives sit on the panels that set research priorities — has drawn heavily on survey work organized by patient-led groups. Much of what the field knows about Post-Exertional Malaise as a phenomenon was characterized first by patients, then confirmed by cardiopulmonary exercise testing in specialist clinics. The patient community has become, in aggregate, a major source of research questions in this field. You are already one of those researchers, whether or not you ever fill out a form.

The people who stay

By the time you reach the end of this book you know who in your life updated their mental model of you and who did not. You have done a hard triage. You have written at least one letter. You have negotiated at least one workplace conversation. You have built a small and tested circle.

I want to say something about the small circle.

The people who stayed — who learned the tokens, who adjusted to canceled plans without collecting it as a grievance, who asked "what would help" instead of "what's wrong with you" — are part of your recovery project, not an accessory to it. A stable relationship with one person who has updated their model of you is, in the data we have on chronic illness trajectories, a stronger predictor of long-term functional outcome than most individual interventions. The mechanism is the isolation loop: being inside a trusted relationship lowers the cumulative stressor load on a system that is already carrying too much.

Dan learned to ask "how many tokens do you have?" before naming a plan. He did not get it right the first time, or the fifth. He got it right consistently around month seven. That question is now part of how the household operates. Catherine's youngest son, without anyone prompting him, asked her the same question last week before suggesting a movie. She said five. They watched the first half, paused, and finished it the next night.

On an ordinary Tuesday evening near the end of that year, Catherine called a friend. Not a doctor's office, not an insurer — a friend, about nothing in particular: a neighbour's new puppy, a recipe, a shared memory from a college apartment neither of them had set foot in for twenty years. Charlie was asleep against her leg. Halfway through the call she realised something had shifted; she was inside the conversation rather than rationing it.

When she hung up she did not crash. She showered sitting down, went to bed at a reasonable hour, and walked Charlie to the end of the driveway the next morning. A year earlier that evening would have cost her the weekend.

This is the shape of a household that has absorbed the illness into its operating system. Nobody has romanticized it. Nobody has pretended the illness is a gift. The family has simply made room for the fact that the resource inside this body is finite on any given day, and they have organized around that fact without resentment.

If you can build this, with one person or with three, it is worth more than any single tool in this book.

What changes when the thermostat moves

People ask me — in emails, in patient communities, and in the research conversations I run — what does it feel like when the thermostat actually starts to renegotiate? What is the tell?

It is almost never a single dramatic moment. It is a set of small quiet ones.

You walked further than you meant to and did not pay for it.

One morning you woke up, and the first thought in your head was something other than “how bad is today going to be.”

A minor stressor came and went — a late package, a parking ticket, an argument with a teenager — without your heart rate spiking into a range you could feel.

You said yes to something two weeks out, and kept the yes.

Three hours passed and the illness did not cross your mind.

None of these are fireworks. They are the small indicators that the feedback loop has started to deliver different signals to a system that was stuck in a defensive configuration. The thermostat renegotiates when the inputs it receives are more consistent than the inputs it was receiving before — when the threat signal is lower than the safety signal often enough and long enough for the system to update its priors.

Your job has been, for twelve chapters, to change the inputs. To reduce the signal noise. To give the system more days of *safe* than *threatened*. The renegotiation is what happens when the cumulative weight of those inputs crosses some threshold that is different for every person and that nobody can predict in advance.

What you can do is keep producing the inputs.

The honest framing

I said at the start of this book that we would not lie to you about outcomes. I am going to keep that promise at the end.

Some readers will read this book, do the work, and in two years be functionally most of the way back to a life they recognize. Some readers will do the same work and stabilize at a reduced but livable ceiling, with a good floor, and build a different and good life inside it. Some readers will continue to fluctuate for longer than they expected, and the work they do will matter not because it produces a neat arc but because it prevents the worst crashes and preserves the floor during the long climb.

All three of these are outcomes the data supports. None of them is failure. The framing in which the first is "success" and the other two are "not working" is the framing that kills people in Stockdale's sense — that breaks their heart by Christmas. I refuse that framing, and I am asking you to refuse it too.

The project is to build a life that is good inside today's constraints, while today's constraints keep moving. That is the whole of it. The tools, the data, the community, the small circle — all of it serves that one purpose.

This work continues — mine, and far more importantly that of the clinicians, patient-led collaboratives, and academic groups whose findings will keep refining the picture in the years ahead. This book is one artifact of that effort, and a snapshot in time. It is not the end of it.

One thing to be plain about, because it shapes everything you have just read. This book is a plain-language review of the research as it stands — what has been looked at, how well, and where the findings disagree with each other. Science does not yet know most of this with any confidence, and I have not smoothed that over. It leans heavily on patient-led research, which in several corners of this field is still ahead of the formal literature. What it is not is a report on research of our own.

What I hope to add later is information that comes from our own data — more specific, and more useful on a Tuesday morning. I cannot promise that. What I can tell you is what is happening now: we are following an observational cohort of women who share this PEM phenotype, working alongside them, with their active involvement in which questions get asked. If that turns up something useful for people with this phenotype, the next book is where it will go.

If you pull a single thread from the logic of this book, let it be the recognition that effort is paid for with a delay. Until you see that pattern, managing the condition is impossible because you have no idea what to track. Once you recognize the delay, pacing becomes the necessary next move. The objective of that pacing is a stable floor rather than an immediate return to your old ceiling. A low baseline held steady creates the biological conditions for repair, while a higher baseline punctuated by crashes breaks the system down further. Because the one test that measures this threshold precisely sits out of reach for most people, self-calibration is what remains. Your daily observations and your diary are the instruments you actually have. This reliance on your own data matters because of the payback window. Waking up with enough energy to fold a basket of laundry offers no useful planning data for the week, since the bill for any overexertion arrives forty-eight to seventy-two hours later. That delayed invoice is the reason subjective feeling in the moment cannot dictate your schedule, bringing you right back to the necessity of observation and a measured pace.

Optional self-observation prompt

Pick one lever.

Ask the one person you trust what they have noticed about you in the last month. You are gathering a second instrument on a system you are measuring from inside.

Mark a single small thing you have added to your life in the last ninety days. A capacity. A conversation made. A shower without the stool. A yes to a plan that held.

Look at the thermostat metaphor one more time, and remember that the thermostat reads the last weeks of inputs as the best available proxy for what kind of world the body is in right now — it has no access to the calendar, only to the signal stream. Your job is to keep sending better inputs.

That is the project.

You are already the field researcher. You are already the person in the room with the most data. The work is already under way — keep sending the signal.

The thermostat renegotiates when the feedback loop delivers something new — consistently, and for long enough that the system finally trusts the new signal. Keep sending.

Sources

- Long COVID trajectory literature (2022–present) — European cohort studies, RECOVER program data, Bateman Horne Center and other specialist-clinic reports converge on a wide-distribution recovery shape rather than a U-curve.
- SARS-1 long-horizon follow-up literature (Hong Kong and Toronto cohorts, post-2003) — three-to-four-year follow-up data on the only structurally comparable post-viral illness at scale; most patients improve meaningfully over one to three years, subset stabilizes at reduced baseline; protection from repeated crashes is the strongest predictor of trajectory.

- Collins, J. (2001). *Good to Great*. HarperBusiness. [Stockdale Paradox]
- RECOVER initiative patient-partnered research framework — patient representatives on priority-setting panels; PEM characterized initially through patient-led surveys and later confirmed by CPET in specialist clinics.

Reading list

A short note on how to read this

What follows is a working shelf rather than an exhaustive bibliography — the papers and books we kept returning to while writing this one, arranged so a curious reader can find the ground under any claim we made. Each chapter has a short **Tier 1** list (the handful of references that earn the ground) and a longer **Tier 2** list (further evidence for readers who want to keep going).

A practical note on reading medical papers. The most load-bearing information usually sits in two places: the last paragraph of the introduction (what the authors think they are doing) and the first paragraph of the discussion (what they think they found). Methods and tables deserve attention only if the conclusions interest you. Cohort size, who the controls are, how long the follow-up ran, and whether the measure matches the clinical question — these are the four questions that decide whether a paper is saying what its headline says. Be suspicious of press releases; they frequently overstate effect size. A conservative read of three papers usually beats an enthusiastic read of thirty.

Patient-led research is flagged where it appears. This body of work is not an afterthought to the field — it is, in several domains, the leading edge.

Chapter 0 — Introduction. What this book is, and what it is not

Tier 1

- **Callard F, Perego E (2021).** "How and why patients made Long Covid." *Social Science & Medicine* 268:113426. DOI: 10.1016/j.socscimed.2020.113426 The founding historical record of how patients organised the term "Long COVID" into existence in spring 2020, before any institution used the phrase.
- **Davis HE, Assaf GS, McCorkell L, et al. (2021).** "Characterizing Long COVID in an international cohort: 7 months of symptoms and their impact." *EClinicalMedicine* 38:101019. DOI: 10.1016/j.eclinm.2021.101019 The first large patient-led survey characterising the symptom set across 3,762 respondents and 203 distinct symptoms; anchor citation for the book's patient-science framing.
- **Institute of Medicine, Committee on the Diagnostic Criteria for ME/CFS (2015).** *Beyond Myalgic Encephalomyelitis / Chronic Fatigue Syndrome: Redefining an Illness.* Washington, DC: National Academies Press. DOI: 10.17226/19012 The report that named post-exertional malaise as the cardinal feature rather than an optional symptom; foundation of current diagnostic frameworks.
- **U.S. Department of Health and Human Services (2022).** *National Research Action Plan on Long COVID*, with the Infection-Associated Chronic Conditions (IACC) umbrella definition. The formal HHS scoping that gathers LC, ME/CFS, POTS, MCAS, post-Lyme, post-EBV, and related conditions into one working category.
- **Patient-Led Research Collaborative (PLRC).** Ongoing publications and reports at patientresearchcovid19.com. The community research group whose 2020 survey work set the methodological template for the rest of the field. https://patientresearchcovid19.com

Tier 2

- **National Academies of Sciences, Engineering, and Medicine (2024).** *A Long COVID Definition: A Chronic, Systemic Disease State with Profound Consequences.* Washington, DC: National Academies Press. Update to the 2015 IOM work, now covering Long COVID specifically. https://www.nationalacademies.org/projects/HMD-HSP-23-01/publication/27768

Chapter 1 — Fatigue is not the right word

Tier 1

- **Singh I, Joseph P, Heerdt PM, Cullinan M, Lutchmansingh DD, Gulati M, Possick JD, Systrom DM, Waxman AB (2022).** "Persistent exertional intolerance after COVID-19: insights from invasive cardiopulmonary exercise testing." *Chest* 161(1):54–63. DOI: 10.1016/j.chest.2021.08.010 Invasive CPET in post-acute LC showing impaired peripheral oxygen extraction at near-normal cardiac output — the muscle-side metabolic lesion that undergirds the whole chapter.
- **Durstenfeld MS, Sun K, Tahir P, et al. (2022).** "Use of cardiopulmonary exercise testing to evaluate Long COVID-19 symptoms in adults: a systematic review and meta-analysis." *JAMA Network Open* 5(10):e2236057. DOI: 10.1001/jamanetworkopen.2022.36057 Pooled CPET analysis confirming reduced exercise capacity across LC cohorts independent of resting cardiopulmonary function.
- **Keller BA, Pryor JL, Giloteaux L (2014).** "Inability of Myalgic Encephalomyelitis / Chronic Fatigue Syndrome patients to reproduce VO_2peak indicates functional impairment." *Journal of Translational Medicine* 12:104. DOI: 10.1186/1479-5876-12-104 The Workwell

Foundation two-day CPET work showing the day-2 drop that is one of the most specific objective signatures of PEM.

- **Stevens S, Snell C, Stevens J, Keller B, Van Ness JM (2018).** "Cardiopulmonary exercise test methodology for assessing exertion intolerance in ME/CFS." *Frontiers in Pediatrics* 6:242. DOI: 10.3389/fped.2018.00242 The practical manual for how to run the two-day CPET without harming the patient.
- **Swank Z, Senussi Y, Manickas-Hill Z, Yu XG, Li JZ, Alter G, Walt DR (2023).** "Persistent circulating SARS-CoV-2 spike is associated with post-acute COVID-19 sequelae." *Clinical Infectious Diseases* 76(3):e487–e490. DOI: 10.1093/cid/ciac722 The paper that names the molecular hook behind "viral persistence" — spike fragments still circulating months after acute infection.
- **National Institute for Health and Care Excellence (2021).** *Myalgic encephalomyelitis (or encephalopathy) / chronic fatigue syndrome: diagnosis and management* (NG206). The current UK clinical guideline, relevant for PEM definition, harm caution around graded exercise, and pacing as first-line activity management. https://www.nice.org.uk/guidance/ng206

Tier 2

- **Nalbandian A, Sehgal K, Gupta A, et al. (2021).** "Post-acute COVID-19 syndrome." *Nature Medicine* 27(4):601–615. DOI: 10.1038/s41591-021-01283-z Clinical overview of organ-system manifestations during the early LC literature.
- **Stein SR, Ramelli SC, Grazioli A, et al. (2022).** "SARS-CoV-2 infection and persistence in the human body and brain at autopsy." *Nature* 612(7941):758–763. DOI: 10.1038/s41586-022-05542-y Autopsy data confirming widespread viral persistence in tissues beyond the respiratory tract.

- **Peter RS, Nieters A, Göpel S, et al. (2025).** "Persistent symptoms and clinical findings in adults with post-acute sequelae of COVID-19/post-COVID-19 syndrome in the second year after acute infection: a population-based, nested case-control study." *PLoS Medicine* 22(1):e1004511. DOI: 10.1371/journal.pmed.1004511 Nested case-control study within the EPILOC project: 982 participants with post-COVID syndrome against 576 matched controls, with CPET and neurocognitive testing; useful companion read on LC phenotyping.

Chapter 2 — The thermostat that won't turn off

Tier 1

- **Raj SR (2013).** "Postural tachycardia syndrome (POTS)." *Circulation* 127(23):2336–2342. DOI: 10.1161/CIRCULATIONAHA.112.144501 Compact clinical review of POTS that establishes diagnostic criteria, subtypes, and first-line management.
- **Freeman R, Wieling W, Axelrod FB, et al. (2011).** "Consensus statement on the definition of orthostatic hypotension, neurally mediated syncope and the postural tachycardia syndrome." *Clinical Autonomic Research* 21(2):69–72. DOI: 10.1007/s10286-011-0119-5 The diagnostic-threshold document the book leans on for HR-rise and BP-drop numbers.
- **Dani M, Dirksen A, Taraborrelli P, et al. (2021).** "Autonomic dysfunction in 'long COVID': rationale, physiology and management strategies." *Clinical Medicine* 21(1):e63–e67. DOI: 10.7861/clinmed.2020-0896 Bridges POTS-literature to LC and articulates why the autonomic lens changed the picture.
- **Bateman L, Bested AC, Bonilla HF, et al. (2021).** "Myalgic encephalomyelitis / chronic fatigue syndrome: essentials of diagnosis and management." *Mayo Clinic Proceedings* 96(11):2861–2878. DOI:

10.1016/j.mayocp.2021.07.004 US ME/CFS Clinician Coalition / Bateman Horne Center consensus — the clinical-care guide the book cross-references for HUA and pacing.

- **Larsen NW, Stiles LE, Miglis MG (2021).** "Preparing for the long-haul: autonomic complications of COVID-19." *Autonomic Neuroscience* 235:102841. DOI: 10.1016/j.autneu.2021.102841 Expert overview of how POTS-like dysautonomia manifests after COVID; readable for a motivated non-specialist.

Tier 2

- **Shaw BH, Stiles LE, Bourne K, et al. (2019).** "The face of postural tachycardia syndrome — insights from a large cross-sectional online community-based survey." *Journal of Internal Medicine* 286(4):438–448. DOI: 10.1111/joim.12895 Patient-survey data (n=4,835) showing that 77% of POTS patients were initially told they had an anxiety disorder — the statistical anchor under the chapter's "system stuck in high alert" frame.
- **Badran BW, Huffman SM, Dancy M, et al. (2022).** "Supervised, at-home, self-administered transcutaneous auricular vagus nerve stimulation to manage long COVID symptoms: a pilot randomized controlled trial." *Bioelectronic Medicine* 8(1):13. DOI: 10.1186/s42234-022-00094-y Device-literature taVNS pilot; grounding for the "medical-device research is separate from wellness marketing" discussion.
- **Corrado J, Iftekhar N, Halpin S, et al. (2024).** "Heart rate variability biofeedback for Long COVID dysautonomia: results of a feasibility study (HEARTLOC)." *Advances in Rehabilitation Science and Practice* 13. DOI: 10.1177/27536351241227261 HRV-biofeedback feasibility data in LC dysautonomia.
- **Mauro M, Cegolon L, Bestiaco N, Zulian E, Larese Filon F (2024).** "Heart rate variability modulation through slow-paced breathing

in health care workers with Long COVID." *American Journal of Medicine*. DOI: 10.1016/j.amjmed.2024.05.021 Case-control study on slow-paced breathing; low-cost intervention evidence.

Chapter 3 — Five diagnoses, one problem

Tier 1

- **Mathew GT, Novak P. (2026).** "Prevalence of Central Sensitization in Postural Tachycardia Syndrome." *JAMA Network Open* 9(1):e2553694. DOI: 10.1001/jamanetworkopen.2025.53694 Case-control study of 305 patients with confirmed POTS; 264 (86.6%) met criteria for central sensitization syndrome. Underpins the "same underlying process, many diagnostic labels" frame.
- **Weinstock LB, Brook JB, Walters AS, Goris A, Afrin LB, Molderings GJ (2021).** "Mast cell activation symptoms are prevalent in long-COVID." *International Journal of Infectious Diseases* 112:217–226. DOI: 10.1016/j.ijid.2021.09.043 Compares LC symptom patterns to MCAS symptom patterns; the source the chapter's mast-cell paragraph rests on, with corrected reading that the paper reports severity and symptom-count overlap rather than a prevalence percentage.
- **Miller AJ, Stiles LE, Sheehan T, et al. (2020).** "Prevalence of hypermobile Ehlers-Danlos syndrome in postural orthostatic tachycardia syndrome." *Autonomic Neuroscience* 224:102637. DOI: 10.1016/j.autneu.2020.102637 The 31% hEDS-in-POTS statistic and what it means for the connective-tissue-plus-autonomic picture.
- **Khalsa SS, Adolphs R, Cameron OG, et al. (2018).** "Interoception and mental health: a roadmap." *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging* 3(6):501–513. DOI: 10.1016/j.bpsc.2017.12.004 Accessible anatomical anchor for "interoceptive

mismatch" — the insular cortex, and why the body's report is accurate even when the input has changed.

- **Salvucci F, Codella R, Coppola A, Zacchei I, et al. (2023).** "Antihistamines improve cardiovascular manifestations and other symptoms of long-COVID attributed to mast cell activation." *Frontiers in Cardiovascular Medicine* 10:1202696. DOI: 10.3389/fcvm.2023.1202696 Fourteen treated patients against thirteen controls; complete resolution in 29%. Small and open-label — Open-label n=27 of fexofenadine plus famotidine at standard doses; roughly one-third of patients showed full resolution. Small, non-randomised, clinician-supervised territory — included because the pairing appears in current clinical practice and is referenced directly in-chapter.
- **Vernino S, Hopkins S, Bryarly M, et al. (2024).** "Randomized controlled trial of intravenous immunoglobulin for autoimmune POTS (iSTAND)." *Clinical Autonomic Research* 34(1):153–163. DOI: 10.1007/s10286-024-01020-9 The first placebo-controlled IVIg trial in autoimmune POTS; neutral primary endpoint, and the anchor of the book's hedged IVIg paragraph.

Tier 2

- **Oaklander AL, Herzog ZD, Downs HM, Klein MM (2013).** "Objective evidence that small-fiber polyneuropathy underlies some illnesses currently labeled as fibromyalgia." *Pain* 154(11):2310–2316. DOI: 10.1016/j.pain.2013.06.001 The small-fibre-neuropathy overlap with fibromyalgia.
- **Ray JC, Pham X, Foster E, et al. (2022).** "The prevalence of headache disorders in postural tachycardia syndrome: a systematic review and meta-analysis." *Cephalalgia* 42(11–12):1274–1287. DOI: 10.1177/03331024221095153 Headache co-occurrence with POTS.
- **Sjögren P, Bragée B, Britton S (2024).** "Successful subcutaneous immunoglobulin therapy in a case series of patients with

ME/CFS." *Clinical Therapeutics* 46(7):597–600. DOI: 10.1016/j.clinthera.2024.05.010 Case-series signal for subcutaneous Ig in ME/CFS.

- **Stein E, Heindrich C, Wittke K, et al. (Scheibenbogen group, 2025).** "Efficacy of repeated immunoadsorption in patients with post-COVID ME/CFS and elevated β2-adrenergic receptor autoantibodies." *Lancet Regional Health — Europe* 49:101161. DOI: 10.1016/j.lanepe.2024.101161 Charité group's immunoadsorption data in autoantibody-positive post-COVID ME/CFS.
- **McAlpine L, Zubair AS, Joseph P, Spudich S (2024).** "Case-control study of individuals with small fiber neuropathy after COVID-19." *Neurology Neuroimmunology & Neuroinflammation* 11(3):e200244. DOI: 10.1212/NXI.0000000000200244 Small-fibre neuropathy identified in a defined post-COVID subgroup.
- **Ayoubkhani D, Bermingham C, Pouwels KB, et al. (2022).** "Trajectory of long covid symptoms after covid-19 vaccination: community based cohort study." *BMJ* 377:e069676. DOI: 10.1136/bmj-2021-069676 Cohort evidence on post-vaccination LC symptom course.
- **Bowe B, Xie Y, Al-Aly Z (2022).** "Acute and postacute sequelae associated with SARS-CoV-2 reinfection." *Nature Medicine* 28(11):2398–2405. DOI: 10.1038/s41591-022-02051-3 Cumulative risk with successive infections; relevant for the chapter's reinfection paragraph.

Chapter 4 — Why standing is harder than walking

Tier 1

- **Thieben MJ, Sandroni P, Sletten DM, et al. (2007).** "Postural orthostatic tachycardia syndrome: the Mayo Clinic experience." *Mayo Clinic Proceedings* 82(3):308–313. DOI: 10.4065/82.3.308 Clinical description of POTS subtypes as the Mayo group worked them out; useful for the neuropathic / hyperadrenergic / hypovolemic / autoimmune split.

- **Fu Q, VanGundy TB, Galbreath MM, et al. (2010).** "Cardiac origins of the postural orthostatic tachycardia syndrome." *Journal of the American College of Cardiology* 55(25):2858–2868. DOI: 10.1016/j.jacc.2010.02.043 Twenty-seven patients against sixteen controls; blood and plasma volume about a fifth lower, left-ventricular mass about 16% smaller. Characterises the reduced stroke volume and cardiac-reserve side of POTS — a companion to the peripheral-pooling picture.
- **Haensch C-A, Tosch M, Katona I, Weis J, Isenmann S (2014).** "Small-fiber neuropathy with cardiac denervation in postural tachycardia syndrome." *Muscle & Nerve* 50(6):956–961. DOI: 10.1002/mus.24245 The anatomical hook under the neuropathic POTS subtype.
- **Uppal J, Baker JR, Hira R, et al. (2025).** "Physiological and clinical comparison of active stand and head-up tilt tests in Postural Orthostatic Tachycardia Syndrome (POTS)." *Autonomic Neuroscience* 260:103281. DOI: 10.1016/j.autneu.2025.103281 Sixty patients did both tests on the same day. Hemodynamic trajectories were broadly similar, but 74% met the POTS heart-rate criterion on active stand against 98% on tilt. Read it for the limit as much as for the ground for the NASA Lean Test home-equivalent.
- **Bateman Horne Center.** *NASA Lean Test Instructions.* Patient handout. The protocol Ch 04 and Ch 05 use directly; a one-page procedure any reader can run at home. https://batemanhornecenter.org/

Tier 2

- **Benarroch EE (2012).** "Postural tachycardia syndrome: a heterogeneous and multifactorial disorder." *Mayo Clinic Proceedings*

87(12):1214–1225. DOI: 10.1016/j.mayocp.2012.08.013 Mechanism review — useful if the subtype split raises more questions.

- **Garland EM, Celedonio JE, Raj SR (2015).** "Postural tachycardia syndrome: beyond orthostatic intolerance." *Current Neurology and Neuroscience Reports* 15(9):60. DOI: 10.1007/s11910-015-0583-8 Symptoms beyond the heart-rate response.
- **Fu Q, Levine BD (2018).** "Exercise and non-pharmacological treatment of postural orthostatic tachycardia syndrome." *Autonomic Neuroscience* 215:20–27. DOI: 10.1016/j.autneu.2018.07.001 Graded reconditioning protocol — for readers whose clinician brings it up; read alongside the Ch 06 pacing caveats.
- **Wallukat G, Hohberger B, Wenzel K, et al. (2021).** "Functional autoantibodies against G-protein coupled receptors in patients with persistent Long-COVID-19 symptoms." *Journal of Translational Autoimmunity* 4:100100. DOI: 10.1016/j.jtauto.2021.100100 Autoimmune POTS biomarker evidence.
- **Medinger G, Altmann D (2022).** *The Long COVID Handbook*. London: Hutchinson Heinemann. Patient-plus-researcher co-authored overview, including the Imperial College autonomic work; repeatedly useful across the book.

Chapter 5 — Tools that work every day

Tier 1

- **Vernino S, Bourne KM, Stiles LE, et al. (2021).** "Postural orthostatic tachycardia syndrome (POTS): State of the science and clinical care from a 2019 NIH Expert Consensus Meeting — Part 1." *Autonomic Neuroscience* 235:102828. DOI: 10.1161/CIRCULATIONAHA.112.144501 The consensus that sets the sodium range at 3–10 g daily and says the optimal dose is unsettled. Practical first-line management review that the tools chapter leans on for salt-loading, compression, ivabradine, midodrine, and pyridostigmine.

- **Sheldon RS, Grubb BP, Olshansky B, et al. (2015).** "2015 Heart Rhythm Society expert consensus statement on the diagnosis and treatment of postural tachycardia syndrome, inappropriate sinus tachycardia, and vasovagal syncope." *Heart Rhythm* 12(6):e41–e63. DOI: 10.1016/j.hrthm.2015.03.029 The consensus the chapter's beta-blocker and ivabradine paragraphs track.
- **Bourne KM, et al. (2021).** "Compression Garment Reduces Orthostatic Tachycardia and Symptoms in Patients With Postural Orthostatic Tachycardia Syndrome." *Journal of the American College of Cardiology* 77(3):285–296. DOI: 10.1016/j.jacc.2020.11.040 The evidence base for why abdominal-plus-thigh compression beats stockings alone — the splanchnic-bed story in clinical form.
- **El-Sayed H, Hainsworth R (1996).** "Salt supplement increases plasma volume and orthostatic tolerance in patients with unexplained syncope." *Heart* 75(2):134–140. DOI: 10.1136/hrt.75.2.134 Classic evidence on salt-loading and orthostatic tolerance.
- **World Health Organization (2006).** *Oral rehydration salts: production of the new ORS.* The formulation document behind the glucose-plus-sodium pairing the chapter recommends; useful for understanding why plain water fails in POTS. https://www.who.int/publications/i/item/WHO-FCH-CAH-06.1

Tier 2

- **Tani H, Singer W, McPhee BR, et al. (2000).** "Splanchnic-mesenteric capacitance bed in the postural tachycardia syndrome (POTS)." *Autonomic Neuroscience* 86(1–2):107–113. DOI: 10.1161/CIRCULATIONAHA.112.144501 Meal-timing and blood-pressure work; context for the small-meals recommendation.

- **Hitch D, Wrench J, eds. (2026).** *The Rehabilitation and Management of Long COVID: A Handbook for Clinical Practice.* London: Routledge. DOI: 10.4324/9781003528104 Clinical reference translated for practical reading; strong on allied-health referral pathways (speech-language, dietetic, cognitive rehab).

Chapter 6 — Pacing

Tier 1

- **National Institute for Health and Care Excellence (2021).** *Myalgic encephalomyelitis (or encephalopathy) / chronic fatigue syndrome: diagnosis and management* (NG206). The guideline that formally replaced graded exercise with pacing as first-line activity management in ME/CFS and related conditions. https://www.nice.org.uk/guidance/ng206
- **Fluge Ø, Mella O, Bruland O, et al. (2016).** "Metabolic profiling indicates impaired pyruvate dehydrogenase function in myalgic encephalomyelitis / chronic fatigue syndrome." *JCI Insight* 1(21):e89376. DOI: 10.1172/jci.insight.89376 The pyruvate-dehydrogenase downregulation finding that explains why "exercise doesn't help" at the cell-biology level.
- **van Campen CLMC, Verheugt FWA, Rowe PC, Visser FC (2020).** "Cerebral blood flow is reduced in myalgic encephalomyelitis / chronic fatigue syndrome during head-up tilt testing even in the absence of hypotension or tachycardia." *Clinical Neurophysiology Practice* 5:50–58. DOI: 10.1016/j.cnp.2020.01.003 Objective evidence of reduced cerebral blood flow during upright posture — the mechanism under "my brain stops working when I stand too long."
- **Casson S, Jones MD, Cassar J, et al. (2023).** "The effectiveness of activity pacing interventions for people

with chronic fatigue syndrome: a systematic review and meta-analysis." *Disability and Rehabilitation* 45(23):3788–3802. DOI: 10.1080/09638288.2022.2135776 Fourteen studies; fatigue Hedges' g −0.52 (95% CI −0.73 to −0.32). Pair it with the scoping review below, which reads the same literature more cautiously. Pooled evidence on pacing outcomes.

Tier 2

- **Seeley M-C, O'Brien H, Wilson G, et al. (2025).** "Novel brain SPECT imaging unravels abnormal cerebral perfusion in patients with postural orthostatic tachycardia syndrome and cognitive dysfunction." *Scientific Reports* 15(1):3487. DOI: 10.1038/s41598-025-87748-4 Fifty-six participants; abnormal cerebral blood flow in 61%, present even while supine. Imaging companion to the Van Campen work, using SPECT.
- **White PD et al. (2011).** "Comparison of adaptive pacing therapy, cognitive behaviour therapy, graded exercise therapy, and specialist medical care for chronic fatigue syndrome (PACE): a randomised trial." *The Lancet* 377(9768):823–836. DOI: 10.1016/S0140-6736(11)60096-2 The PACE trial, included here because the field's relationship to it still matters; read alongside Wilshire et al. 2018.
- **Wilshire CE et al. (2018).** "Rethinking the treatment of chronic fatigue syndrome — a reanalysis and evaluation of findings from a recent major trial of graded exercise and CBT." *BMC Psychology* 6:6. DOI: 10.1186/s40359-018-0218-3 The reanalysis that reframed PACE's claimed effect sizes and helped shift UK guidance.

Chapter 7 — Food as a tool, not a diet

Tier 1

- **Wong AC, Devason AS, Umana IC, Cox TO, Dohnalová L, et al. (2023).** "Serotonin reduction in post-acute sequelae of viral infection." *Cell* 186(22):4851–4867.e20. DOI: 10.1016/j.cell.2023.09.013 The peripheral-serotonin-reduction paper that sparked the field's current tryptophan / kynurenine conversation; read alongside Sommen 2024 for replication context.
- **Mathé P, Götz V, Stete K, et al. (2024).** "No reduced serum serotonin levels in patients with post-acute sequelae of COVID-19." *Infection*. DOI: 10.1007/s15010-024-02397-5 Negative replication of the peripheral-serotonin-reduction finding in an independent cohort.
- **Anderson GM, Cook EH, Blakely RD, et al. (2024).** "Long COVID-19 and peripheral serotonin: a commentary and reconsideration." *Journal of Inflammation Research* 17:2169–2172. DOI: 10.2147/JIR.S456000 Commentary threading the conflicting serotonin data; useful for the "the story is still being worked out" framing.
- **Liu Q, Mak JWY, Su Q, Yeoh YK, et al. (2022).** "Gut microbiota dynamics in a prospective cohort of patients with post-acute COVID-19 syndrome." *Gut* 71(3):544–552. DOI: 10.1038/s41591-021-01283-z Microbiome changes associated with LC; the empirical ground for the chapter's fibre paragraph.
- **O'Kelly B, Vidal L, McHugh T, Woo J, Avramovic G, Lambert JS (2022).** "Safety and efficacy of low dose naltrexone in a long covid cohort: an interventional pre-post study." *Brain, Behavior, & Immunity — Health* 24:100485. DOI: 10.1016/j.bbih.2022.100485 The LC-specific LDN interventional study most often cited in current clinical practice.
- **Sasso EM, Eaton-Fitch N, Smith P, Muraki K, Marshall-Gradisnik S (2025).** "Low-dose naltrexone restored TRPM3 ion channel function

in natural killer cells from long COVID patients.” *Frontiers in Molecular Biosciences* 12:1582967. DOI: 10.3389/fmolb.2025.1582967 Mechanism-of-action evidence for LDN at the ion-channel level — the closest the field has to a molecular rationale.

- **Byambasuren O, Atkins T, Baptista S, Glasziou P, Chakraborty S (2026).** “Effect of low-dose naltrexone for long COVID: a systematic review and meta-analysis.” *BMJ Open* 16(7):e111253 (preprint 2025, medRxiv). DOI: 10.1136/bmjopen-2025-111253 Systematic review of LDN evidence as of early 2025; read together with Soliman 2025 below.
- **Du A, Nguyen ADK (2025).** “Does Low-Dose Oral Naltrexone Alleviate Symptoms of Long COVID? A Systematic Review and Meta-Analysis.” *COVID* (MDPI) 5(12):198. DOI: 10.3390/covid5120198 Published meta-analytic companion to Tomas 2025; hedged-positive signal.
- **Salvucci F, et al. (2023).** *Frontiers in Cardiovascular Medicine* 10:1202696 — full reference as listed under Chapter 3. DOI: 10.3389/fcvm.2023.1202696 Primary-source for the H1-plus-H2 combination pairing discussed in both chapters.

Tier 2

- **Yano JM, Yu K, Donaldson GP, et al. (2015).** “Indigenous bacteria from the gut microbiota regulate host serotonin biosynthesis.” *Cell* 161(2):264–276. DOI: 10.1016/j.cell.2015.02.047 Establishes that gut microbes regulate host serotonin production; explains why dietary fibre is part of the chapter’s logic.
- **Shukla AP, Iliescu RG, Thomas CE, Aronne LJ (2015).** “Food order has a significant impact on postprandial glucose and insulin levels.” *Diabetes Care* 38(7):e98–e99. DOI: 10.2337/dc15-0429 The empirical ground under the meal-order recommendation.

- **Zeevi D, Korem T, Zmora N, et al. (2015).** "Personalized nutrition by prediction of glycemic responses." *Cell* 163(5):1079–1094. DOI: 10.1016/j.cell.2015.11.001 Individual glycaemic-response variability; context for why a single diet doesn't fit everyone.
- **Estruch R, Ros E, Salas-Salvadó J, et al. (2018).** "Primary prevention of cardiovascular disease with a Mediterranean diet supplemented with extra-virgin olive oil or nuts" (PREDIMED). *New England Journal of Medicine* 378(25):e34. DOI: 10.1056/NEJMoa1800389 The Mediterranean-dietary-pattern evidence the chapter's food framework leans on.
- **Castro-Marrero J, Sáez-Francàs N, Santillo D, Alegre J (2017).** "Treatment and management of chronic fatigue syndrome / myalgic encephalomyelitis: all roads lead to Rome." *British Journal of Pharmacology* 174(5):345–369. DOI: 10.1111/bph.13702 Broad pharmacology overview including CoQ10 and related cofactor work.
- **Barletta MA, Marino G, Spagnolo B, et al. (2023).** "Coenzyme Q10 plus alpha-lipoic acid for chronic COVID syndrome." *Clinical and Experimental Medicine* 23(3):667–678. DOI: 10.1007/s10238-022-00871-8 CoQ10-plus-ALA combination trial — the evidence current supplement recommendations lean on; hedge appropriately.
- **Singh A, D'Amico D, Andreux PA, et al. (2022).** "Urolithin A improves muscle strength, exercise performance, and biomarkers of mitochondrial health in a randomized trial in middle-aged adults." *Cell Reports Medicine* 3(5):100633. DOI: 10.1016/j.xcrm.2022.100633 Urolithin A trial in non-LC adults; context for why the LC evidence is still sparse.

Chapter 8 — Sleep architecture, not sleep hours

Tier 1

- **Besedovsky L, Lange T, Born J (2012).** "Sleep and immune function." *Pflügers Archiv — European Journal of Physiology* 463(1):121–137. DOI: 10.1007/s00424-011-1044-0 The landmark review that established how slow-wave sleep consolidates immune memory — why the immune system also "sleeps poorly" when sleep architecture is broken.
- **Nedergaard M, Goldman SA (2016).** "Brain drain." *Scientific American* 314(3):44–49. DOI: 10.1038/scientificamerican0316-44 Accessible introduction to the glymphatic system and nighttime brain waste clearance.
- **Imeri L, Opp MR (2009).** "How (and why) the immune system makes us sleep." *Nature Reviews Neuroscience* 10(3):199–210. DOI: 10.1038/nrn2576 Cytokines, sleep regulation, and the overlap with sickness behavior.
- **Walker MP, van der Helm E (2009).** "Overnight therapy? The role of sleep in emotional brain processing." *Psychological Bulletin* 135(5):731–748. DOI: 10.1037/a0016570 How REM sleep metabolises emotional weight; grounds the chapter's "sleep does emotional work" paragraph.
- **Harding EC, Franks NP, Wisden W (2019).** "The temperature dependence of sleep." *Frontiers in Neuroscience* 13:336. DOI: 10.3389/fnins.2019.00336 Temperature-regulation physiology of sleep; practical implications for bedroom-environment recommendations.

Tier 2

- **Thakkar MM (2011).** "Histamine in the regulation of wakefulness." *Sleep Medicine Reviews* 15(1):65–74. DOI: 10.1016/j.smrv.2010.06.004 Why mast-cell-driven histamine surges at night matter for sleep fragmentation in MCAS overlap.

- **Goldstein AN, Walker MP (2014).** "The role of sleep in emotional brain function." *Annual Review of Clinical Psychology* 10:679–708. DOI: 10.1146/annurev-clinpsy-032813-153716 Companion to Walker & van der Helm 2009.
- **Zaccaro A, Piarulli A, Laurino M, et al. (2018).** "How breath-control can change your life: a systematic review on psycho-physiological correlates of slow breathing." *Frontiers in Human Neuroscience* 12:353. DOI: 10.3389/fnhum.2018.00353 Review of slow-paced-breathing effects on HRV, baroreflex sensitivity, and cortical activity.
- **Cooper VL, Hainsworth R (2008).** "Head-up sleeping improves orthostatic tolerance in patients with syncope." *Clinical Autonomic Research* 18(6):318–324. DOI: 10.1007/s10286-008-0494-8 Eleven of twelve patients improved and plasma volume rose after three to four months of sleeping with the bed head raised. Work on head-of-bed elevation and its effect on overnight plasma-volume regulation.
- **Sachse C, Brockmöller J, Bauer S, Roots I (1999).** "Functional significance of a C→A polymorphism in intron 1 of the CYP1A2 gene tested with caffeine." *British Journal of Clinical Pharmacology* 47(4):445–449. DOI: 10.1046/j.1365-2125.1999.00898.x The CYP1A2 polymorphism and its effect on caffeine clearance — why the same afternoon coffee has different consequences for different people.
- **Stavem K, Ghanima W, Olsen MK, Gilboe HM, Einvik G (2021).** "Persistent symptoms 1.5–6 months after COVID-19 in non-hospitalised subjects." *Thorax* 76(4):405–407. DOI: 10.1136/thoraxjnl-2020-216377 Sleep-disturbance prevalence in early LC cohorts.

Chapter 9 — Data over feelings

Tier 1

- **Bateman Horne Center.** *Clinical Care Guide for ME/CFS and Long COVID*, including the Hours of Upright Activity (HUA) measure. Free to download at batemanhornecenter.org. The clinical-care document under the HUA definition used in-chapter. https://batemanhornecenter.org/clinical-care-guide/
- **Jason LA, So S, Brown AA, Sunnquist M, Evans M (2015).** "Test-retest reliability of the DePaul Symptom Questionnaire." *Fatigue: Biomedicine, Health & Behavior* 3(1):16–32. DOI: 10.1080/21641846.2014.978110 Psychometric validation of the instrument the chapter recommends for symptom tracking; useful to know the test-retest figures before committing to a diary system.
- **Jason LA, Sunnquist M, Brown A, Evans M, Newton JL (2014).** "Are ME and CFS different illnesses? A preliminary analysis." *Journal of Health Psychology* 21(1):3–15. DOI: 10.1177/1359105313520335 The DePaul group's work on diagnostic-criteria discrimination; useful groundwork for thinking about which symptom constellations the DSQ is good at detecting.
- **Stone AA, Shiffman S, Schwartz JE, Broderick JE, Hufford MR (2002).** "Patient non-compliance with paper diaries." *BMJ* 324(7347):1193–1194. DOI: 10.1136/bmj.324.7347.1193 The recall-bias evidence — why symptom tracking has to be near-real-time to be reliable; practical implication for the diary design the chapter recommends.
- **da Silva ALG, Vieira LP, Dias LS, et al. (2023).** "Impact of long COVID on the heart rate variability at rest and during deep breathing." *Scientific Reports* 13:22695. DOI: 10.1038/s41598-023-50276-0 HRV-evidence work specific to LC; grounds the chapter's HRV paragraph.

Tier 2

- **Stone AA, Shiffman S (2002).** "Capturing momentary, self-report data: a proposal for reporting guidelines." *Annals of Behavioral Medicine* 24(3):236–243. DOI: 10.1207/S15324796ABM2403_09 Methodological companion to the 2002 BMJ paper above.
- **Ruijgt TM, Slaghekke A, Ellens A, Janssen KW, Wüst RCI (2026).** "Wearable Heart Rate Variability Monitoring, Autonomic Dysfunction and Post-exertional Malaise in Long COVID: An Observational Study." *Sports Medicine*, online first. DOI: 10.1007/s40279-026-02487-4 Current work on wearable HRV data and PEM thresholds.
- **Lank GK, Budhiraja S, Gaelen JI, et al. (2026).** "Characterizing Neuro-PASC outcome with the mobile Neuro-COVID recovery care companion application." *BMC Neurology* 26(1):24. DOI: 10.1186/s12883-025-04577-8 Sixty-three patients on an app-based symptom tracker across three months; the improvers were the ones whose recovery scores fluctuated most.

Chapter 10 — Talking to your doctor

Tier 1

- **Baptista SN, Atkins T, Chakraborty S, Bakhit M, Glasziou P, Byambasuren O (2026).** "Candidate treatments for long COVID: a narrative review of expert and patient-driven priorities." *Frontiers in Medicine* 13:1734600. DOI: 10.3389/fmed.2026.1734600 The source for the "only six of fourteen have any LC-specific randomised-trial evidence" frame around the navigator paragraph — essential if you want to see where the treatment evidence currently stands.
- **Bramante CT, et al. (2023).** "Outpatient treatment of COVID-19 and incidence of post-COVID-19 condition over 10 months (COVID-

OUT): a multicentre, randomised, quadruple-blind, parallel-group, phase 3 trial." *The Lancet Infectious Diseases* 23(10):1119–1129. DOI: 10.1016/S1473-3099(23)00299-2 The metformin / acute-phase prophylaxis trial the chapter describes, with the explicit window caveat for readers already past acute illness.

- **Bateman L, Bested AC, Bonilla HF, et al. (2021).** "Myalgic encephalomyelitis / chronic fatigue syndrome: essentials of diagnosis and management." *Mayo Clinic Proceedings* 96(11):2861–2878. DOI: 10.1016/j.mayocp.2021.07.004 The clinical-care consensus document to put in front of a doctor who is new to the condition.
- **U.S. ME/CFS Clinician Coalition.** Clinician handouts available at mecfscliniciancoalition.org. Printable one-pagers suitable for a first appointment — the physical artefact the chapter's "hand this to your doctor" phrase refers to. https://mecfscliniciancoalition.org
- **Shaw BH, Stiles LE, Bourne K, et al. (2019).** "The face of postural tachycardia syndrome — insights from a large cross-sectional online community-based survey." *Journal of Internal Medicine* 286(4):438–448. DOI: 10.1111/joim.12895 Patient-survey data (n=4,835) including the 77%-misdiagnosed-as-anxiety finding the chapter references.

Tier 2

- **Mathew GT, Novak P. (2026).** "Prevalence of Central Sensitization in Postural Tachycardia Syndrome." *JAMA Network Open* 9(1):e2553694. DOI: 10.1001/jamanetworkopen.2025.53694 Mechanism citation useful for the doctor-conversation framing around "system stuck in high alert."
- **RECOVER-AUTONOMIC (ivabradine arm, Appendix B).** Registry entry https://clinicaltrials.gov/study/NCT06305806; design and rationale in Fudim M, Novak P, Taub PR, et al. 2026. "Design and rationale of RECOVER-AUTONOMIC: a randomized platform

trial evaluating interventions for Long COVID postural orthostatic tachycardia syndrome." *American Heart Journal* 296:107384. DOI: 10.1016/j.ahj.2026.107384 Randomised, placebo-controlled, 181 participants; the trial completed in December 2025 and results had not been published when this book was written.

Chapter 11 — Psychological support

Tier 1

- **van der Kolk B (2014).** *The Body Keeps the Score: Brain, Mind, and Body in the Healing of Trauma.* New York: Viking. A 40-year clinical trauma psychiatrist's synthesis of how the nervous system holds threat; the book's frame for "being believed changes your biology" uses van der Kolk for the medial-prefrontal-watchtower picture, without adopting his more speculative serotonin and smoke-detector claims.
- **Dantzer R (2001).** "Cytokine-induced sickness behavior: mechanisms and implications." *Annals of the New York Academy of Sciences* 933:222–234. DOI: 10.1111/j.1749-6632.2001.tb05827.x The classic sickness-behavior review — why immune signals produce the slowing, withdrawal, and loss of appetite pattern.
- **Capuron L, Miller AH (2011).** "Immune system to brain signaling: neuropsychopharmacological implications." *Pharmacology & Therapeutics* 130:226–238. DOI: 10.1016/j.pharmthera.2011.01.014 Interferon-α-induced depression literature — direct evidence that immune signals can produce depressive symptoms without any psychological mediation.
- **Stephan KE, Manjaly ZM, Mathys CD, et al. (2016).** "Allostatic self-efficacy: a metacognitive theory of dyshomeostasis-induced fatigue and depression." *Frontiers in Human Neuroscience* 10:550. DOI:

10.3389/fnhum.2016.00550 The predictive-coding model of fatigue as a signal about the brain's confidence in its own body-forecasts; the paper under the book's "being heard by a clinician can lift fatigue within minutes" frame.

- **Callard F, Perego E (2021).** "How and why patients made Long Covid." *Social Science & Medicine* 268:113426. DOI: 10.1016/j.socscimed.2020.113426 Reappears here as the citation under the borrowed-reality / peer-community paragraph — the psychological function of a community whose members have the same illness.

Tier 2

- **Guedj E, Campion JY, Dudouet P, et al. (2021).** "^{18}F-FDG brain PET hypometabolism in patients with long COVID." *European Journal of Nuclear Medicine and Molecular Imaging* 48(9):2823–2833. DOI: 10.1007/s00259-021-05215-4 The first systematic FDG-PET description of cerebral hypometabolism in LC.
- **Zhu Y, Quan P, Yamazaki T, Norweg A, Natelson B, Xu X (2025).** "Metabolic neuroimaging of ME/CFS and Long-COVID." *Immunometabolism* 7(4):e00068. DOI: 10.1097/IN9.0000000000000068 Subsequent-cohort confirmation and synthesis of FDG-PET findings in post-infectious illness.
- **Visser D, Golla SSV, Palard-Novello X, et al. (2025).** "Varying Levels of Inflammatory Activity in Brain and Body of Patients with Persistent Fatigue and Difficulty Concentrating After COVID-19: A TSPO PET Study." *Journal of Nuclear Medicine* 66(11):1787–1794. DOI: 10.2967/jnumed.124.269297 Forty-seven people scanned at the two-year mark, all high-affinity TSPO binders by genotype. Raised binding appeared in a subset, varied widely between people, and did not correlate with symptom severity — the basis for the inflammatory-subtype reading in Chapter 11.

- **Visser D, Pieperhoff L, Lorenzini L, et al. (2026).** "Decreased functional connectivity in post-COVID syndrome patients with high neuroinflammatory activity." *NeuroImage* 333:121951. DOI: 10.1016/j.neuroimage.2026.121951 Ten of forty-five showed neuroinflammation on [^{18}F]DPA-714 PET — the source of the second figure quoted in Chapter 11.
- **VanElzakker MB, Bues HF, Brusaferri L, et al. (2024).** "Neuroinflammation in post-acute sequelae of COVID-19 (PASC) as assessed by [^{11}C]PBR28 PET correlates with vascular disease measures." *Brain, Behavior, and Immunity* 119:713–723. DOI: 10.1016/j.bbi.2024.04.015 Independent confirmation with a different tracer, and the paper that ties the imaging signal to circulating markers of vascular dysfunction.
- **Nieuwland JM, Nutma E, Philippens IHCHM, et al. (2023).** "Longitudinal positron emission tomography and postmortem analysis reveals widespread neuroinflammation in SARS-CoV-2 infected rhesus macaques." *Journal of Neuroinflammation* 20(1):179. DOI: 10.1186/s12974-023-02857-z Animal model, four macaques. Postmortem staining found TSPO on microglia, astrocytes and small-vessel walls alike — the reason a bright TSPO scan reports immune activity in tissue without naming the cell responsible.
- **Peter RS, Nieters A, Göpel S, et al. (2025).** "Persistent symptoms and clinical findings in adults with post-acute sequelae of COVID-19/post-COVID-19 syndrome in the second year after acute infection: a population-based, nested case-control study." *PLoS Medicine* 22(1):e1004511. DOI: 10.1371/journal.pmed.1004511 The Freiburg case-control work on LC phenotyping (982 cases against 576 matched controls) with CPET and cognitive testing; clinical-characterisation companion to the imaging literature.

Chapter 12 — Explaining to those around you

Tier 1

- **Miserandino C (2003).** "The Spoon Theory." butyoudontlooksick.com. The essay where the fixed-daily-allowance metaphor for chronic illness was first worked out; intellectual ancestor of the book's battery / tokens / usable-hours frames.
- **Callard F, Perego E (2021).** "How and why patients made Long Covid." *Social Science & Medicine* 268:113426. DOI: 10.1016/j.socscimed.2020.113426 Returns here in the patient-built-tools lineage — where much of the current vocabulary for invisible illness was first articulated.
- **Bateman Horne Center.** *ME/CFS & Long COVID Clinical Care Guide*, §17 "Caregiving." The section the chapter's caregiver-burnout paragraph rests on; short, practical, written for the household context. https://batemanhornecenter.org/clinical-care-guide/
- **U.S. Equal Employment Opportunity Commission.** *Americans with Disabilities Act* (1990, as amended). The U.S. statutory basis for workplace accommodation conversations discussed in the workplace-register section of the chapter. https://www.eeoc.gov/statutes/titles-i-and-v-americans-disabilities-act-1990-ada
- **U.S. Department of Justice, Civil Rights Division, and HHS Office for Civil Rights (2021).** *Guidance on "Long COVID" as a Disability Under the ADA, Section 504, and Section 1557.* The 2021 federal guidance that established LC can qualify as a disability; relevant for anyone in the US facing a workplace-accommodation conversation. https://www.hhs.gov/civil-rights/for-providers/civil-rights-covid19/guidance-long-covid-disability/index.html

Tier 2

- **Patient-Led Research Collaborative (PLRC).** Workplace and disability-accommodation resources curated for the LC community at

patientresearchcovid19.com. Patient-led practical materials adjacent to but distinct from the formal statutory guidance above. https://patientresearchcovid19.com

- **Medinger G, Altmann D (2022).** *The Long COVID Handbook.* London: Hutchinson Heinemann. The chapter on household communication is a calmer companion to this chapter; the book as a whole remains useful cross-theme reading.

Conclusion — Recovery as a project

Tier 1

- **Collins J (2001).** *Good to Great: Why Some Companies Make the Leap … and Others Don't.* New York: HarperBusiness. Source of the Stockdale Paradox — the two-things-at-once posture (unwavering faith in eventual outcome plus discipline to confront brutal current facts) the Conclusion borrows as its closing frame.
- **NIH RECOVER Initiative.** Program materials and patient-partnered research framework at recovercovid.org. The ongoing national research program the book sits inside; useful context for the closing "this is one artefact of an ongoing study" framing. https://recovercovid.org

Tier 2

Patient-led research and community resources

A separate list, because this body of work sits outside conventional journal pipelines and deserves to be found directly.

- **Patient-Led Research Collaborative (PLRC) —** patientresearchcovid19.com. The community-research group whose 2020 symptom survey set the methodological template for the rest

of the field. Ongoing publication list, interview archive, and research prioritisation reports. https://patientresearchcovid19.com

- **Body Politic** — the online community that incubated the Patient-Led Research Collaborative in early 2020; archived materials remain a historical record of how the framing came together in real time. https://patientresearchcovid19.com
- **Bateman Horne Center** — batemanhornecenter.org. Clinical-care guides, the NASA Lean Test handout, the HUA protocol, and the caregiver support materials all referenced in this book. Free to download. https://batemanhornecenter.org/
- **U.S. ME/CFS Clinician Coalition** — mecfscliniciancoalition.org. The physician-facing handouts designed for first appointments with a clinician unfamiliar with the condition. https://mecfscliniciancoalition.org
- **Workwell Foundation** — workwellfoundation.org. The clinical group behind the two-day CPET literature and most of the practical pacing work. https://workwellfoundation.org
- **ME Association (UK)** — meassociation.org.uk. A long-running patient-organisation publishing current clinical summaries, trial updates, and the monthly magazine that kept the field together through the decades when it had no institutional home. https://meassociation.org.uk
- **DePaul Center for Community Research** — the DePaul Symptom Questionnaire and the Jason lab's work on diagnostic-criteria discrimination.
- **Long COVID Physio** — longcovid.physio. Physiotherapy-led patient-facing resources written for PEM-aware activity management. https://longcovid.physio

The project will keep reading, and we will note where the evidence moves. This shelf is where it stood in the spring of 2026.

Appendix 1. Symptom diary

How to use this diary

A diary is the cheapest medical instrument you will ever own and the one most likely to change a clinical conversation in your favour. Its job is not to remember everything; its job is to catch what your nervous system cannot — the payback window, the forty-eight to seventy-two hour gap between an effort and its bill, the difference between a five-minute upright phone call and a five-minute upright phone call after a poor night, the slow drift of a baseline that no single day's sensation can read.

Two warnings before you start. The first: more pages do not equal more information. A diary that asks for thirty fields will be abandoned by the second week; one that asks for six can run for six months. The second: a diary is a measurement, not a verdict on you. If a week looks bad on paper, the bad week was already there before you wrote it down — the diary just makes it legible. The point is to find the patterns that are already in your body, not to grade your performance.

For the first two weeks, change nothing. Record only. The patterns surface around days nine to fourteen, and the temptation to act on day three almost always misreads what is actually happening. After two weeks, the diary becomes a planning instrument: pacing decisions (Chapter 6), the data you bring to a clinician (Chapter 10), and the one-pager you sit down and assemble (Chapter 9) all read off this same kitchen-table file.

The weekly template

Print one copy per week. Keep it somewhere you will see it — the kitchen counter, the bedside table, a fridge magnet — not in a folder you will open once.

WEEK OF: ____________________

| Mon | Tue | Wed | Thu | Fri | Sat | Sun |

-——————————|——|——|——|——|——|——|——|

Morning resting HR | | | | | | | |

(supine, before standing)| | | | | | | |

-——————————|——|——|——|——|——|——|——|

Standing HR @ 2 min | | | | | | | |

(after 5 min lying) | | | | | | | |

-——————————|——|——|——|——|——|——|——|

Orthostatic tax | | | | | | | |

(standing − supine) | | | | | | | |

-——————————|——|——|——|——|——|——|——|

Sleep quality | | | | | | | |

(1 unrefreshing → 10 | | | | | | | |

fully refreshed) | | | | | | | |

-——————————|——|——|——|——|——|——|——|

HUA — hours of upright | | | | | | | |

activity (see Chapter 9) | | | | | | | |

-——————————|——|——|——|——|——|——|——|

Energy at midday | | | | | | | |

(1 → 10) | | | | | | | |

-——————————|——|——|——|——|——|——|——|

Cognitive clarity | | | | | | | |

at midday (1 → 10) | | | | | | | |

-——————————|——|——|——|——|——|——|——|

PEM today? (Y/N) | | | | | | | |

If Y, trace to which | | | | | | | |

day's activity (−1, −2, | | | | | | | |

−3 days) | | | | | | | |

-———————————|———|———|———|———|———|———|———|

Activity log (free-text, same page or facing page)

For each block of activity, write one line in this form:

[time] — [what you did] — [body position] — [duration]

Examples that work:

- *09:15 — answered email — seated at desk — 25 min*
- *11:00 — walked to school — upright outdoor — 12 min*
- *14:30 — phone call with sister, difficult — seated — 35 min*
- *18:00 — chopped vegetables for dinner — standing at counter — 15 min*

Examples that do not work (too vague to read patterns from later):

- *did some work*
- *short walk*
- *normal day*

Body position matters because cerebral perfusion drops on standing in this population — Chapter 4 covers the mechanism — and a thirty-minute upright task is a different cost from a thirty-minute reclining task. Cognitive load matters because a difficult email read seated still asks the brain to work under reduced perfusion. Emotional load matters because a sympathetic surge from a hard conversation spends from the same envelope as a flight of stairs. The four spending categories from Chapter 6 — physical, cognitive, orthostatic, emotional/sensory — are the lens. The activity log is where you actually see them.

Triggers and PEM episodes (free-text, end of day)

Two questions to answer before bed:

1. **Did anything today feel disproportionate to what you spent?** Light that felt too bright, conversation that felt too loud, a normal task that felt unusually heavy. (These are early-warning signals; record them whether or not they led anywhere.)
2. **If a PEM episode arrived today, what is the most plausible trigger from the activity log of the previous one to three days?** Write the date and the activity, not just the trigger type. ("Yesterday's school pickup, walked uphill" is more useful than "exertion.")

Food and supplements (one line per day)
A single column tracking *anything new today* — a new supplement, a food eaten outside your usual pattern, a missed meal, a caffeine spike, an alcohol exposure. The point is not nutrition; the point is a reproducible record of variables that change so you can read against them later.

Mood and stress (one line per day)
Two numbers, 1–10. Not a self-evaluation. A reading. Mood (1 = lowest you have been, 10 = best you remember being recently). Stress (1 = calm, 10 = acute). These two numbers, written the same way each day, become the cheapest correlate-tracker you have for the question *did emotional load buy a flare three days later*.

The monthly summary

At the end of each calendar month, transfer the bare numbers from your weekly sheets onto one page. The format is fixed:

MONTH OF: ____________________

PEM episodes this month: ___ (count days marked Y)

Trace pattern: most common trigger window

(−1 day / −2 days / −3 days from PEM): ___

Average HUA: ___ hours

Highest single day: ___ Lowest single day: ___

Average orthostatic tax (standing HR − supine HR): ___ bpm

Highest single day: ___ Lowest single day: ___

Average sleep quality: ___ / 10

Average energy (midday): ___ / 10

Average cognitive clarity (midday): ___ / 10

Days that did NOT produce a delayed worsening

(check Chapter 6 — the "repeatability test"): ___ days

Notable patterns (free-text, max 5 lines):

1. ____
2. ____
3. ____
4. ____
5. ____

One question for next month's clinician conversation:

____

Three months of these summary pages, stapled, are the document that does the most work in a fifteen-minute appointment. The clinician reads numbers more readily than narratives. You are giving them numbers.

Example: a filled-in day

A hypothetical reader's Wednesday entry, six weeks into the diary.

DATE: Wed, 7 May

Morning resting HR (supine): 72 bpm

Standing HR @ 2 min: 104 bpm

Orthostatic tax: +32 bpm

Sleep quality: 5 / 10 (woke at 04:30, back to sleep ~05:30)

Activity log:

07:00 — got up, lay on sofa with electrolyte drink — reclined — 20 min 07:30 — made breakfast — seated at counter — 10 min 08:15 — school *drop-off, walked the dog — upright outdoor — 25 min 09:00 — emails — seated at desk — 40 min 09:45 — lay down, eyes closed — supine — 15 min 10:00 — phone call with insurance, frustrating — seated — 30 min 10:30 — lay down — supine — 20 min 11:00 — short shopping trip, used the cart — upright — 35 min 12:00 — lunch — seated — 20 min 13:00 — read a chapter of a novel — reclined — 40 min 14:30 — picked the children up — upright + driving — 25 min 15:30 — homework supervision, seated at table — seated — 45 min 17:00 — chopped vegetables, sat on a stool — seated at counter — 20 min 18:00 — dinner — seated — 30 min 19:30 — bath, husband took kids — reclined warm water — 25 min 20:30 — bed, audiobook, no screens — reclined — until sleep*

HUA today (cumulative time feet on floor / non-reclined sit): 4.5 hours

Energy at midday: 4 / 10

Cognitive clarity at midday: 5 / 10

PEM today? N — but heavier than Tuesday

Trace if Y: —

New today (food/supplements): started magnesium glycinate, evening dose

Mood: 6 / 10

Stress: 7 / 10 (insurance call)

End-of-day notes: "The insurance call cost more than the school run. I can feel it now in the heaviness behind the eyes. Booking the next call for the morning, before noon, when I have more reserve. Also — the stool at the kitchen counter worked. Dinner did not feel like the bill it usually feels like."

The day is well-documented in roughly twelve minutes of writing across the day, mostly during natural rest pauses. It captures: the orthostatic tax (+32, at the top of the POTS-suggestive band), the cognitive cost of an emotional phone call, a working pacing tool (the kitchen stool), and one new variable (magnesium glycinate) flagged for the next two-week observation window. None of those four findings would have survived a weekly recall — they would have melted into “a tired Wednesday.”

How this diary feeds the next conversations

Three places in this book read off the diary directly.

Chapter 6 — pacing. The repeatability test (“can you do this today and still do tomorrow and the day after”) needs an activity log with body position to be answerable. Once you have two weeks of entries, the highlighted days that did not produce a delayed worsening become your starting baseline.

Chapter 9 — data over feelings. The one-pager you assemble for clinical conversations is condensed from this diary’s weekly and monthly sheets. HUA, orthostatic tax, and standing pulse are the three numbers most likely to be understood on sight by a clinician who has never read this book; the diary is where they come from.

Chapter 10 — talking to your doctor. The twenty-second opening (trigger, pattern, function, one specific ask) needs a function delta — *“I used to walk for an hour; now I can do twenty minutes before standing pulse goes over a hundred and ten”* — and the diary is the source for that sentence. Bring the monthly summary pages, not the weekly sheets, to the appointment. Three stapled summary pages travel better than twelve loose weekly grids and are easier for a clinician to scan in the time they have.

A short note on what to skip

Some diary fields that look useful in principle are not worth the effort:

- **Step counts on a phone or watch.** Steps do not read orthostatic load, do not read cognitive load, and are systematically wrong on days you carry the phone differently.
- **Calorie counts or strict food logs.** Unless food is the question you are testing this week (a histamine probe, an elimination trial — see Chapter 7), a single line for *anything new today* is enough.
- **Long narrative entries every day.** A two-sentence end-of-day note is sustainable. A two-paragraph daily journal is not, for a population whose cognitive reserve is the limiting factor; the diary that runs for six months is the diary that captures the patterns.
- **Symptom-rating every symptom on a long list.** Pick the three to five symptoms that change most for you (for many readers: post-exertional heaviness, brain fog, sleep, lightheadedness on standing, headache). Track those. A flat list of forty checkboxes does not produce information you can act on; it produces a daily reminder of how many things are wrong.

The diary's job is to be readable in one minute, fillable in twelve, and printable in a year as the document that explains your illness in a language the clinic understands. Anything that does not serve those three functions is decoration.

Appendix 2. Doctor summary

What this page is for

A ten-minute appointment has a bandwidth problem, and Chapter 9 explains why: your illness runs in time series, and the visit is built to absorb a single snapshot. This page is the compression. It holds seven blocks, it fits on one side of one sheet, and a clinician can read it in under a minute.

One page. Not two. The limit is the point — two pages get skimmed, and the second one is where the ask usually lives.

Fill it in from your diary rather than from memory. Chapter 9 covers why that difference matters; the short version is that recall reorders events and loses the timing, and timing is the whole diagnostic signal in this illness.

Send it through the clinic's patient portal twenty-four to forty-eight hours before the visit if you can, and bring a printed copy in any case. A clinician who has already seen a structured page spends the appointment on your problem instead of on decoding it.

The template

Print one copy per appointment. Handwriting in the blanks is fine — this page is for reading, not for filing.

1. Main problem, one sentence — with the suspicion and the ask both explicit.

```
Suspected _______________________ following
_______________ in ____________ [month, year]. Requesting
_______________________________________________.
```

This is the sentence the clinician reads first and uses to orient the whole appointment.

2. Red flags today — present or absent.

```
Chest pain: ______ Fainting or near-fainting: ______
Falling oxygen saturation: ______ Leg swelling: ______
Anything new and severe in the last week: ______
```

Answer these even when all the answers are no. It tells the clinician you are not in acute danger, which clears the room for the chronic picture.

3. Timeline — three lines.

```
Before this, I could:
______________________________________________ It
started: ____________ [date], after ____________
[trigger, if known] I have been in the current state for:
____________
```

4. Three patterns, in plain language. Three, not five.

1. ____
2. ____
3. ____

This is the part the clinician will actually remember. Name mechanics rather than feelings: intolerance of standing still, delayed crashes after activity, disrupted sleep.

5. Objective data — one compact line.

```
Supine pulse ______ Standing pulse ______ Orthostatic tax
______ Repeatable across ______ mornings One observation
from the diary showing the delay:
______________________________________________
```

6. Function, before and now.

```
Previously:
_______________________________________________ Currently:
_______________________________________________
```

Numbers alone do not carry a sense of the change. This block does.

7. The ask. One, two, at most three items.

```
1. ______________________________________________________
2. ______________________________________________________
3. ______________________________________________________
```

A specific ask is what keeps the visit from ending with "let's see how you do" by accident.

A filled-in example

This is Catherine's page, written at month eight, the morning after she completed the DePaul Symptom Questionnaire with her diary open. It is one example, not a form to copy — your numbers, patterns and asks will be your own.

Main problem. Suspected post-infectious dysautonomia with post-exertional malaise following COVID in March 2021. Requesting an orthostatic evaluation, a referral to autonomic medicine if indicated, and a conversation about whether low-dose pharmacologic support is appropriate.

Red flags today. Chest pain: no. Fainting or near-fainting: no, though light-headedness on standing daily. Falling oxygen saturation: no. Leg swelling: no. Anything new and severe this week: no.

Timeline. Before this I taught a full classroom load, walked around eight thousand steps a day, and ran a household with two sons and a Labrador. It started in March 2021, six weeks after a mild COVID infection. I have been in the current state for eight months.

Three patterns. Orthostatic intolerance — symptoms on standing still, relieved by lying flat. Delayed crashes at forty-eight to seventy-two hours after both physical and cognitive load. Early-morning wakings, which improved with head-of-bed elevation.

Objective data. Supine sixty-eight, standing one hundred and four at five minutes, orthostatic tax thirty-six. Repeatable across three separate mornings. Diary shows the crash arriving on day two after a long phone call, not on the day of the call. DePaul Symptom Questionnaire attached, PEM score moderate-to-severe.

Function. Previously: full-time teaching, eight thousand steps, showering standing. Currently: working at fifty per cent, pacing ceiling of eighty-seven beats, unable to stand in the shower longer than three minutes.

Ask. Orthostatic vitals measured today. Referral to autonomic medicine if the tachycardia pattern replicates in clinic. A discussion of low-dose pharmacologic options.

What this page does not do

It does not diagnose you, and it is not a medical record. It is a summary you wrote, and a clinician is free to read it, question it, or set it aside.

It also does not replace the conversation. Chapter 10 covers the twenty-second spoken introduction that goes with this page, and the scripts for the four moments where an appointment most often goes sideways. The page buys you the minutes; the chapter is about how to spend them.

Glossary

This is a quick-reference glossary, not a textbook. It covers the technical and clinical terms that surface across the book, with a short plain-language definition for each and a pointer to the chapter where the term does most of its work. Where a term has a locked meaning inside the book (for example, “orthostatic tax” or “Hours of upright activity”), that meaning is the one given here. Use it when you hit a word you do not know, and then go back to the chapter that uses it.

A

Active Stand Test. Clinical version of the home NASA Lean Test. The patient lies supine for several minutes, stands up, and has pulse and blood pressure recorded at timed intervals. Produces heart-rate and stroke-volume trajectories broadly similar to head-up tilt, but detects fewer cases: 74% of POTS patients met the heart-rate criterion on active stand against 98% on tilt (Uppal 2025). A positive result is strong evidence; a negative one does not rule POTS out. Chapter 4.

ADA (Americans with Disabilities Act). 1990 U.S. federal statute, as amended, that requires employers to provide reasonable accommodation for qualifying disabilities. The 2021 DOJ guidance established that Long COVID can qualify. The chapter treats the ADA as a statutory anchor for workplace conversations rather than as legal advice. Chapter 12.

Allostatic load. The cumulative physiological cost of repeated or chronic stress on the body’s regulatory systems. Distinct from acute stress. In this book the concept sits behind the thermostat metaphor: it describes why the system that once returned to baseline now no longer does. Chapters 2, 11.

Allostatic self-efficacy. Stephan and colleagues' predictive-coding model (2016) in which fatigue functions as the brain's running report on its confidence in its own forecasts about bodily homeostasis. Low self-efficacy lifts fatigue; being heard and helped by a clinician can demonstrably lift it within minutes. Chapter 11.

ANS (Autonomic Nervous System). The background regulation layer controlling heart rate, blood pressure, vessel tone, digestion, temperature, and sweating. It runs without conscious input. ELCs often present with ANS dysregulation, most visibly as orthostatic intolerance and heart-rate variability changes. Chapters 2, 4, 5.

ATP (Adenosine Triphosphate). The cell's short-term energy currency. Aerobic metabolism yields roughly thirty molecules per glucose; anaerobic yields two. The difference explains why muscles that lose aerobic capacity rely on anaerobic pathways far earlier than expected, and why exertion produces disproportionate fatigue. Chapters 1, 6, 7.

Autoantibody. An antibody produced by the body that binds to one of the body's own structures. Long COVID research has documented autoantibodies against adrenergic and muscarinic G-protein coupled receptors that plausibly contribute to autonomic dysregulation (Wallukat 2021). Chapter 3.

B

BBB (Blood-Brain Barrier). The selective tight-junction layer separating brain tissue from the bloodstream. Many molecules, including serotonin, cannot cross it, which is why peripheral serotonin measurements do not speak directly to brain serotonin levels. Chapter 7.

Boom-and-bust. A repeating pattern in ELCs: a good day leads to overspending on activity, which produces a multi-day crash, which leads to underspending and a new good day. Pacing is the operational tool for breaking the cycle. Chapter 6.

C

CAP (cranio-cervical associated pathology / community-acquired). Abbreviation listed here because readers sometimes encounter it in clinical notes without context. The book itself does not use this acronym as a technical term.

Central sensitization. Measurable change in how the central nervous system handles incoming signals — descending pain inhibition is weakened and ordinary stimuli are amplified. Documented in POTS cohorts by Novak and colleagues (2026). Appears in ELCs as part of the "system stuck in high alert" picture. Chapters 3, 10.

CCI / AAI / TCS (cranio-cervical instability, atlantoaxial instability, tethered cord syndrome). Structural conditions sometimes discussed in the connective-tissue-hypermobility neighbourhood around POTS and hEDS. The book mentions them only in passing; diagnosis and management require specialist evaluation and fall outside this book's scope. Chapter 3.

Central venous pooling. Redistribution of blood into peripheral veins during standing, reducing the volume returning to the heart. In POTS, the pooling is exaggerated, which forces the heart to race to maintain brain perfusion. Chapters 2, 4.

Cerebral hypometabolism. Reduced glucose uptake in specific brain regions visible on FDG-PET. Consistently documented in Long COVID cohorts, concentrated in olfactory-limbic and brainstem regions, and correlated with cognitive symptom severity (Guedj 2021; Zhu 2025). Chapter 11.

CPET (Cardiopulmonary Exercise Test). Graded exercise test measuring breath-by-breath gas exchange, heart rate, and workload. A two-day CPET protocol exposes PEM through a day-2 drop in peak oxygen uptake, which is one of the most specific objective signatures of the condition. Chapters 1, 6, 9.

D

DAO (Diamine Oxidase). The gut-lining enzyme that breaks down dietary histamine. When DAO capacity is overwhelmed or reduced, histamine loads enter the bloodstream and may contribute to MCAS-adjacent symptoms. Chapter 7.

Deconditioning. Loss of cardiovascular and muscular capacity from prolonged inactivity, reversible by graded movement. Distinct from PEM. Some LC research uses deconditioning to explain exertional intolerance; the two-day CPET evidence argues against deconditioning alone as a complete explanation. Chapters 1, 4, 6.

DePaul Symptom Questionnaire (DSQ). A validated multi-domain questionnaire (developed by Leonard Jason's group) that captures PEM, sleep, cognitive symptoms, pain, and orthostatic symptoms across a six-month window. Used in research and, increasingly, in clinical settings. Chapter 9.

Dysautonomia. Umbrella term for disrupted autonomic-nervous-system regulation. POTS, orthostatic hypotension, and neurally-mediated hypotension are the subtypes most relevant to ELCs. Chapters 2, 3, 4.

E

EDS / hEDS (Ehlers-Danlos Syndrome / hypermobile Ehlers-Danlos Syndrome). A family of connective-tissue disorders; hEDS is the most common form. Roughly 31% of POTS patients meet hEDS criteria (Miller 2020). Loose connective tissue in vessel walls and the gastrointestinal tract plausibly contributes to the orthostatic and gastrointestinal symptom overlap. Chapters 1, 3.

ELC (Energy-Limiting Conditions). Patient-literature umbrella term for conditions characterised by reduced energy envelope and post-exertional worsening: ME/CFS, Long COVID, POTS, and adjacent post-infectious illnesses. Used throughout this book as a reader-facing synonym for the clinical IACC umbrella. Used across the book.

F

Fibromyalgia. Chronic widespread pain with sleep, fatigue, and cognitive features. Roughly 41% of fibromyalgia patients show small-fibre polyneuropathy on biopsy (Oaklander 2013), which suggests neuropathic mechanisms the field is still working out. Chapters 3, 5, 7, 11.

FDG-PET (^{18}F-fluorodeoxyglucose positron emission tomography). Nuclear-medicine imaging that tracks glucose uptake across the brain. Long COVID cohorts show hypometabolism in specific regions at scan windows months to years after initial infection. See cerebral hypometabolism. Chapter 11.

FUNCAP. A functional-capacity scale used in ME/CFS and Long COVID research. FUNCAP-55 serves as the primary endpoint in the LIFT LDN trial. Appears in the reading list rather than the main text.

G

Glymphatic system. The nighttime brain-waste-clearance pathway running along perivascular spaces around cerebral vessels. Most active during slow-wave sleep, when the system flushes amyloid-beta, tau fragments, and metabolic by-products. Dependent on adequate slow-wave sleep rather than total sleep hours. Chapter 8.

H

H1 / H2 antagonists (antihistamines). H1 blockers (for example, fexofenadine) reduce classical allergic-type histamine signaling. H2 blockers (for example, famotidine) reduce the gastric and vascular arms. The combination is used in some clinical protocols for Long COVID patients with MCAS-consistent symptoms (Salvucci 2024, n=27, open-label). Chapters 3, 7.

HRV (Heart Rate Variability). The beat-to-beat variation in time between consecutive heartbeats. A proxy for autonomic balance: higher variability at rest generally indicates better autonomic flexibility. In ELCs it is a useful signal but a poor dictator, and is best read alongside three other inputs rather than alone. Chapters 8, 9.

HUA (Hours of Upright Activity). The cumulative daily time each day with feet on the floor — standing, walking, or sitting without legs elevated. Healthy adults live at twelve to sixteen hours; people with moderate ME/CFS are often below five; severe disease often below two. Developed at Bateman Horne Center. The book’s primary-home definition is in Chapter 9.

I

IACC (Infection-Associated Chronic Conditions). The U.S. Department of Health and Human Services umbrella term for chronic conditions triggered by prior infection, including Long COVID, ME/CFS, POTS, MCAS, post-Lyme syndrome, and post-EBV conditions. The book treats the IACC frame as the current best institutional scoping. Chapter 0.

IBS (Irritable Bowel Syndrome). A symptom-defined gastrointestinal disorder with altered motility, bloating, and pain, often co-occurring with post-infectious dysautonomia. In the book's context, IBS features commonly overlap with POTS and with MCAS-adjacent mast-cell symptoms. Chapters 3, 7.

Interoception. The brain's perception of internal bodily signals, anchored in the insular cortex. Chapter 3 introduces interoception in order to name one reason ELC symptoms can feel overwhelming even when traditional tests come back normal.

Interoceptive mismatch. A useful frame from the Khalsa and colleagues review (2018): the brain is reading the body correctly, but the input itself has changed. The report is not a perceptual error; the signal is unusual. Chapter 3.

IOM 2015 / NASEM 2024 ME/CFS diagnostic framework. The Institute of Medicine (now NASEM) 2015 report — updated in 2024 — that identifies PEM as a cardinal, rather than optional, feature of ME/CFS. The framework requires fatigue, PEM, unrefreshing sleep, and either cognitive impairment or orthostatic intolerance. Chapter 0.

IVIg (Intravenous Immunoglobulin). Pooled human-donor immunoglobulin infused into the bloodstream as an immune-modulating therapy. The iSTAND trial (Vernino 2024) was placebo-controlled and did not reach a positive primary endpoint for autoimmune POTS; subgroup signals remain under investigation. Chapter 3.

K

Kin-protection hypothesis. The evolutionary frame for sickness behavior: behavioural slowing, withdrawal, and appetite loss reduce onward transmission within a group. The cost to the individual is tolerable at population scale. The book pairs this frame with the trade-offs frame in Chapter 11 without assigning blame to the body.

L

LC (Long COVID). The post-acute syndrome following SARS-CoV-2 infection. The term was coined by patients in May 2020 and entered the medical literature from there. Used throughout the book as a working synonym for the LC subset of the broader IACC family.

LDN (Low-Dose Naltrexone). Naltrexone at 0.5–6 mg daily, roughly a tenth of the dose used for opioid-use disorder. Proposed mechanisms include microglial modulation and partial restoration of TRPM3 ion-channel function in natural-killer cells (Sasso 2025). Clinical evidence in LC is small, short-duration, and mostly non-randomised. Chapter 7.

M

Mast cell activation. Mast cells releasing mediators (histamine, tryptase, prostaglandins, leukotrienes, cytokines) outside the context of ordinary allergic triggers. MCAS is the clinical label when the pattern meets formal criteria. Chapters 3, 7, 8.

MCAS (Mast Cell Activation Syndrome). A clinical syndrome of inappropriate mast-cell mediator release producing multi-system symptoms (skin, gastrointestinal, respiratory, cardiovascular, neurological). Diagnostic criteria remain debated; the Weinstock 2021 finding is that LC symptom patterns overlap with MCAS rather than that a fixed percentage of LC patients have MCAS. Chapters 0, 3, 7.

ME/CFS (Myalgic Encephalomyelitis / Chronic Fatigue Syndrome). The decades-old diagnostic category most structurally similar to Long COVID. The clinical and patient community whose prior work the current post-infectious-illness field stands on. Used throughout the book.

Microclots. Very small fibrin-containing clots proposed as contributors to impaired microcirculation in Long COVID. Not visible on standard coagulation tests; the mechanism is promising but the evidence remains preliminary. Chapters 0, 1, 11.

Muscle pump. The contraction of leg muscles during walking that squeezes blood back toward the heart. Idle during quiet standing, which is why standing still is harder than walking for patients with orthostatic intolerance. Chapters 4, 5.

N

NASA Lean Test. A ten-minute home-administered stand test. The patient rests supine for five minutes, then stands with heels fifteen to twenty centimetres out from a wall while pulse and blood pressure are recorded at timed intervals. Developed from astronaut orthostatic-intolerance work; adopted by Bateman Horne Center. Chapters 4, 5, 9.

NICE (National Institute for Health and Care Excellence). UK clinical-guideline body. NICE NG206 (2021) replaced graded exercise therapy with pacing as first-line activity management in ME/CFS, a shift the book references repeatedly. Chapters 1, 6.

O

OI (Orthostatic Intolerance). Umbrella term for upright-symptom conditions: POTS, orthostatic hypotension (OH), and neurally-mediated hypotension (NMH). Distinguishable by pulse and blood-pressure patterns during a standing or tilt test. Chapters 2, 4, 5.

Orthostatic hypotension (OH). A drop in systolic blood pressure of at least 20 mmHg or diastolic of at least 10 mmHg within three minutes of standing. An OI subtype distinct from POTS. Chapter 4.

Orthostatic tax. The book's working term for the derived home measurement: standing pulse minus supine pulse, typically measured five to ten minutes after standing. Locked in Chapter 4 as the single most useful Long-COVID home metric. Carries through the symptom diary, and the data chapter.

ORS (Oral Rehydration Salts). The WHO-formulated sodium-glucose powder designed to maximise intestinal fluid absorption via sodium-glucose co-transport. More effective than plain water or sports drinks for restoring plasma volume in POTS. Chapter 5.

P

Pacing. The system of accounting and scheduling activity to stay under the PEM threshold. Current first-line activity management in the NICE ME/CFS guideline and the Bateman Horne Center clinical-care guide. The opposite of graded exercise as an operational strategy. Chapters 5–9, 11, 12.

PAMP (Pathogen-Associated Molecular Pattern). A molecular feature the innate immune system recognises as a sign of infection. Swank 2023 found that circulating SARS-CoV-2 spike protein behaves as a persistent PAMP in Long COVID patients, keeping the immune system partly activated months after the acute infection has cleared. Chapter 1.

Payback window. The book's term for the interval in which the cost of an effort actually arrives: most often 48 to 72 hours after the trigger, sometimes as early as a day out. Locked in Chapter 6 as the unit of planning, and used in that sense throughout. NICE guidance describes a wider range of twelve to seventy-two hours for the onset of PEM. Chapters 1, 6, 9, and the symptom diary.

PEM (Post-Exertional Malaise). The delayed worsening of symptoms after physical, cognitive, or emotional effort. Onset is most often 48 to 72 hours after the trigger — the window this book plans around — although NICE guidance describes a wider range of twelve to seventy-two hours; recovery takes days to weeks. The cardinal feature of ME/CFS and PEM-type Long COVID per the IOM / NASEM framework. Chapters 1, 6, 9.

PEM severity spectrum. The range from brief symptom flares lasting hours to multi-week collapses requiring bed-rest. Severity correlates with how far and how long activity exceeded the patient's current energy envelope. NICE NG206 (2021) provides formal criteria. Chapter 1.

PESE (Post-Exertional Symptom Exacerbation). Clinical synonym for PEM used in some chart notes and rehabilitation literature. Same phenomenon, different vocabulary. Chapter 10.

POTS (Postural Orthostatic Tachycardia Syndrome). A sustained heart-rate rise of at least 30 beats per minute (40 in adolescents) within ten minutes of standing, without a matching blood-pressure drop, accompanied by orthostatic symptoms. Diagnostic criteria from the Freeman 2011 consensus. Chapters 2, 4, 5.

POTS subtypes. Neuropathic (small-fibre denervation in lower limbs), hypovolemic (reduced plasma volume), hyperadrenergic (elevated norepinephrine), and autoimmune / post-infectious (G-protein-coupled-receptor autoantibodies). Mechanism determines which first-line interventions make sense. Chapters 4, 5.

Predictive coding. The neuroscientific framework under Stephan 2016′s fatigue model: the brain constantly generates forecasts about the body and updates them against incoming signals. Fatigue is a signal that forecasts have stopped being reliable. Chapter 11.

PROMIS (Patient-Reported Outcomes Measurement Information System). A family of validated questionnaires produced by the U.S. National Institutes of Health. PROMIS-Fatigue is the primary endpoint in the RECOVER-TLC LDN trial and appears in the reading list.

R

RECOVER. The NIH-funded U.S. national research initiative on post-acute sequelae of SARS-CoV-2 infection. Includes the RECOVER-AUTONOMIC trial network whose ivabradine arm reported out in 2026, and the Neuro-PASC and pediatric arms. A reference frame for where the evidence stands. Chapters 0, 10.

Recall bias. The systematic distortion that arises when patients report symptoms retrospectively rather than in real time. Retrospective accounts diverge from near-real-time entries by twenty to forty percent. Relevant for any diary-based symptom tracking. Chapter 9.

REM sleep (Rapid Eye Movement sleep). The dream-rich sleep stage that consolidates episodic memory and metabolises emotional weight, reducing next-day amygdala reactivity. REM fragmentation is common in ELCs and contributes to the characteristic daytime emotional fragility. Chapter 8.

S

SF-36. A widely used short-form health-survey questionnaire. The SF-36 Physical Functioning subscale appeared as an endpoint in Stein 2025

(Charité immunoadsorption) and Bonilla 2024 (LDN + NAD+ pilot). Reading-list reference.

Sickness behavior. The coordinated behavioural program triggered by immune signals: behavioural slowing, appetite loss, social withdrawal, and sleep change. Normally self-limiting during acute infection, the program can persist in chronic inflammation (Dantzer 2001; Capuron and Miller 2011). Chapter 11.

Slow-wave sleep. The deep, delta-band sleep stage where growth-hormone release peaks, tissue repair runs, glymphatic flushing occurs, and immune memory consolidates. Amount of slow-wave sleep matters more than total sleep hours for recovery. Chapter 8.

SLP / RD (Speech-Language Pathologist / Registered Dietitian). Two allied-health referral categories useful in ELC management: SLPs for cognitive rehabilitation and swallowing or voice symptoms; RDs for the food-as-a-tool work and for nutritional strategies that support pacing. Reading list for Chapter 5.

SOCRATES. A clinical mnemonic (Site, Onset, Character, Radiation, Associated symptoms, Timing, Exacerbating / Easing factors, Severity) for structured symptom description. Adapted in Chapter 10 for systemic illness, where timing and function often do more diagnostic work than location.

Spike protein. The viral surface protein of SARS-CoV-2. In some Long COVID patients, spike fragments remain in circulation for months after the acute infection, measurable with ultrasensitive assays (Swank 2023). Implicated in ongoing immune activation and mechanism debates around viral persistence. Chapter 1.

Splanchnic bed. The venous reservoir serving the gut and abdominal organs. Holds roughly a quarter of circulating blood volume and expands in dysautonomia, drawing blood away from the central circulation.

Abdominal compression outperforms leg-only compression because of this distribution. Chapters 5, 7.

T

taVNS (transcutaneous auricular Vagus Nerve Stimulation). A small skin electrode applied over the auricular branch of the vagus nerve. Pilot trials have tested taVNS in Long COVID and similar conditions (Badran 2022). The book distinguishes device-literature taVNS from consumer "vagal tone" marketing. Chapter 2.

Tilt table test. The clinical gold-standard assessment for orthostatic intolerance: the patient is strapped to a motorised table that tilts head-up while pulse and blood pressure are recorded. The active stand test and the home NASA Lean Test follow a similar pattern but are less sensitive, and tilt remains the more sensitive of the two for diagnosis (Uppal 2025). Chapters 2, 4, 6.

Trade-offs frame. A second frame on the sickness-behavior programme, operating on afternoon-scale rather than evolutionary-scale: the immune system and brain run their own energy budgets, and fatigue is what a body looks like while making real-time trade-offs between competing protective priorities. Paired with the kin-protection hypothesis in Chapter 11 without blaming the body.

TSPO-PET. Positron-emission-tomography imaging using tracers that bind the translocator protein, expressed by activated glial and immune cells in the brain. Visser 2025 and Nieuwland 2023 used TSPO-PET to document widespread low-grade neuroinflammation in Long COVID patients up to two years after infection. Chapter 11.

V

Vagus nerve. The tenth cranial nerve, carrying bidirectional signals between brain and body. Most fibres are inbound to the brain, and the largest inputs come from the lungs, heart, and airways — not primarily from the gut, as much popular literature suggests. Chapters 2, 7, 8.

Viral persistence. The presence of viral fragments (and, less commonly, live virus) in tissues after negative swab tests. Documented at autopsy in multiple organs (Stein 2022). Distinct from active viral replication; keeps the immune system partly activated and produces ongoing signal into sickness-behavior pathways. Chapters 1, 3, 7, 11.

VO_2 peak / VO_2max. Peak oxygen uptake measured during cardiopulmonary exercise testing. The day-2 drop in VO_2 peak on a two-day CPET is one of the most specific objective signatures of PEM (Keller 2014). Long COVID cohorts show reduced VO_2 peak independent of resting cardiopulmonary function. Chapters 1, 6.

Abbreviations at a glance

- **ADA** — Americans with Disabilities Act https://www.eeoc.gov/statutes/titles-i-and-v-americans-disabilities-act-1990-ada
- **ANS** — Autonomic Nervous System
- **ATP** — Adenosine Triphosphate
- **BBB** — Blood-Brain Barrier
- **CCI / AAI / TCS** — Cranio-Cervical Instability / Atlantoaxial Instability / Tethered Cord Syndrome
- **CPET** — Cardiopulmonary Exercise Test
- **DAO** — Diamine Oxidase
- **DSQ** — DePaul Symptom Questionnaire
- **EDS / hEDS** — Ehlers-Danlos Syndrome / hypermobile Ehlers-Danlos Syndrome

- **ELC** — Energy-Limiting Condition(s)
- **FDG-PET** — ^{18}F-fluorodeoxyglucose Positron Emission Tomography
- **FUNCAP** — Functional Capacity scale
- **H1 / H2** — Histamine-receptor 1 / Histamine-receptor 2 antagonists
- **HRV** — Heart Rate Variability
- **HUA** — Hours of Upright Activity
- **IACC** — Infection-Associated Chronic Conditions
- **IBS** — Irritable Bowel Syndrome
- **IOM / NASEM** — Institute of Medicine / National Academies of Sciences, Engineering, and Medicine
- **IVIg** — Intravenous Immunoglobulin
- **LC** — Long COVID
- **LDN** — Low-Dose Naltrexone
- **MCAS** — Mast Cell Activation Syndrome
- **ME/CFS** — Myalgic Encephalomyelitis / Chronic Fatigue Syndrome
- **NASA Lean Test** — home-adapted ten-minute orthostatic stand test
- **NICE** — National Institute for Health and Care Excellence (UK) https://www.nice.org.uk/guidance/ng206
- **OH** — Orthostatic Hypotension
- **OI** — Orthostatic Intolerance
- **ORS** — Oral Rehydration Salts
- **PAMP** — Pathogen-Associated Molecular Pattern
- **PEM** — Post-Exertional Malaise
- **PESE** — Post-Exertional Symptom Exacerbation
- **POTS** — Postural Orthostatic Tachycardia Syndrome
- **PROMIS** — Patient-Reported Outcomes Measurement Information System
- **RECOVER** — NIH Researching COVID to Enhance Recovery initiative
- **REM** — Rapid Eye Movement (sleep stage)
- **SF-36** — Short-Form 36 health survey

- **SLP / RD** — Speech-Language Pathologist / Registered Dietitian
- **SOCRATES** — Site, Onset, Character, Radiation, Associated, Timing, Exacerbating/Easing, Severity (clinical mnemonic)
- **taVNS** — transcutaneous auricular Vagus Nerve Stimulation
- **TSPO-PET** — Translocator Protein Positron Emission Tomography
- **VO_2 peak / VO_2max** — peak oxygen uptake